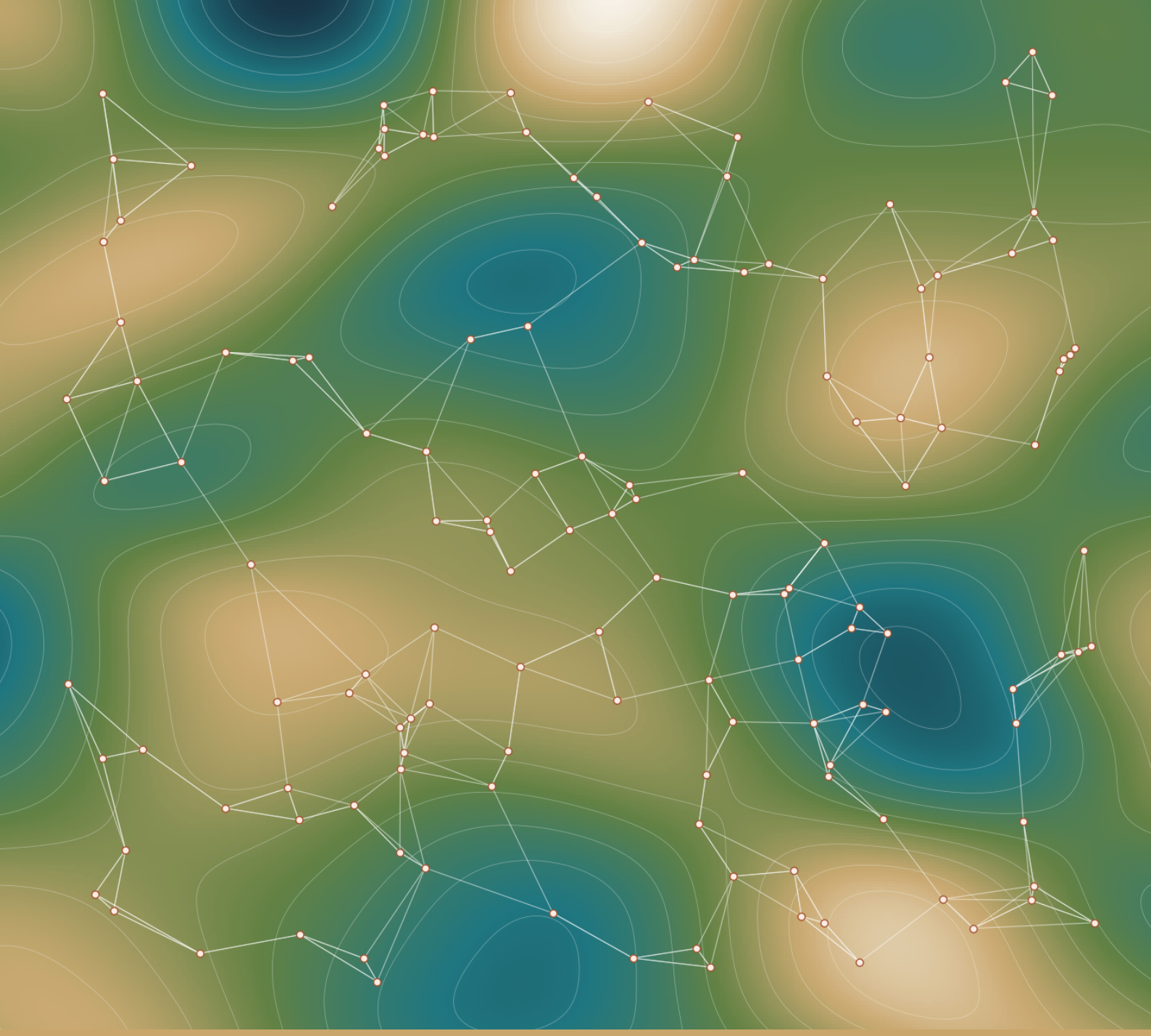


VISTA-Sindh

The 100-Day Research and Development Execution Plan

Village Intelligence for Spatiotemporal Tracking and Anticipation of Socioecological Decline in Sindh, Pakistan

VERSION 1.0 | 1 NOVEMBER 2026

Execution manual: contains a plan, definitions and checklists. It contains no results and no predetermined accuracy targets.

Table of contents

1. How to use this document . . . 7

2. Executive summary . . . 8

3. Project overview . . . 9

3.1 Research question and sub-questions . . . 9

3.2 Study region . . . 9

3.3 Objectives mapped to execution days . . . 9

3.4 Hypotheses (not established facts) . . . 9

4. Research methodology and definitions . . . 11

4.1 Design principles . . . 11

4.2 Unit of analysis . . . 11

4.3 Definitions used throughout . . . 12

4.4 Forecast design and temporal protocol . . . 12

5. Village inventory methodology . . . 14

5.1 Obtaining the inventory . . . 14

5.2 Matching village records to settlement footprints . . . 14

5.3 Duplicate village names . . . 14

5.4 Missing coordinates . . . 14

5.5 Consistent identifiers . . . 14

5.6 Settlement-core, resource-zone and connectivity-zone boundaries . . . 14

5.7 Measuring geographic coverage and matching accuracy . . . 15

6. Dataset catalogue and epistemic status . . . 16

7. Indicator handbook: formulas, interpretation, normalisation, limits . . . 22

8. AI architecture: VISTA-Net . . . 24

8.1 Input tensors and feature engineering . . . 24

8.2 Mathematical specification . . . 24

8.3 Graph construction . . . 25

9. Training strategy, baselines, ablations and reproducibility . . . 26

9.1 Model ladder (eight classes) . . . 26

9.2 Training procedure and hyper-parameters . . . 26

9.3 Ablation experiments (pre-registered) . . . 26

9.4 Reproducibility requirements . . . 26

10. Fallback architectures and decision rules . . . 28

11. Scientific validation framework . . . 29

11.1 Metrics by target type . . . 29

11.2 Spatial and temporal validation designs . . . 29
11.3 Leakage audit checklist (signed on Day 90) . . . 30
11.4 Statistical comparison and reporting policy . . . 30

12. Software implementation and code . . . 31

12.1 Stack . . . 31
12.2 Recommended directory structure . . . 31
12.3 Executable pseudocode and code modules . . . 31

Part B. Daily execution plans (Days 1-100) . . . 35

Phase 1: Research foundation and project setup (Days 1-10) . . . 35
Day 1: Project charter and scope lock . . . 35
Day 2: Literature protocol and first harvest . . . 35
Day 3: Evidence synthesis I: Pakistan and Sindh . . . 36
Day 4: Evidence synthesis II: methods and novelty matrix . . . 36
Day 5: Hypotheses, definitions and pre-analysis plan . . . 37
Day 6: Software environment, GEE access and quotas . . . 37
Day 7: Weekly review 1 and repository scaffold . . . 38
Day 8: Dataset inventory and asset verification . . . 38
Day 9: Methodology blueprint and compute budget . . . 39
Day 10: MILESTONE 1: protocol v1.0 and smoke test . . . 39
Phase 2: Village inventory and geospatial database (Days 11-25) . . . 40
Day 11: PBS Mouza Census 2020: acquire and parse Sindh tables . . . 40
Day 12: Administrative boundaries and crosswalks . . . 40
Day 13: Settlement evidence layers: footprints, OSM, GeoNames . . . 41
Day 14: Weekly review 2 and pilot settlement clustering . . . 41
Day 15: Province-wide settlement polygons . . . 42
Day 16: Names: attach, normalise and transliterate . . . 42
Day 17: Deh boundaries and assignment . . . 43
Day 18: Village identifier system and registry . . . 43
Day 19: Duplicate detection and resolution . . . 44
Day 20: Missing coordinates and quality classes . . . 44
Day 21: Weekly review 3 and validation sample design . . . 45
Day 22: Execute inventory validation . . . 45
Day 23: Village zones: core, resource and connectivity . . . 46
Day 24: Village atlas v0.1 and coverage report . . . 46
Day 25: MILESTONE 2: inventory freeze v1.0 . . . 47
Phase 3: Satellite data acquisition and processing (Days 26-40) . . . 47
Day 26: GEE zonal-statistics framework . . . 47
Day 27: Sentinel-2 preprocessing and cloud masking . . . 47
Day 28: Weekly review 4 and Sentinel-2 province run (batch 1) . . . 48
Day 29: Sentinel-2 province run (batch 2) and QC . . . 48
Day 30: Landsat Collection 2 harmonisation . . . 49
Day 31: Landsat extraction and cross-sensor calibration . . . 49
Day 32: Sentinel-1 SAR processing . . . 50
Day 33: Dynamic World integration . . . 50
Day 34: MODIS vegetation, LST and ET products . . . 51

Day 35: Weekly review 5 and CHIRPS rainfall . . . 51
Day 36: ERA5-Land climate variables and SPEI . . . 52
Day 37: Surface water: JRC GSW and S1/S2 extension . . . 52
Day 38: DEM, terrain and hydrology covariates . . . 53
Day 39: Exposure, accessibility and settlement context layers . . . 53
Day 40: MILESTONE 3: harmonised environmental dataset v0.9 . . . 54
Phase 4: Environmental indicator engineering (Days 41-55) . . . 55
Day 41: Indicator catalogue and data model . . . 55
Day 42: Weekly review 6 and spectral indices . . . 55
Day 43: Seasonal normalisation framework . . . 55
Day 44: Resource-zone sensitivity and vegetation productivity . . . 56
Day 45: Rainfall anomaly indicators . . . 56
Day 46: Land-surface temperature anomalies . . . 57
Day 47: Surface-water persistence and wetland dynamics . . . 57
Day 48: Flood frequency and persistence . . . 58
Day 49: Weekly review 7 and agricultural activity proxies . . . 58
Day 50: Settlement expansion and contraction signals . . . 59
Day 51: Environmental stress indicators (event-based) . . . 59
Day 52: Environmental recovery indicators . . . 60
Day 53: Indicator validation I: internal and cross-sensor . . . 60
Day 54: Indicator validation II: external reference data . . . 61
Day 55: MILESTONE 4: indicator dataset v1.0 and handbook . . . 61
Phase 5: Environmental trajectory analysis (Days 56-65) . . . 62
Day 56: Weekly review 8 and exploratory analysis with ecological regions . . . 62
Day 57: Long-term trend analysis . . . 62
Day 58: Seasonal decomposition . . . 63
Day 59: Persistent stress detection . . . 63
Day 60: Recovery trajectory identification . . . 64
Day 61: Change-point detection . . . 64
Day 62: Environmental clustering and typology . . . 64
Day 63: Weekly review 9 and spatial autocorrelation . . . 65
Day 64: Regional comparisons and target definition . . . 65
Day 65: MILESTONE 5: forecasting dataset v1.0 and splits . . . 66
Phase 6: Advanced AI development (Days 66-80) . . . 67
Day 66: Evaluation harness and data loaders . . . 67
Day 67: Baselines 1-2: climatology, persistence, autoregression . . . 67
Day 68: Baselines 3-4: Random Forest and XGBoost . . . 67
Day 69: LSTM forecaster . . . 68
Day 70: Weekly review 10 and Temporal Transformer . . . 68
Day 71: Village graph construction . . . 69
Day 72: Graph neural network baseline . . . 69
Day 73: Spatial feature encoder and chip pipeline . . . 70
Day 74: Climate encoder and cross-modal attention fusion . . . 70
Day 75: Village Graph Attention Network integration . . . 71
Day 76: Multivariate heads and uncertainty . . . 71
Day 77: Weekly review 11 and full model assembly . . . 72
Day 78: Hyper-parameter selection . . . 72

Day 79: Full training runs on primary split . . . 73
Day 80: MILESTONE 6: model zoo frozen . . . 73
Phase 7: Validation and scientific experiments (Days 81-90) . . . 74
Day 81: Spatially blocked cross-validation . . . 74
Day 82: District-held-out and ecological-region transfer . . . 74
Day 83: Temporal holdout and extreme-event testing . . . 75
Day 84: Weekly review 12 and ablation studies . . . 75
Day 85: Error analysis . . . 76
Day 86: Uncertainty calibration . . . 76
Day 87: Out-of-distribution detection and area of applicability . . . 77
Day 88: Comparison across ecological regions . . . 77
Day 89: Independent validation against reference observations . . . 77
Day 90: MILESTONE 7: validation package and claims audit . . . 78
Phase 8: Final maps, manuscript and research release (Days 91-100) . . . 79
Day 91: Weekly review 13 and province-wide inference . . . 79
Day 92: Provincial environmental maps . . . 79
Day 93: Village-level prediction and uncertainty maps . . . 79
Day 94: Scientific figures and tables . . . 80
Day 95: Reproducibility checks and release engineering . . . 80
Day 96: Manuscript I: Methods and supplementary methods . . . 81
Day 97: Manuscript II: Results and abstract draft . . . 81
Day 98: Weekly review 14 and manuscript III: Introduction and literature positioning . . . 82
Day 99: Manuscript IV: Discussion, limitations, availability and ethics . . . 82
Day 100: MILESTONE 8: submission package and release . . . 83

13. Milestones and required deliverables . . . 84

14. Daily productivity system and 100-day calendar . . . 86
14.1 Weekly rhythm . . . 86
14.2 Daily time blocks (5-7 focused hours) . . . 86
14.3 Weekly review checklist (every 7th day) . . . 87
14.4 Automation versus manual scientific decisions . . . 87
14.5 Slippage policy and cut list . . . 87

15. Risk register and mitigation . . . 88

16. Novelty assessment and literature positioning . . . 89
16.1 Positioning as a methodological contribution . . . 89
16.2 Search protocol (summary) . . . 90

17. Research deliverables, feasibility and manuscript plan . . . 91
17.1 What is completed within 100 days versus future extensions . . . 91
17.2 Manuscript blueprint . . . 91
17.3 Planned figures and tables . . . 91

18. References and useful links . . . 92
Useful links . . . 94

Appendix A. Project charter template 95

Appendix B. Asset verification list (Day 8) 96

1. How to use this document

This is an execution manual, not a project description. Each of the 100 days in Part B is a self-contained work order. Open it in the morning, read the day's page, execute, tick the checklist, and log the result. Sections 1-12 explain the science and the design that the daily plans refer to; Sections 13-18 collect milestones, the productivity system, risks, novelty assessment, deliverables and references.

Reading the daily pages

Field	Meaning
Main objective	The one outcome that must exist by the end of the day.
Action steps	3-7 concrete activities in the order to perform them.
Technical instructions	Datasets, parameters, formulas, modules and algorithms to use, including known traps.
Expected output	The tangible files or results (named, so they can be checked).
Completion checklist	2-5 measurable acceptance criteria. If one fails, the day is not done.
Time split	R = core research / scientific decisions, C = coding or data processing, D = documentation, Q = quality control (hours). Totals are 5-7 h.
Mode	Auto = can be scripted and run unattended; Mixed = scripted with human checks; Manual = requires scientific judgement (decision named).
Dependencies / Fallback	Which earlier days must be complete, and what to do if the planned dataset or method fails.

Status of this plan. It contains no results and no accuracy targets. Dataset IDs and URLs were checked where noted in the text (PBS documents and the literature items marked 'verified'); all Earth Engine asset IDs are treated as hypotheses until Day 8 runs the verification script. Where a statement is a hypothesis it is labelled as such.

2. Executive summary

VISTA-Sindh asks whether multimodal spatiotemporal AI can predict persistent environmental stress and recovery trajectories at the level of individual rural settlements across Sindh, with reliable skill on *geographically unseen* villages and *future* years. The plan builds, in 100 days and with free data (Earth Engine, Python, open-source ML), (i) a reproducible village inventory and zone geometry, (ii) a harmonised 2001-2025 environmental data store, (iii) individually defined stress and recovery indicators (no composite score), (iv) a ladder of eight forecasting models from seasonal climatology to a graph-attention multimodal network, (v) a leakage-audited spatial and temporal validation programme with uncertainty quantification, and (vi) provincial maps, a reproducible repository and a manuscript draft.

Three design choices make the plan different from a conventional regional application. First, validation is built around **spatial blocks, district hold-out and future years** from the start; random pixel or village splits are used only to demonstrate leakage. Second, the proposed architecture is treated as a **hypothesis tested against simpler baselines** with ablations, and the plan states fallback tiers if the full model cannot be completed. Third, every indicator is classified as a direct observation, a satellite-derived proxy or a modelled estimate, and the manuscript claims are limited accordingly.

> **What was verified while preparing this plan (PBS Mouza Census 2020, Sindh provincial report).** Pakistan: 49,463 mouzas in total and 148,380 settlements. Sindh: 5,976 dehs (rural 5,211; partly urban 199; forest populated 18; forest unpopulated 42; urban 298; unpopulated 208), of which 5,428 are rural populated. In Sindh a Deh is the Mouza, a Taluka is a Tehsil. Four Karachi districts were not covered. **The updated record-level mouza list is the sampling frame and is not published**; the report publishes tabulations at taluka, district and province level. Consequence: the village inventory cannot be a simple download. It is constructed from settlement evidence (footprints, OSM, GeoNames, GHSL) and, if obtained, deh boundaries; the PBS tables serve as aggregated reference data. The final number of villages is a **measured milestone on Day 25**, not an assumption.

What the 100 days can and cannot deliver

Realistic within 100 days	Future extensions (explicitly not promised)
Village inventory v1.0 with measured coverage and matching accuracy; zones; atlas.	Field-based ground truth campaigns; household or welfare linkage.
Harmonised 2001-2025 store from free sources; indicators with documented limits.	Real-time operational forecasting service; prospective (live) evaluation over several seasons.
Eight-model comparison, ablations, spatial/temporal/event tests, uncertainty calibration.	Causal attribution of stress or recovery to interventions or policies.
Provincial and village maps with uncertainty and applicability masks.	Sub-village (household) monitoring; drinking-water quality, migration or yield estimation (not identifiable from this data).
Reproducible repository and a submission-ready manuscript draft.	Guaranteed publication. Acceptance depends on peer review and cannot be promised.

3. Project overview

3.1 Research question and sub-questions

Primary question. Can multimodal spatiotemporal AI predict persistent environmental stress and recovery trajectories at the individual village level across diverse ecological regions of Sindh, with reliable generalisation to geographically unseen villages and future years?

The question is decomposed into three testable sub-questions: **Q1** Are persistence and recovery measurable from satellite proxies at village scale with documented reliability? **Q2** Do models beat seasonal climatology out-of-sample in space *and* time, and where not? **Q3** Do graph structure and multimodal fusion add skill beyond strong tabular and sequence baselines?

3.2 Study region

All rural areas of Sindh: Tharparkar, Umerkot, Badin, Sujawal, Dadu, Jamshoro, Sanghar, Mirpurkhas, Khairpur, Jacobabad and all other districts. The province spans hyper-arid desert (Thar), the Kirthar foothills, canal-irrigated plains of the Indus Basin Irrigation System, riverine (kacho) lands, and the Indus delta and coast, with flood regimes (e.g. 2010, 2011, 2020, 2022), drought episodes (e.g. Tharparkar) and waterlogging/salinity problems. Urban cores are excluded; peri-urban villages are flagged. Karachi's urban districts are not part of the PBS Mouza Census.

3.3 Objectives mapped to execution days

#	Objective (from brief)	Days	Primary artefact
1	Reproducible geospatial inventory of rural villages	11-25	village_inventory_v1.0, zones, atlas
2	Reconstruct historical environmental conditions	26-40	env_store_v0.9 (2001-2025)
3	Village-level environmental indicators	41-55	feature_store_v1.0, indicator handbook
4	Persistent stress and recovery patterns	56-65	persistence/recovery tables, typology
5	Multimodal spatiotemporal forecasting model	66-80	VISTA-Net + model zoo
6	Village connectivity graph	71, 75	graph_v1.pt, edge reliability audit
7	Comparison with conventional ML/DL baselines	67-72, 79-84	baseline and ablation tables
8	Spatial and temporal generalisation experiments	81-83, 88	CV, LODO, event-withheld results
9	Prediction uncertainty	76, 86-87	calibration, coverage, OOD/AoA
10	Province-wide stress and recovery maps	91-93	maps, atlas v1.0
11	Manuscript and reproducible repository	94-100	manuscript, GitHub v1.0, Zenodo

3.4 Hypotheses (not established facts)

The following hypotheses are registered on Day 5 with falsification criteria. They are expectations to be tested; any of them may fail.

ID	Hypothesis	Falsified if
H1	Persistent vegetation stress is more frequent and spatially clustered than expected from rainfall autocorrelation alone.	Observed run-length frequencies are within the surrogate-series envelope (Day 59).
H2	Recovery time differs by ecoregion and irrigation access.	Log-rank/Cox effects are null with CIs spanning no effect (Day 60).
H3	At a 3-month horizon, lag-based models beat seasonal climatology on unseen blocks and future years.	Skill score CI includes 0 in most regions (Days 81-83).
H4	Skill decays with horizon; at 12 months, gains over climatology are small or absent.	Skill does not decay (unexpected).
H5	Transformers/LSTMs improve on gradient boosting only if data volume suffices; equality is plausible.	Sequence models do not beat or tie XGBoost within the 1-SE rule.
H6	Graph attention adds skill beyond neighbour-averaged features.	Edge-shuffle and neighbour-mean controls match the GAT (Day 72, 84).
H7	Multimodal fusion helps flood-related targets more than vegetation targets.	Ablations show no target-specific pattern.

ID	Hypothesis	Falsified if
H8	Conformal/ensemble intervals are near-nominal under blocked CV but lose coverage under region transfer; OOD scores track error.	Coverage holds under transfer, or OOD scores do not predict error (Days 86-87).

4. Research methodology and definitions

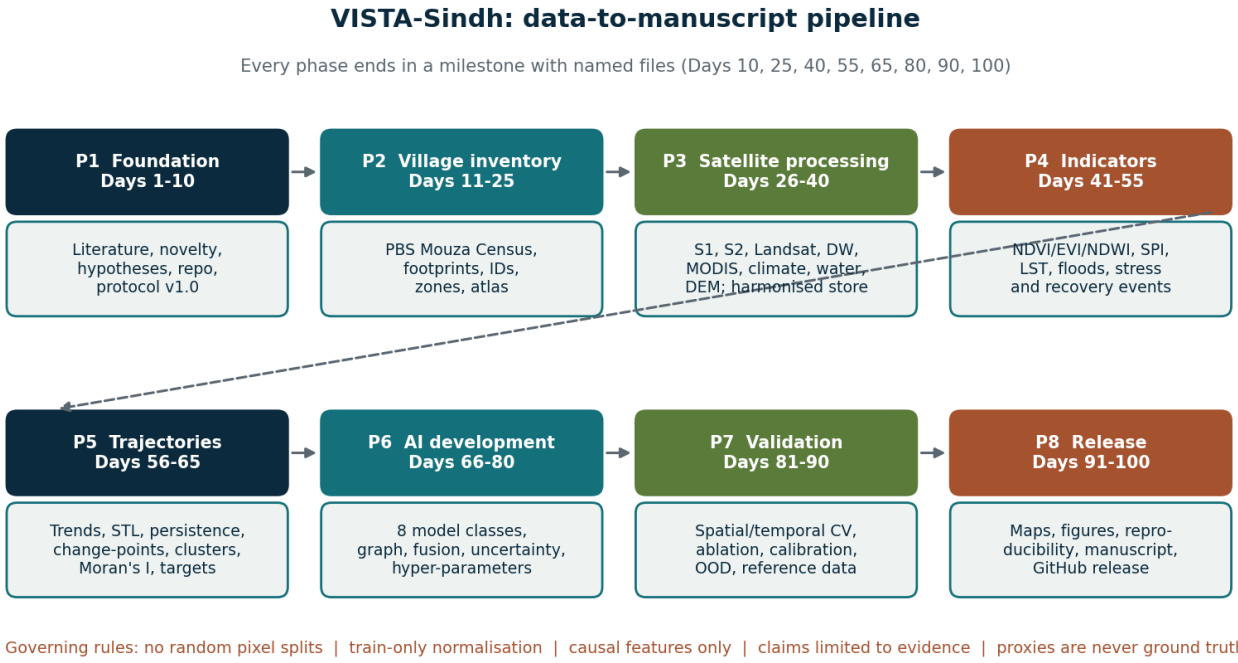


Figure 1. End-to-end pipeline with the eight phases and milestones.

4.1 Design principles

- **Leakage-safe by construction.** Climatologies, scalers, SPI/SPEI parameters, thresholds and cluster fits use training-period data only; features carry an availability date; losses never see neighbour labels.
- **Generalisation first.** Headline results come from unseen spatial blocks, unseen districts and future years; random-split numbers appear only as leakage diagnostics.
- **No composite score.** Stress and recovery are defined per indicator as events with explicit thresholds and sensitivity ranges.
- **Proxies are proxies.** Vegetation greenness is not yield; water signal is not water quality; built-up change is not migration; GRACE is not village groundwater.
- **Simplicity ladder.** Models are added only if they earn their complexity (one-standard-error rule, ablations).
- **Everything logged.** Decisions, deviations and all trials are recorded; no silent edits to frozen protocols.

4.2 Unit of analysis

PBS publishes the **deh/mouza** (revenue unit) as the statistical frame, but a deh may contain several settlements, or none. The analytic unit of VISTA-Sindh is therefore the **settlement-based village**: a settlement polygon derived from building/settlement evidence, identified by a persistent VID, linked to a deh where boundaries can be obtained. This distinction is stated in the manuscript; counts of dehs and counts of villages are never conflated.

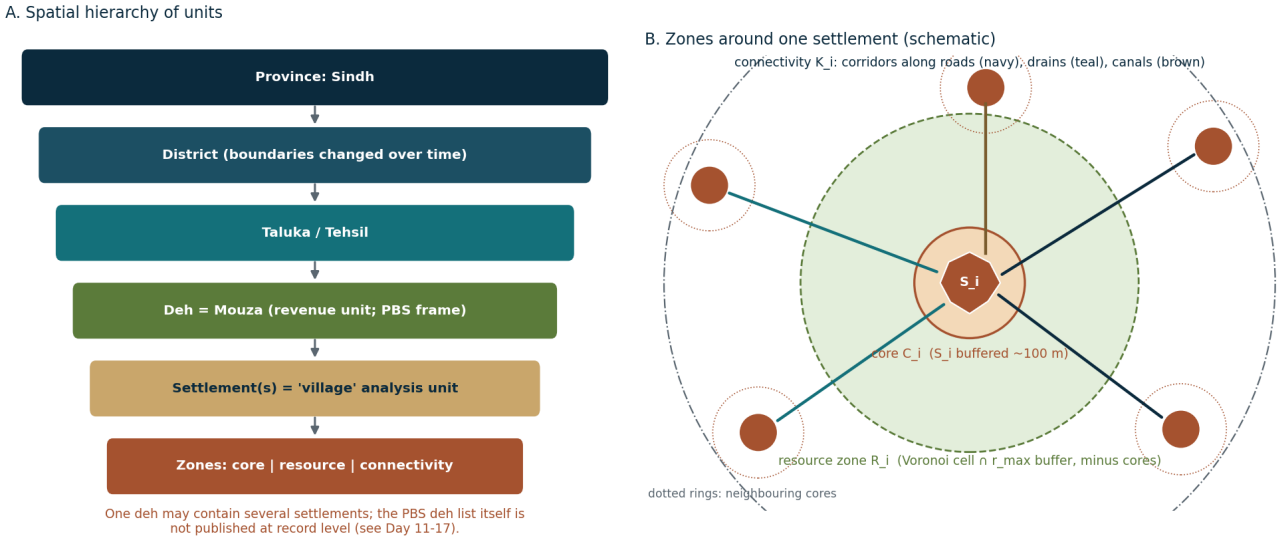


Figure 2. Spatial hierarchy and village zones (schematic). Settlement S_i, core C_i, resource zone R_i and connectivity zone K_i.

4.3 Definitions used throughout

Term	Operational definition (v0.9; frozen on Day 5/55)
Season	Rabi = Nov-Apr, Kharif = May-Oct (modelling convention; monsoon is mainly Jul-Sep).
Forecast origin t0	Last month whose data are available when the forecast is issued. Features may use only information with availability date <= t0.
Horizon h	3, 6 and 12 months ahead; label windows start after t0.
Anomaly	Standardised deviation from a village x calendar-month climatology fitted on training years (with shrinkage of the SD toward regional pooled value).
Stress event E(k)	Indicator k crosses its threshold in a season, e.g. z_NDVI <= -1.0; thermal z_LST >= +1.0; SPI-3 <= -1; inundation fraction >= 5% core or >= 20% resource zone. Thresholds are parameters with sensitivity ranges.
Persistent stress	Stress event for >= k consecutive seasons (default k = 3; sensitivity 2-4), compared against surrogate-series expectation.
Recovery	After a stress spell, anomaly returns to z >= -0.5 within m seasons (default m = 2); time-to-recovery is right-censored when not observed.
Resilience metrics	Recovery rate, deficit area, rolling AR(1) and variance (exploratory hypotheses, not validated early-warning indicators).

4.4 Forecast design and temporal protocol

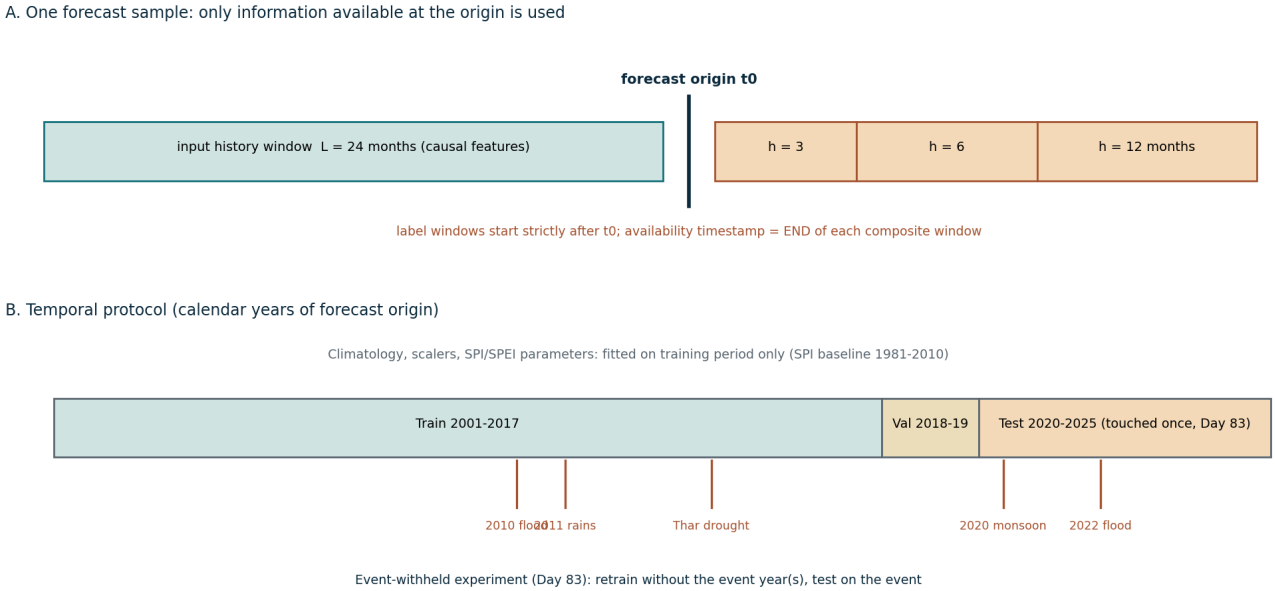


Figure 3. Sample construction (A) and temporal protocol (B). The test years and spatial test blocks are touched once, on Day 83.

Retrospective forecasting caveat. All experiments are hindcasts. Several inputs are produced in hindsight: reanalysis (ERA5-Land) assimilates observations and is re-run with later model versions; CHIRPS and MODIS products are reprocessed; gap-filled MODIS products (e.g. ET/GPP-GF) and annual embedding products use information after the nominal date; composites are stamped with the window start. VISTA-Sindh addresses this by (a) using availability = window end, (b) excluding non-causal gap-filled products from model inputs, (c) using causal smoothing/filling only, (d) lagging annual embeddings by one year, and (e) stating that true prospective validation (real-time forecasts with versioned inputs) is a future extension.

5. Village inventory methodology

This section answers the seven inventory questions of the brief. The total number of villages is not assumed; it is measured on Day 25 and reported with its quality classes.

5.1 Obtaining the inventory

Source	Provides	Access / caveat
PBS Mouza Census 2020 (Sindh provincial report; country report)	Tabulations by taluka/district: mouza status counts, settlements per mouza, irrigation source, water-table depth, groundwater taste, disasters in last 5 years, facilities. Reference period Jan-Mar 2020.	Public PDF (pbs.gov.pk/mouza-census). Record-level mouza list = sampling frame, not published: request from PBS Agricultural Census Wing.
Deh boundaries (Sindh Board of Revenue / Survey of Pakistan)	Revenue-unit polygons for resource-zone clipping and deh assignment.	Formal request (use Sindh right-to-information law; verify current statute). May be unavailable: proxies defined on Day 17.
Administrative boundaries (OCHA COD-AB via HDX, geoBoundaries, GADM, OSM)	Province, district, taluka, UC polygons; P-codes.	Open; reconcile naming and district history (Day 12).
Building footprints (Microsoft, Overture, Google Open Buildings)	Primary evidence for settlement polygons.	Open licences vary; verify Pakistan coverage and licence terms.
OSM places and landuse (Geofabrik Pakistan); GeoNames PK	Names and point locations.	Incomplete in rural Sindh; ODbL attribution.
GHSL (built-up, SMOD), WorldPop	Cross-checks and urban masking.	Modelled products; resolution 10-100 m.

5.2 Matching village records to settlement footprints

Footprints are clustered into settlement polygons (buffer 40 m, dissolve; DBSCAN as a comparison; min. 5 buildings; parameters selected on Day 14 with a sensitivity table). Records from lists or gazetteers are matched to polygons within the same taluka by the score below; candidates are ranked and top-3 are kept.

$$\text{score}(r,p) = 0.5\,\text{sim}_{name}(r,p) + 0.3\,\text{agree}_{admin}(r,p) + 0.2\exp(-d(r,p)/d_0)$$

The weights and thresholds are tuned on 200 hand-verified pairs (Day 20), then frozen. Name similarity uses normalised tokens (Goth/Gawn/Basti/Deh/Mohalla removed or unified) with fuzzy token matching; transliteration variants are handled by a documented variant dictionary.

5.3 Duplicate village names

Three cases are distinguished: (a) same name, different places (kept; disambiguated by VID and admin context), (b) several detections of the same settlement (merged if centroid distance < 150 m and similarity >= 0.85), (c) neighbouring places with similar names (kept). Ambiguous pairs enter a manual review queue (300 pairs, Day 19); every merge is logged.

5.4 Missing coordinates

Class	Definition	Use in modelling
Q1	Direct coordinates or footprint-derived polygon with consistent source agreement.	Training and prediction
Q2	Name + admin match to a polygon above the validated score threshold.	Training and prediction
Q3	Administrative-level assignment only (taluka/UC).	Inventory only; excluded from models
Q4	Unmatched record.	Inventory only; reported as coverage loss

5.5 Consistent identifiers

VID = 'SD' + district code + taluka code + H3 cell (resolution 10) of the centroid + two-character disambiguator. IDs never encode names. A versioned registry stores crosswalks to OSM/GeoNames/PBS codes and a 'superseded_by' field; re-running the pipeline must reproduce identical IDs (unit test, Day 18).

5.6 Settlement-core, resource-zone and connectivity-zone boundaries

$$C_i = \text{buffer}(S_i,\ 100\,\text{m})$$

$$R_i = (\text{Vor}_i \cap \text{buffer}(S_i,\ r_{max})) \setminus \bigcup_j C_j$$

$$K_i = \bigcup_{j \in \mathcal{N}_8(i)} \text{buffer}(\text{corridor}_{ij},\ \leq 5\,\text{km})$$

Default r_max = 2 km (3-5 km in the Thar), to be justified by the Day-44 sensitivity analysis. The resource zone is masked for permanent water and urban areas and clipped to deh boundaries if available. Connectivity corridors follow roads, canals and drains toward the 8 nearest neighbours. These are scientific choices: they are documented, tested for sensitivity and never presented as natural boundaries.

5.7 Measuring geographic coverage and matching accuracy

$$\text{Coverage}_g = \frac{\sum_{i \in g} a_i}{A_g^{ref}} \qquad \text{Precision} = \frac{TP}{TP + FP} \qquad \text{Recall} = \frac{TP}{TP + FN}$$

g is a taluka or district; the sum runs over Q1-Q2 villages in g, with a_i the area (or population) represented by village i; A_ref the PBS-reported rural populated area/mouza count or a GHSL-based reference. Precision (commission) and recall (omission) are estimated from the stratified sample of 600 settlements and 150 random 1-km cells (Days 21-22) with Wilson 95% intervals; boundary quality by IoU on 100 polygons; name-matching accuracy on 300 verified pairs; inter-rater agreement by Cohen's kappa. PBS counts are comparison data, not ground truth.

6. Dataset catalogue and epistemic status

Every dataset is described by the same fields. 'Epistemic status' separates **direct observations** (e.g. radar backscatter, reflectance), **satellite-derived proxies** (indices, classifications, flood maps) and **modelled estimates** (reanalysis, interpolated rainfall, population disaggregation, GRACE inversions). Earth Engine asset IDs are verified on Day 8 and may change.

Explicit warnings. (1) GRACE/GRACE-FO has an effective resolution of hundreds of kilometres and mixes all water-storage components: it must not be used as a village-level groundwater measurement. (2) Satellite imagery is not direct evidence of drinking-water quality, migration/displacement or agricultural yield. NDVI-based productivity is a greenness proxy; built-up change is a surface signal; water indices detect surface water, not its quality. (3) PBS reports are respondent-based aggregates and are used only as aggregated reference data.

Overview table

Dataset	Epistemic status	Resolution	Coverage	GEE
Sentinel-1 SAR (C-band)	Direct observation	10 m	2014/2015 onward over Pakistan	Yes
Sentinel-2 MSI	Direct observation	10-20 m	2017 regionally, consistent from late 2018	Yes
Landsat Collection 2 Level 2 (TM, ETM+, OLI, OLI-2)	Direct observation	30 m	1984/85-present	Yes
Dynamic World	Satellite-derived proxy	10 m	2015-06 onward	Yes
MODIS vegetation, LST and evapotranspiration	Satellite-derived proxies	250 m	2000/2001-present	Yes
JRC Global Surface Water	Satellite-derived proxy	30 m	1984-03 to 2021-12	Yes
CHIRPS rainfall	Modelled estimate	0.05 degrees	1981-present	Yes
ERA5-Land	Modelled estimate	0.1 degrees	1950-present	Yes
NASA GRACE / GRACE-FO	Modelled/inverted estimate of terrestrial water storage anomaly	About 300 km effective; gridded at 1 degree	2002-2017 and 2018-present	Yes
SRTM / Copernicus DEM / MERIT Hydro	Direct observation	30 m	Global	Yes
GHSL population and built-up	Modelled/derived estimate	100 m	1975-2030	Yes
WorldPop	Modelled estimate	100 m	2000-2020	Yes
OpenStreetMap	Crowd-sourced observation	Vector	Global; rural Sindh completeness uneven	No
PBS Mouza Census 2020 (Sindh)	Direct but respondent-based observation, tabulated	Taluka	Rural populated Sindh dehs; four Karachi districts not covered	No
Sindh and federal government geospatial/irrigation resources	Mixed	Varies	Varies; formal request required	No
Supporting: building footprints, boundaries, GeoNames	Derived proxies	Vector	Pakistan-wide	Yes
Optional: satellite embeddings, night lights, accessibility	Modelled/derived	10 m	Embeddings from 2017	Yes

Field	6.1 Sentinel-1 SAR (C-band)
Epistemic type	Direct observation (backscatter); water/flood maps are derived proxies
Official / access links	sentinels.copernicus.eu/web/sentinel/missions/sentinel-1 | EE: developers.google.com/earth-engine/datasets/catalog/COPERNICUS_S1_GRD
Earth Engine ID	COPERNICUS/S1_GRD (verify Day 8)
Variables to extract	VV, VH, VH-VV, monthly median and minimum VH, inundation fraction, revisit count
Spatial resolution	10 m (IW GRD), zonal means per zone
Temporal resolution	6-12 day revisit; S1B lost Dec 2021 (revisit drops)
Historical coverage	2014/2015 onward over Pakistan (verify)
Limitations	Wind-roughened water, wet sand, rice paddies and flooded vegetation confuse water mapping; urban double-bounce; speckle; sampling change after 2021.
Preprocessing	One orbit direction per area; dB conversion; speckle reduction (focal median/multi-temporal); optional slope normalisation near hills.
Intended model inputs	Water/flood indicators, VH dynamics (crop-cycle cross-check), x_ts

Field	6.2 Sentinel-2 MSI
Epistemic type	Direct observation (surface reflectance); indices are derived proxies
Official / access links	sentinels.copernicus.eu/web/sentinel/missions/sentinel-2 | EE: .../catalog/COPERNICUS_S2_SR_HARMONIZED
Earth Engine ID	COPERNICUS/S2_SR_HARMONIZED + GOOGLE/CLOUD_SCORE_PLUS/V1/S2_HARMONIZED
Variables to extract	B2,B3,B4,B8,B11,B12; NDVI, EVI, MNDWI, NDMI; valid-pixel count; chips
Spatial resolution	10-20 m
Temporal resolution	5-day (2 satellites)
Historical coverage	2017 regionally, consistent from late 2018 (verify over Sindh)
Limitations	Monsoon cloud; bright sand/salt crust flagged as cloud; short record vs Landsat; atmospheric correction errors in dust.
Preprocessing	Scale 1e-4; Cloud Score+ mask (threshold tuned per ecoregion); monthly/16-day medians; n_valid export.
Intended model inputs	Spatial chips x_img; NDVI/EVI/water in x_ts (2017+)

Field	6.3 Landsat Collection 2 Level 2 (TM, ETM+, OLI, OLI-2)
Epistemic type	Direct observation (reflectance, surface temperature); indices are proxies
Official / access links	usgs.gov/landsat-missions/landsat-collection-2 | EE catalog LANDSAT_LC08_C02_T1_L2 etc.
Earth Engine ID	LANDSAT/LT05, LE07, LC08, LC09 /C02/T1_L2
Variables to extract	SR bands, ST_B10, QA_PIXEL; NDVI, EVI, MNDWI, NDMI; seasonal composites
Spatial resolution	30 m (thermal resampled)
Temporal resolution	16-day per satellite
Historical coverage	1984/85-present (2000-2025 used)
Limitations	ETM+ SLC-off stripes after 2003; cross-sensor differences; cloud; thermal accuracy over salt flats.
Preprocessing	Scale factors (0.0000275, -0.2); QA masking; Roy et al. (2016) harmonisation; sensor-mix covariate.
Intended model inputs	Long-term 30 m cross-checks, seasonal chips, validation of MODIS series

Field	6.4 Dynamic World
Epistemic type	Satellite-derived proxy (deep-learning land-cover probabilities)
Official / access links	dynamicworld.app | EE catalog GOOGLE_DYNAMICWORLD_V1
Earth Engine ID	GOOGLE/DYNAMICWORLD/V1
Variables to extract	Probabilities: water, crops, trees, grass, shrub_and_scrub, built, bare, flooded_vegetation
Spatial resolution	10 m
Temporal resolution	Per Sentinel-2 scene (about 5 days)
Historical coverage	2015-06 onward (coverage over Sindh to verify)
Limitations	Confuses salt crust, dust, fallow, bright sand with built/bare; paddy vs water; no labels for Sindh-specific classes.
Preprocessing	Monthly mean probabilities (not hard labels); n_valid.
Intended model inputs	Crop/water/built fractions as proxies in x_ts and context

Field	6.5 MODIS vegetation, LST and evapotranspiration
Epistemic type	Satellite-derived proxies (NDVI/EVI, LST); ET/GPP are modelled
Official / access links	lpdaac.usgs.gov/products/mod13q1v061 | .../mod11a2v061 | .../mod16a2gfv061
Earth Engine ID	MODIS/061/MOD13Q1, MOD11A2, MYD11A2, MOD16A2GF, MOD17A2HGF
Variables to extract	NDVI, EVI, SummaryQA; LST_Day_1km, LST_Night_1km; ET, PET; GPP (optional)
Spatial resolution	250 m (VI), 1 km (LST), 500 m (ET)
Temporal resolution	16-day (VI), 8-day (LST, ET)
Historical coverage	2000/2001-present
Limitations	Mixed pixels for small villages; clear-sky bias in LST; LST != air temperature; gap-filled ET/GPP use non-causal filling.
Preprocessing	Scale factors; QA mask; availability = composite end date; resource-zone means only; GF products descriptive only.
Intended model inputs	Long vegetation/LST backbone 2001-2025 in x_ts

Field	6.6 JRC Global Surface Water
Epistemic type	Satellite-derived proxy (Landsat-based water classification)
Official / access links	global-surface-water.appspot.com | EE catalog JRC_GSW1_4_MonthlyHistory
Earth Engine ID	JRC/GSW1_4/MonthlyHistory, GlobalSurfaceWater, YearlyHistory
Variables to extract	Monthly water/valid, occurrence, recurrence, seasonality
Spatial resolution	30 m
Temporal resolution	Monthly
Historical coverage	1984-03 to 2021-12 (v1.4; verify any update)
Limitations	Ends 2021 so does not include 2022 floods; gaps from cloud/no-data; underestimates wet-season flooding; turbidity.
Preprocessing	Keep water and valid counts; cross-compare with S1/S2 water for 2016-2021; static layers as covariates.
Intended model inputs	Water persistence indicators; static wetland covariates

Field	6.7 CHIRPS rainfall
Epistemic type	Modelled estimate (satellite IR + station blend)
Official / access links	chc.ucsb.edu/data/chirps | EE catalog UCSB-CHG_CHIRPS_DAILY
Earth Engine ID	UCSB-CHG/CHIRPS/DAILY
Variables to extract	Daily precipitation; monthly totals, rainy days, RX1day, RX5day, dry spells, SPI
Spatial resolution	0.05 degrees (about 5 km)
Temporal resolution	Daily
Historical coverage	1981-present
Limitations	Coarse for village scale; weak station density in Thar; errors in intense convective rain; versions revised.
Preprocessing	Aggregate to monthly/seasonal; SPI parameters fitted on 1981-2010; document product version.
Intended model inputs	x_clim sequences; SPI; rainfall anomalies

Field	6.8 ERA5-Land
Epistemic type	Modelled estimate (reanalysis)
Official / access links	cds.climate.copernicus.eu (ERA5-Land) | EE catalog ECMWF_ERA5_LAND_MONTHLY_AGGR
Earth Engine ID	ECMWF/ERA5_LAND/MONTHLY_AGGR (also HOURLY)
Variables to extract	t2m, skin temperature, total precipitation, PET, soil water (3 layers), runoff
Spatial resolution	0.1 degrees (about 9 km)
Temporal resolution	Hourly/monthly
Historical coverage	1950-present
Limitations	Coarse; model biases in arid zones; PET sign/definition; produced in hindsight (assimilation).
Preprocessing	Fix PET sign; SPEI on 1981-2010; cell-id covariate (many villages share a cell).
Intended model inputs	x_clim, soil-moisture and water-balance features

Field	6.9 NASA GRACE / GRACE-FO
Epistemic type	Modelled/inverted estimate of terrestrial water storage anomaly (regional)
Official / access links	grace.jpl.nasa.gov | podaac.jpl.nasa.gov/GRACE | EE catalog NASA_GRACE_MASS_GRACE_LAND_V04 (verify)
Earth Engine ID	NASA/GRACE/MASS_GRACE_LAND_V04 (verify)
Variables to extract	Liquid water equivalent thickness anomaly (regional context only)
Spatial resolution	About 300 km effective; gridded at 1 degree
Temporal resolution	Monthly
Historical coverage	2002-2017 and 2018-present (with a gap of about 11 months)
Limitations	NOT a village-level groundwater measurement; mixes snow, surface water, soil moisture, groundwater; leakage and scaling errors.
Preprocessing	Optional province/basin-scale covariate shared by all villages in a region; anomalies only; do not interpret at village scale.
Intended model inputs	Optional regional context variable; excluded from village-level claims

Field	6.10 SRTM / Copernicus DEM / MERIT Hydro
Epistemic type	Direct observation (radar topography) with vertical errors; hydrology layers derived
Official / access links	usgs.gov (SRTM 1 arc-sec) | spacedata.copernicus.eu (Copernicus DEM) | hydro.iis.u-tokyo.ac.jp/~yamadai/MERIT_Hydro
Earth Engine ID	USGS/SRTMGL1_003, COPERNICUS/DEM/GLO30, MERIT/Hydro/v1_0_1
Variables to extract	Elevation, slope, TWI, HAND, upstream area, flow direction
Spatial resolution	30 m (DEM), about 90 m (MERIT Hydro)
Temporal resolution	Static (2000 / 2011-2015 acquisitions)
Historical coverage	Global
Limitations	Metre-scale vertical error in flat delta; DSM includes surface features; HAND coarse in canal-modified terrain.
Preprocessing	Single DEM choice; zonal statistics in UTM 42N; distances to river/canals/coast.
Intended model inputs	x_st static covariates; graph drainage edges

Field	6.11 GHSL population and built-up
Epistemic type	Modelled/derived estimate (census disaggregation; Landsat/Sentinel-2 built-up)
Official / access links	ghsl.jrc.ec.europa.eu | EE catalog JRC_GHSL_P2023A_GHS_POP
Earth Engine ID	JRC/GHSL/P2023A/GHS_POP, GHS_BUILT_S, GHS_SMOD
Variables to extract	Population, built-up surface, settlement model class
Spatial resolution	100 m (10 m built-up 2018)
Temporal resolution	5-year epochs 1975-2030
Historical coverage	1975-2030 (2025/2030 are projections)
Limitations	Modelled; omissions in sparse rural settlements; epochs interpolated; not census counts.
Preprocessing	Use last available epoch only (no look-ahead interpolation); urban masking.
Intended model inputs	Context/exposure covariates; settlement-change indicator

Field	6.12 WorldPop
Epistemic type	Modelled estimate (population disaggregation)
Official / access links	worldpop.org | EE catalog WorldPop_GP_100m_pop
Earth Engine ID	WorldPop/GP/100m/pop (verify years)
Variables to extract	Population counts per 100 m
Spatial resolution	100 m
Temporal resolution	Annual
Historical coverage	2000-2020 (verify)
Limitations	Modelled from census + covariates; inherits covariate artefacts; uncertainty not village-scale.
Preprocessing	Zonal sums; flag as modelled; causal epoch use.
Intended model inputs	Context covariates only (never outcomes)

Field	6.13 OpenStreetMap
Epistemic type	Crowd-sourced observation (variable completeness)
Official / access links	openstreetmap.org | download.geofabrik.de/asia/pakistan.html
Earth Engine ID	No (download PBF; Overpass API)
Variables to extract	Place nodes, landuse, buildings, roads, waterways/canals
Spatial resolution	Vector
Temporal resolution	Continuous edits; use dated extracts
Historical coverage	Global; rural Sindh completeness uneven
Limitations	Incomplete rural roads and canals; edit dates unknown; ODbL licence (attribution, share-alike).
Preprocessing	Dated Geofabrik extract; completeness audit (Day 71); edge-type masking if unreliable.
Intended model inputs	Names, graph road/canal edges, distances

Field	6.14 PBS Mouza Census 2020 (Sindh)
Epistemic type	Direct but respondent-based observation, tabulated (taluka/district)
Official / access links	pbs.gov.pk/mouza-census | .../mouza_census_2020/mouza_census_2020_sindh_report.pdf (verified)
Earth Engine ID	No (PDF tables; parse)
Variables to extract	Mouza counts by status; settlements per mouza; irrigation source; water-table depth; groundwater taste; disasters in last 5 years; facilities
Spatial resolution	Taluka, district, province (record level not published)
Temporal resolution	Single reference: Jan-Mar 2020 (disaster window about 2015-2019)
Historical coverage	Rural populated Sindh dehs; four Karachi districts not covered
Limitations	Aggregates only; respondent knowledge (enumerator/patwari); one time point; ecological-fallacy risk when compared with village indicators.
Preprocessing	Table extraction (Day 11); taluka crosswalk (Day 12); Spearman comparisons (Days 49, 54, 89).
Intended model inputs	Reference/validation only, not model input

Field	6.15 Sindh and federal government geospatial/irrigation resources
Epistemic type	Mixed (administrative records, canal/barrage data); access varies
Official / access links	Sindh Irrigation Dept (irrigation.sindh.gov.pk), Sindh Bureau of Statistics (sbos.sindh.gov.pk), Sindh Board of Revenue, Survey of Pakistan, IRSA (irsa.gov.pk), PMD (pmd.gov.pk), SUPARCO (suparco.gov.pk) - verify URLs and access
Earth Engine ID	No
Variables to extract	Deh boundaries, canal command areas, barrage discharges, gauge rainfall, crop statistics
Spatial resolution	Varies
Temporal resolution	Varies
Historical coverage	Varies; formal request required
Limitations	Often not openly licensed or digital; can take weeks; may be restricted.
Preprocessing	Submit requests on Day 11; if unavailable use OSM canals/MERIT drainage and state the limitation (resulting scientific limitation: irrigation connectivity edges become less reliable).
Intended model inputs	Deh clipping; irrigation edges; gauge validation if obtained

Field	6.16 Supporting: building footprints, boundaries, GeoNames
Epistemic type	Derived proxies (ML-extracted footprints) and administrative data
Official / access links	github.com/microsoft/GlobalMLBuildingFootprints | overturemaps.org | sites.research.google/open-buildings | data.humdata.org | geoboundaries.org | download.geonames.org/export/dump
Earth Engine ID	Open Buildings v3 (verify Pakistan); others no
Variables to extract	Building polygons, admin polygons, populated places
Spatial resolution	Vector
Temporal resolution	Static snapshots (acquisition years vary)
Historical coverage	Pakistan-wide (completeness varies)
Limitations	Missed small/mud huts in dunes; shadows/flooded areas; acquisition years differ; licences differ.
Preprocessing	Deduplicate across sources; clustering to settlement polygons (Day 14-15).
Intended model inputs	Inventory; zone geometry

Field	6.17 Optional: satellite embeddings, night lights, accessibility
Epistemic type	Modelled/derived (foundation-model embeddings; radiance; travel time)
Official / access links	deepmind.google AlphaEarth Foundations (verified) | EE: GOOGLE/SATELLITE_EMBEDDING/V1/ANNUAL (verify) | NOAA VIIRS DNB | Oxford MAP accessibility
Earth Engine ID	GOOGLE/SATELLITE_EMBEDDING/V1/ANNUAL; NOAA/VIIRS/DNB/ANNUAL_V21; Oxford/MAP/accessibility_to_cities_2015_v1_0
Variables to extract	64-d annual embedding; night radiance; travel time to cities
Spatial resolution	10 m (embedding), about 500 m (VIIRS), 1 km (accessibility)
Temporal resolution	Annual
Historical coverage	Embeddings from 2017 (verify)
Limitations	Embeddings summarise a whole year and use temporal context: leakage risk if used at origins inside that year; opaque features; night lights confounded by non-economic light.
Preprocessing	Use only embeddings of years strictly before the origin year; treat as an optional comparison/fallback encoder.
Intended model inputs	Tier-2 spatial encoder alternative; ablation

7. Indicator handbook: formulas, interpretation, normalisation, limits

The indicators are computed per zone and time step and then standardised using climatologies fitted on training years only (shrinkage: sigma_shrunk = w sigma_v + (1-w) sigma_region, w = n/(n+5)). Seasonal normalisation therefore removes the expected annual cycle, but also treats long-term change as anomaly; trends are analysed separately (Day 57). No arbitrary composite score is constructed: stress and recovery are defined event by event.

7.1 NDVI

$$\mathrm{NDVI} = \frac{\rho_{NIR} - \rho_{Red}}{\rho_{NIR} + \rho_{Red}}$$

Interpretation. Greenness/photosynthetic activity proxy. **Seasonal normalisation.** Village x calendar-month z-score or robust z from training years; seasonal integral for productivity. **Limitations.** Saturates in dense canopy; soil background in sparse vegetation (Thar); not yield; sensor differences.

7.2 EVI

$$\mathrm{EVI} = 2.5\,\frac{\rho_{NIR} - \rho_{Red}}{\rho_{NIR} + 6\rho_{Red} - 7.5\rho_{Blue} + 1}$$

Interpretation. Canopy greenness with reduced soil/atmosphere effects. **Seasonal normalisation.** As NDVI. **Limitations.** Needs blue band; noisy under thin cloud/dust; coefficients assume specific band definitions.

7.3 NDWI and MNDWI (surface water)

$$\mathrm{NDWI} = \frac{\rho_G - \rho_{NIR}}{\rho_G + \rho_{NIR}} \qquad \mathrm{MNDWI} = \frac{\rho_G - \rho_{SWIR1}}{\rho_G + \rho_{SWIR1}}$$

Interpretation. Open-water signal; MNDWI better separates water from built-up and soil. **Seasonal normalisation.** Water fraction compared with month-specific baseline. **Limitations.** Turbidity, shadows, wet soil, paddies; not water quality.

7.4 Surface-water persistence

$$\mathrm{Pers}_{v,y} = \frac{1}{12}\sum_{m=1}^{12} \mathbf{1}[w_{v,m} \geq \tau]$$

Interpretation. Share of months in a year in which the resource zone holds water above fraction tau (default 0.05). **Seasonal normalisation.** Anomaly vs training-year persistence; seasonal class (permanent/seasonal/ephemeral). **Limitations.** Cloud-limited monsoon; JRC ends 2021; irrigation water counts as water.

7.5 Land-surface temperature anomaly

$$Z^{LST}_{v,m,y} = \frac{LST_{v,m,y} - \mu_{v,m}}{\sigma_{v,m}}$$

Interpretation. Surface thermal departure under clear sky. **Seasonal normalisation.** mu, sigma from training years with SD shrinkage to ecoregion values. **Limitations.** Skin temperature, not air temperature; clear-sky bias; emissivity of salt/sand.

7.6 Rainfall anomaly and SPI

$$\mathrm{SPI}_k = \Phi^{-1}(F_\Gamma(P_k;\, a, b)) \quad \text{(mixed zero handling)}$$

Interpretation. Meteorological drought/wetness at accumulation scale k (1, 3, 6, 12 months). SPEI replaces P by P - PET. **Seasonal normalisation.** Gamma parameters fitted on 1981-2010. **Limitations.** Coarse CHIRPS/ERA5-Land; stationarity assumption; PET uncertainty.

7.7 Vegetation productivity anomaly

$$P_{v,s} = \sum_{t \in s} \max\left(\mathrm{NDVI}_{v,t} - \mathrm{NDVI}^{base}_{v},\, 0\right)\Delta t, \quad z^{P}_{v,s} = \frac{P_{v,s} - \mu^{P}_{v,s}}{\sigma^{P}_{v,s}}$$

Interpretation. Season-integrated greenness above a dry-season baseline. **Seasonal normalisation.** z by season type (Rabi/Kharif) from training years. **Limitations.** Greenness proxy, not crop yield; rain-fed and irrigated systems differ.

7.8 Flood frequency and persistence

$$F_v = \sum_y \mathbf{1}\left[\max_{t \in y} f_{v,t} \geq \theta_f\right], \qquad D_{v,y} = \sum_t f_{v,t}^{\geq \theta_f} \Delta t_t / n_{rev}$$

Interpretation. Years with inundation of core/resource zone and normalised duration (days between observations over revisit count). **Seasonal normalisation.** Event-based relative to dry-season baseline; no seasonal z. **Limitations.** SAR errors (wind, paddies, flooded vegetation); S1B loss; pre-2015 sparse optical.

7.9 Agricultural activity proxies

$$\mathrm{CI}_{v,y} = \#\,\mathrm{peaks}(\mathrm{NDVI}_{v,t})_{prom \geq 0.1}, \qquad \mathrm{Fallow}_{v,y} = \frac{1}{2}\sum_s \mathbf{1}\left[\max_{t \in s} \mathrm{NDVI} < \theta\right]$$

Interpretation. Cropping-intensity and fallow indicators of greenness cycles. **Seasonal normalisation.** Season-specific; causal smoothing. **Limitations.** Does not identify crops or yield; cloud limits Kharif peaks.

7.10 Settlement expansion/contraction

$$\Delta A_v = \frac{A_v^{core}(t_2) - A_v^{core}(t_1)}{A_v^{core}}, \quad \mathrm{class} = \mathrm{sign}(\Delta A) \text{ if } |\Delta A| > 2\,\sigma_{ctrl}$$

Interpretation. Net change of built-up signal in the core zone. **Seasonal normalisation.** Noise floor from stable control polygons. **Limitations.** Classifier noise; flood/demolition effects; NOT evidence of migration.

7.11 Stress events (one per indicator)

$$E_{v,s}^{(k)} = \mathbf{1}\left[z_{v,s}^{(k)} \text{ crosses } \theta_k\right]$$

Interpretation. Individual events: vegetation z <= -1; thermal z >= +1; SPI-3/SPEI-3 <= -1; flood fractions above thresholds. **Seasonal normalisation.** Thresholds fixed from training years; sensitivity grid. **Limitations.** Thresholds are conventions; no composite or weighted sum is built.

7.12 Recovery indicators

$$T_r = \min\{k \geq 1 :\ z_{s+k} \geq -0.5\} \ \text{(right censored)}, \quad R_{rate} = \frac{z_{s+T_r} - z_{min}}{T_r}$$

Interpretation. Time to recovery, rate, depth and deficit area after stress spells. **Seasonal normalisation.** Defined on anomalies (climate-conditional); interpreted with rainfall anomalies. **Limitations.** Rainfall-driven recovery is not land recovery; censoring; few spells in some regions.

8. AI architecture: VISTA-Net

VISTA-Net is the *proposed* architecture. It is justified only if it improves on simpler models in the ablations (Days 80-84). The components map to the required elements: spatial encoder, temporal Transformer, climate encoder, cross-modal attention fusion, village graph attention network, multivariate forecasting heads and uncertainty estimation.

VISTA-Net: multimodal spatiotemporal architecture (proposed; tested against simpler fallbacks)

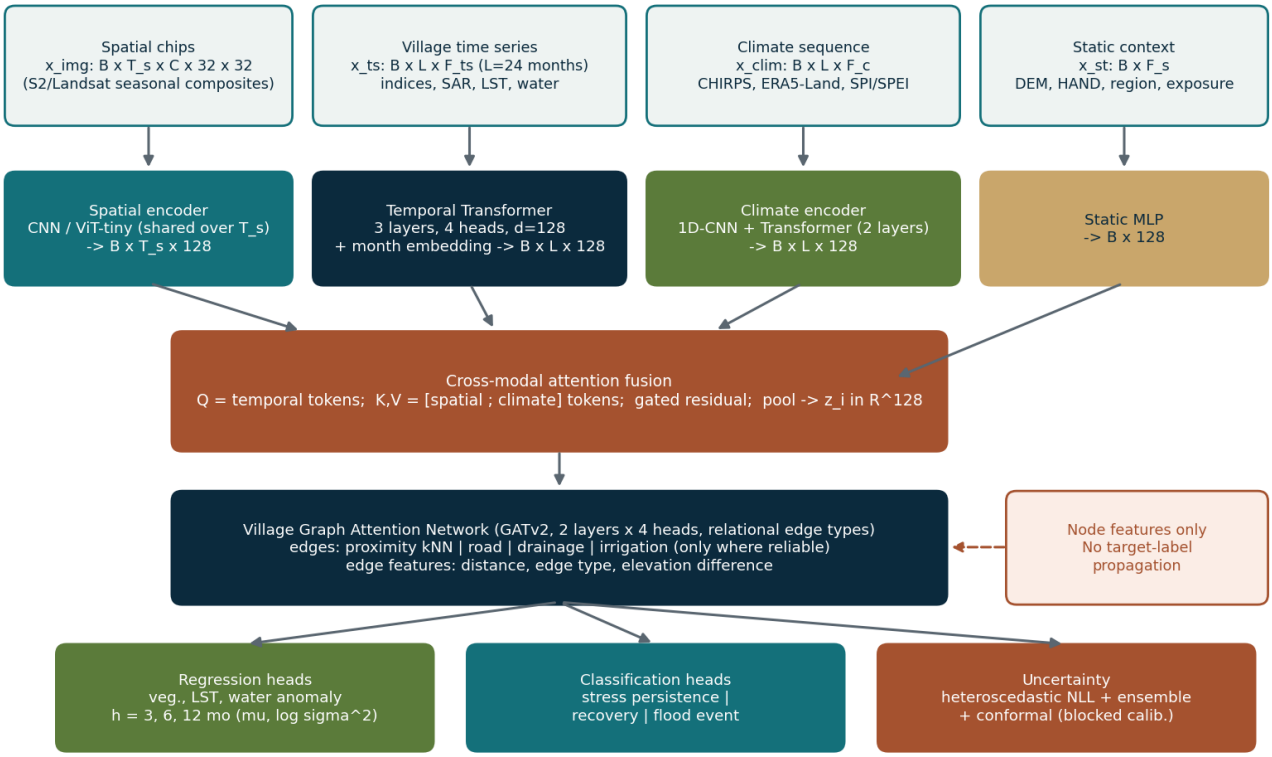


Figure 4. VISTA-Net architecture with tensor shapes (B = batch, L = 24 months, T_s = 8-16 seasonal chip steps).

8.1 Input tensors and feature engineering

Tensor	Shape	Content	Notes
x_img	[B, T_s, C, 32, 32]	Seasonal composites around the settlement (C = 8: B2,B3,B4,B8,B11,B12, NDVI, VH)	Chip subset policy (Day 9/73); flips/rot90 only
x_ts	[B, L, F_ts]	Monthly anomalies: NDVI, EVI, water fraction, LST_day/night, VH, DW fractions, valid counts, age-since-observation	F_ts about 30-40; causal fill
x_clim	[B, L, F_c]	CHIRPS rain, ERA5-Land t2m, soil water, PET, SPI/SPEI (1/3/6/12)	F_c about 12-16
x_st	[B, F_s]	Elevation, slope, HAND, TWI, distances (river, canal, coast, road), ecoregion one-hot, GHSL/WorldPop context	F_s about 25
masks	[B, L], [B, T_s]	Valid-observation masks	Mask-aware attention/loss
y_reg	[B, H=3, K=3]	Vegetation, LST, water anomalies at 3/6/12 months	z-space
y_cls	[B, 3]	Stress persistence, recovery (conditional), flood-event next monsoon	Conditional masks for T5/T6
edge_index/attr/type	[2, E], [E, F_e], [E]	Graph edges and features	Static geography only

8.2 Mathematical specification

Spatial encoder. A shared CNN (or ViT-tiny) maps each chip to a token; Tier-2 uses an MLP over zonal statistics instead.

$$s_{i,\tau} = f_{\text{CNN}}\left(X^{img}_{i,\tau}\right) \in \mathbb{R}^{128}, \quad S_i = [s_{i,1}, \ldots, s_{i,T_s}]$$

Temporal Transformer. Inputs are projected, month-of-year and positional embeddings are added, and a 3-layer encoder produces tokens H^t; padding masks handle missing months.

$$H^t_i = \text{TE}\left(W_{in} x^{ts}_{i,1:L} + E_{pos} + E_{month}\right), \qquad \text{Attn}(Q,K,V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}} + M\right)V$$

Climate encoder and cross-modal fusion. Climate tokens H^c come from a 1D-CNN plus a 2-layer Transformer. Temporal tokens query the concatenated spatial and climate tokens; a gate blends the attended and original representation.

$$A_i = \text{CA}(Q = H^t_i,\ K = V = [S_i; H^c_i]), \quad g_i = \sigma(W_g[H^t_i; A_i]), \quad z_i = \text{pool}(g_i \odot A_i + (1 - g_i) \odot H^t_i)$$

Village graph attention. Relational GATv2 over the village graph with edge features e_ij (distance, edge type, elevation difference):

$$\alpha_{ij} = \text{softmax}_j(a^\top \text{LeakyReLU}(W[h_i \,\|\, h_j \,\|\, e_{ij}])), \qquad h^{(l+1)}_i = \text{LN}\left(h^{(l)}_i + \sigma\Big(\sum_{j \in \mathcal{N}(i)} \alpha_{ij} W h^{(l)}_j\Big)\right)$$

Heads and losses. Regression heads output mean and log-variance; classification heads output logits. Multi-task loss with fixed weights (uncertainty weighting is an ablation):

$$\mathcal{L} = \sum_{h,k}\left[\tfrac{1}{2}\log\sigma^2_{hk} + \frac{(y_{hk} - \mu_{hk})^2}{2\sigma^2_{hk}}\right] + \lambda_c \sum_c \text{BCE}_w(y_c, \hat{y}_c) + \lambda_r \|\theta\|^2$$

Uncertainty. Total predictive variance combines aleatoric (heteroscedastic head) and epistemic (ensemble) components; interval calibration uses conformalised quantile regression on spatially blocked calibration data.

$$\sigma^2_{tot} = \frac{1}{M}\sum_{m=1}^{M} \sigma^2_m + \frac{1}{M}\sum_{m=1}^{M} (\mu_m - \bar{\mu})^2, \qquad C(x) = \left[\hat{q}_{lo}(x) - Q,\ \hat{q}_{hi}(x) + Q\right]$$

8.3 Graph construction

Nodes are Q1-Q2 villages. Edge types: proximity (kNN, k = 8, within 15 km), road (network distance <= 10 km on OSM), drainage (directed upstream to downstream within a catchment using MERIT Hydro/OSM waterways) and irrigation (shared canal reach within 5 km, only where reliable data exist). Reliability is audited on Day 71; unreliable types are masked, never imputed. Edges are built from static geography only, and message passing uses node *features*, never target labels of validation/test nodes.

9. Training strategy, baselines, ablations and reproducibility

9.1 Model ladder (eight classes)

#	Model	Inputs	Role	Days
1	Seasonal climatology	Village x month climatology (train years)	Reference for skill scores; class prevalence for classification	67
2	Persistence / AR / ridge-ARX	Last anomaly; lags 1-12; SPI	Linear and autoregressive baseline	67
3	Random Forest	Engineered lag/rolling/static features	Non-linear tabular baseline	68
4	XGBoost	Same features; Optuna-tuned	Strong tabular baseline (likely hard to beat)	68
5	LSTM	x_ts, x_clim, x_st	Sequence baseline	69
6	Temporal Transformer	x_ts, x_clim, x_st	Attention baseline	70
7	Graph neural network	Node features + village graph	Graph baseline (GraphSAGE/GCN/GATv2)	72
8	VISTA-Net (proposed)	All tensors + graph	Multimodal fusion + graph + uncertainty	73-77

9.2 Training procedure and hyper-parameters

Item	Default	Search range / rule
Optimiser	AdamW, lr 1e-3, weight decay 1e-4	lr 1e-4 to 3e-3 (log); wd 1e-5 to 1e-2
Schedule	Cosine decay, 5% warm-up	Fixed
Batch size	512 (sequence models); neighbour/cluster mini-batches for graph	256-1024
Temporal Transformer	d = 128, 4 heads, 3 layers, ff 256, dropout 0.1	d 64-192; layers 2-4; dropout 0.05-0.3
LSTM	2 layers, hidden 128	hidden 64-256
GATv2	2 layers x 4 heads, k = 8 proximity edges	k 4-12; layers 1-3
Losses	Gaussian NLL (regression); weighted BCE or focal (classification)	Task weights 1 by default; uncertainty weighting as ablation
Regularisation	Early stopping on validation blocks (patience 10), gradient clipping 1.0, mixed precision	Fixed
Seeds	3 per model (5 for ensembles)	Report mean +/- SE
Tuning budget	Optuna with ASHA; 30 trials (neural), 40 (trees); equal effort across models	Full search log released
Selection	One-standard-error rule on validation blocks; test years/blocks untouched	Documented on Day 80

9.3 Ablation experiments (pre-registered)

Family	Ablations
Architecture	No spatial encoder; no climate encoder; cross-attention replaced by concatenation; GAT replaced by MLP; no uncertainty head; fixed vs learned task weights
Graph	No edges; proximity only; + road; + drainage; + irrigation; shuffled edges; distance-only attention; neighbour-mean features with XGBoost
Modalities	Without Sentinel-1; without Dynamic World; without DEM/static; without MODIS LST; with/without lagged AlphaEarth embeddings
Data design	Random split vs spatial block (leakage diagnostic); with/without valid-count features; causal fill vs mask only

9.4 Reproducibility requirements

- Single config per run (YAML), config hash stored with outputs; seeds fixed; deterministic algorithms where possible.
- Data manifests with SHA-256, source, date and version; Earth Engine export manifests (asset IDs, dates, parameters).

- Locked environment (conda-lock or requirements.lock; Docker image); continuous integration for tests including the leakage tests.
- Public release: code (Apache-2.0), inventory and feature store (CC BY 4.0 subject to upstream licences), trained weights, model cards, split manifest.
- Three reproducibility levels are declared: (1) analysis from released feature store, (2) re-extraction from Earth Engine, (3) re-training. Level 1 is guaranteed.

10. Fallback architectures and decision rules

The full VISTA-Net is ambitious for an independent researcher. The plan therefore defines four tiers. Each tier is a complete, publishable research product on its own; higher tiers add components only if lower tiers are finished and validated.

Tier	Content	Drops	Trigger to adopt	Scientific consequence
Tier 3 (full)	VISTA-Net with chips, climate encoder, cross-attention, relational GAT, ensembles + conformal	Nothing	Day 77 end-to-end model trains stably and runs within compute budget	Full test of the hypothesis
Tier 2	Zonal-statistics spatial encoder (+ optional lagged embeddings), Transformer, GAT (proximity + road), ensembles, conformal	Raw chips; irrigation edges if unreliable	Chip export or training infeasible (Day 73) or GAT edges incomplete (Day 71)	Cannot test benefit of raw spatial texture; graph test remains
Tier 1	Tabular + LSTM/Transformer; graph only as neighbour-mean features; ensembles + conformal	Graph attention network	GNN training unstable or no benefit over neighbour features (Days 72, 75)	Graph benefit tested only via neighbour-mean features
Tier 0	Climatology, AR, RF, XGBoost; full inventory, indicators, validation framework, uncertainty via quantile/conformal	Deep sequence and graph models	Time or compute shortage after Day 68	Benchmark/diagnostic paper; deep models listed as future work

Rule. If a simpler tier is within one standard error of a higher tier on validation blocks, the simpler tier is reported as the main model. A null result for the graph or fusion component is a valid finding.

11. Scientific validation framework

No accuracy target is fixed in advance. The framework specifies *how* performance is measured, which metric suits which target, and how leakage is prevented.

11.1 Metrics by target type

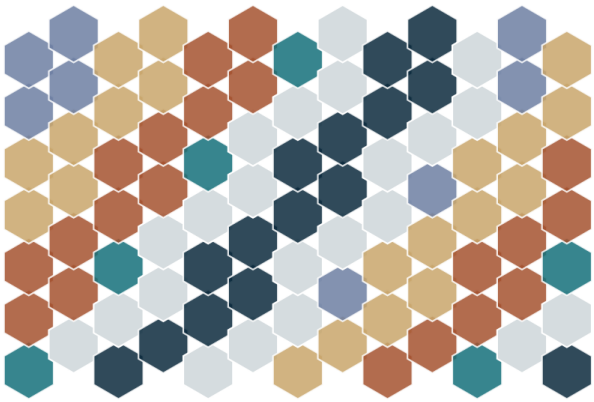
Target	Primary metrics	When appropriate / caution
Continuous anomalies (T1-T3: vegetation, LST, water at h = 3, 6, 12 months)	MAE, RMSE, skill score vs climatology, R-squared (secondary), CRPS for probabilistic forecasts, per-horizon curves	RMSE penalises large errors; MAE robust. R-squared can be inflated by seasonality: use anomaly space and report skill vs climatology instead.
Rare events (T4 flood next monsoon)	AUPRC, precision, recall, F1 at a pre-set operating point, AUROC (secondary), Brier score	Few event years: AUROC alone is misleading under imbalance; report prevalence and CIs; do not tune threshold on test.
Conditional binary (T5 persistence, T6 recovery)	AUROC, AUPRC, F1, calibration (ECE), Brier	Evaluated only on villages currently in stress; reference = prevalence or persistence baseline.
Uncertainty	PICP (coverage), MPIW (width), CRPS, ECE, error by uncertainty decile, sparsification	Check coverage separately by region, year and under transfer.
Maps / spatial	Moran's I of residuals; error maps; block-wise metrics	Residual autocorrelation indicates unresolved spatial structure.

$$\mathrm{MAE} = \frac{1}{n}\sum|y_i - \hat{y}_i|, \quad \mathrm{RMSE} = \sqrt{\frac{1}{n}\sum(y_i - \hat{y}_i)^2}, \quad \mathrm{SS} = 1 - \frac{\mathrm{MSE}_{model}}{\mathrm{MSE}_{clim}}$$

$$\mathrm{F1} = \frac{2PR}{P+R}, \quad \mathrm{PICP} = \frac{1}{n}\sum \mathbf{1}[y_i \in C(x_i)], \quad \mathrm{ECE} = \sum_b^{n_b} \frac{n_b}{n} |\mathrm{acc}_b - \mathrm{conf}_b|$$

11.2 Spatial and temporal validation designs

Spatially blocked CV: hexagonal blocks, folds balanced by ecological region, buffer blocks (grey) removed from training



Dark = held-out test blocks. Coloured = other folds (train). Grey = excluded buffer >= autocorrelation range (Day 63).

Figure 5. Spatially blocked cross-validation with buffer blocks. Block size and buffer are set from the semivariogram range (Day 63).

Design	Procedure	Question answered
Spatially blocked CV	H3 hexagon blocks, 5 folds balanced by ecoregion, buffer >= autocorrelation range removed from training	Skill on unseen places within the province
Leave-one-district-out	>= 10 districts held out with buffer; distribution of held-out performance	Skill on whole unseen districts; worst-case behaviour
Ecological-region transfer	Train on region A, test on region B (matrix)	Transfer across ecological regimes
Temporal hold-out	Train 2001-2017, validate 2018-2019, test 2020-2025 (locked); rolling-origin on validation years	Skill in future years
Event-withheld	Retrain without 2010-2011 floods or 2013-2014 drought, test on them	Skill in unseen extreme regimes

Design	Procedure	Question answered
Independent reference	PBS taluka aggregates, 2022 flood products/damage tables, blind visual labels, gauges	Agreement with non-training observations

11.3 Leakage audit checklist (signed on Day 90)

Check	Test
Random splits of neighbouring pixels/villages	No random splits in headline experiments; split manifest uses blocks; automated test of block disjointness and buffers.
Spatial autocorrelation	Block size >= variogram range; residual Moran's I reported.
Temporal order	All training origins earlier than validation/test origins; label windows start after t0; assertions in code.
Normalisation / climatology	Fitted on training years/samples only (scalers, anomaly baselines, SPI/SPEI parameters, thresholds, cluster fits).
Neighbour information	Graph features only; no neighbour labels; edge-shuffle control.
Products using future data	Availability = window end; no gap-filled MODIS inputs; causal smoothing; embeddings lagged one year; reanalysis/hindsight caveat stated.
Hyper-parameters	Tuned on validation blocks; test set evaluated once; search log released.
Label construction	Target windows disjoint from input windows; no target is a function of an input.
Multiple comparisons	Holm correction for ablations; stated primary vs exploratory analyses.

11.4 Statistical comparison and reporting policy

- Block bootstrap (resampling blocks, 1,000 draws) for CIs; paired comparisons across the same blocks.
- Effect sizes before p-values; adjust for multiplicity; report all models including losers.
- No fabricated or pre-announced accuracy figures; all numbers come from scripts in the repository.
- Claims limited by the claims-evidence matrix (Day 90).

12. Software implementation and code

12.1 Stack

Layer	Tools
Earth observation	Google Earth Engine (Python API, geemap), rasterio, rioxarray, xarray, zarr
Vector/geospatial	geopandas, shapely 2, pyogrio, DuckDB spatial, h3-py, osmium/osmnx, QGIS LTR
Data/statistics	pandas, NumPy, SciPy, statsmodels, PySAL (esda, libpysal), scikit-gstat, lifelines, ruptures, tslearn
Machine learning	scikit-learn, XGBoost, PyTorch, PyTorch Geometric, Optuna, MLflow, SHAP
Visualisation	Matplotlib, cartopy or geopandas plotting, folium/lonboard (interactive)

12.2 Recommended directory structure

```
VISTA-Sindh/
|-- README.md  LICENSE  CITATION.cff  Makefile  environment.yml  requirements.lock
|-- configs/                    # YAML: paths, GEE assets, seasons, thresholds, model configs
|-- data/
|   |-- raw/                    # downloaded sources + manifest.csv (sha256)
|   |-- processed/
|   |-- village_inventory/      # admin/, settlements/, registry/, zones/
|   |-- satellite/              # s1/, s2/, landsat/, modis/, dynamicworld/
|   |-- climate/                # chirps/, era5land/, grace/ (optional context)
|   `-- reference/              # PBS tables, flood products, validation samples
|-- notebooks/                  # exploration only (outputs stripped)
|-- src/
|   |-- preprocessing/          # gee_zonal.py, s2.py, landsat.py, s1.py, modis.py, climate.py
|   |-- village_mapping/        # settlements.py, matching.py, ids.py, zones.py
|   |-- feature_engineering/    # registry.py, indices.py, anomaly.py, stress.py, recovery.py
|   |-- temporal_analysis/      # trends.py, stl.py, persistence.py, changepoints.py, clustering.py
|   |-- graph_construction/     # proximity.py, road.py, drainage.py, irrigation.py, build_graph.py
|   |-- models/                 # data.py, baselines.py, lstm.py, transformer.py, gnn.py, vista_net.py
|   |-- validation/             # spatial_cv.py, metrics.py, conformal.py, ood.py, leakage_tests.py
|   `-- visualization/          # maps.py, figures.py
|-- experiments/                # run configs, optuna studies, registry.csv
|-- outputs/                    # maps/  figures/  tables/  predictions/
|-- documentation/              # protocol, pre-analysis plan, data cards, model cards, decisions
|-- manuscript/                 # sections, bibliography, supplement, submission/
`-- tests/                      # unit tests incl. leakage and ID-stability tests
```

12.3 Executable pseudocode and code modules

The following code is syntax-checked; the numerical modules (conformal, skill/bootstrap) were run on synthetic data. Earth Engine, PyG and shapely modules need your environment and must be tested on the pilot taluka before scaling.

Zonal time series export (Day 26)

```
# src/preprocessing/gee_zonal.py  (Day 26) - yearly multi-band stack reduced per village-zone chunk
import ee, time, pandas as pd
ee.Initialize(project="YOUR_CLOUD_PROJECT")

def s2_year_stack(year, zone_geom):
    """Monthly median NDVI/EVI/MNDWI for one year as a 36-band image (Cloud Score+ masked)."""
    s2 = (ee.ImageCollection("COPERNICUS/S2_SR_HARMONIZED")
          .filterBounds(zone_geom)
          .linkCollection(ee.ImageCollection("GOOGLE/CLOUD_SCORE_PLUS/V1/S2_HARMONIZED"), ["cs_cdf"])
          .map(lambda im: im.updateMask(im.select("cs_cdf").gte(0.6))))
    def idx(im):
        sr = im.divide(10000)
        ndvi = sr.normalizedDifference(["B8", "B4"]).rename("ndvi")
        evi = sr.expression("2.5*(N-R)/(N+6*R-7.5*B+1)",
                            {"N": sr.select("B8"), "R": sr.select("B4"), "B": sr.select("B2")}).rename("evi")
        mndwi = sr.normalizedDifference(["B3", "B11"]).rename("mndwi")
        return ee.Image.cat([ndvi, evi, mndwi]).copyProperties(im, ["system:time_start"])
    bands = []
    for m in range(1, 13):
        start = ee.Date.fromYMD(year, m, 1)
        comp = s2.filterDate(start, start.advance(1, "month")).map(idx).median()
        bands.append(comp.rename([f"ndvi_{m:02d}", f"evi_{m:02d}", f"mndwi_{m:02d}"]))
    return ee.Image.cat(bands)

def export_chunk(fc_asset, chunk_ids, year, folder="vista_s2"):
    zones = ee.FeatureCollection(fc_asset).filter(ee.Filter.inList("vid", chunk_ids))
    img = s2_year_stack(year, zones.geometry())
    red = ee.Reducer.mean().combine(ee.Reducer.count(), sharedInputs=True)
    table = img.reduceRegions(collection=zones, reducer=red, scale=10, tileScale=8)
    task = ee.batch.Export.table.toDrive(collection=table, description=f"s2_{year}_{chunk_ids[0]}",
                                         folder=folder, fileFormat="CSV")
    task.start()
    return task.id
```

Village zones (Day 23)

```
# src/village_mapping/zones.py  (Day 23) - core and resource zones in UTM 42N (shapely 2.x)
import geopandas as gpd
from shapely import voronoi_polygons, MultiPoint
from shapely.ops import unary_union
from shapely.strtree import STRtree

def make_zones(settl: gpd.GeoDataFrame, taluka: gpd.GeoDataFrame, r_max=2000.0, core_buf=100.0):
    s = settl.to_crs(32642).copy()
    boundary = taluka.to_crs(32642).geometry.union_all()
    cent = gpd.GeoDataFrame(geometry=s.geometry.centroid, crs=32642).reset_index(names="sid")
    cells = gpd.GeoDataFrame(geometry=list(voronoi_polygons(MultiPoint(list(cent.geometry)), extend_to=boundary).geoms),
                             crs=32642)
    vor = cells.sjoin(cent, predicate="contains").set_index("sid").geometry     # one Voronoi cell per settlement
    s["core"] = s.geometry.buffer(core_buf)
    tree = STRtree(s["core"].values)
    res = []
    for pos, (i, row) in enumerate(s.iterrows()):
        area = vor.loc[i].intersection(row.geometry.buffer(r_max)).intersection(boundary)
        hit = [h for h in tree.query(area, predicate="intersects") if h != pos]
        res.append(area.difference(unary_union(s["core"].values[hit])) if hit else area)
    s["resource"] = res
    return s
```

Spatial blocks and folds (Days 65, 81)

```
# src/validation/spatial_cv.py  (Days 65, 81) - H3 blocks, buffers, ecoregion-balanced folds
import h3, numpy as np, pandas as pd

def assign_blocks(df, res=4):
    df = df.copy()
    df["block"] = [h3.latlng_to_cell(la, lo, res) for la, lo in zip(df.lat, df.lon)]
    return df

def make_folds(df, n_folds=5, seed=42):
    rng = np.random.default_rng(seed)
    blocks = df.groupby("block").agg(region=("ecoregion", lambda x: x.mode().iat[0]), n=("vid", "size")).reset_index()
    blocks["fold"] = -1
    for reg, g in blocks.groupby("region"):
        order = g.sample(frac=1.0, random_state=int(rng.integers(1e9))).sort_values("n", ascending=False)
        for k, idx in enumerate(order.index):
            blocks.loc[idx, "fold"] = k % n_folds
    return df.merge(blocks[["block", "fold"]], on="block")

def train_mask(df, test_fold, buffer_rings=1):
    test_blocks = set(df.loc[df.fold == test_fold, "block"])
    excluded = set(test_blocks)
    for b in test_blocks:
        excluded |= set(h3.grid_disk(b, buffer_rings))
    return ~df["block"].isin(excluded)
```

Cross-modal fusion and heads (Days 74-77)

```
# src/models/vista_net.py  (Days 74-77) - fusion skeleton with ablation switches
import torch, torch.nn as nn

class CrossModalFusion(nn.Module):
    def __init__(self, d=128, heads=4, use_cross=True):
        super().__init__()
        self.use_cross = use_cross
        self.attn = nn.MultiheadAttention(d, heads, batch_first=True)
        self.gate = nn.Linear(2 * d, d)
        self.concat = nn.Sequential(nn.Linear(2 * d, d), nn.GELU())
        self.norm = nn.LayerNorm(d)

    def forward(self, h_t, kv, kv_mask=None):
        # h_t: [B, L, d]  temporal tokens;  kv: [B, T_s + L, d]  spatial + climate tokens
        if self.use_cross:
            a, _ = self.attn(h_t, kv, kv, key_padding_mask=kv_mask)
            g = torch.sigmoid(self.gate(torch.cat([h_t, a], dim=-1)))
            out = g * a + (1 - g) * h_t
        else:  # concat control: pooled kv appended to every temporal token
            pooled = kv.mean(dim=1, keepdim=True).expand_as(h_t)
            out = self.concat(torch.cat([h_t, pooled], dim=-1))
        return self.norm(out).mean(dim=1)  # z: [B, d]

class Heads(nn.Module):
    def __init__(self, d=128, horizons=3, targets=3, n_cls=3):
        super().__init__()
        self.reg = nn.Linear(d, horizons * targets * 2)   # mu and log-variance
        self.cls = nn.Linear(d, n_cls)
        self.h, self.k = horizons, targets
    def forward(self, z):
        r = self.reg(z).view(-1, self.h, self.k, 2)
        return r[..., 0], r[..., 1].clamp(-6, 4), self.cls(z)

def gaussian_nll(mu, logvar, y, mask):
    loss = 0.5 * (logvar + (y - mu) ** 2 * torch.exp(-logvar))
    return (loss * mask).sum() / mask.sum().clamp(min=1)
```

Relational GATv2 module (Day 75)

```
# src/models/graph_module.py  (Day 75) - relational GATv2 with edge features (PyTorch Geometric)
import torch, torch.nn as nn
from torch_geometric.nn import GATv2Conv

class VillageGAT(nn.Module):
    def __init__(self, d=128, edge_dim=6, heads=4, layers=2, dropout=0.1):
        super().__init__()
        self.convs = nn.ModuleList([GATv2Conv(d, d // heads, heads=heads, edge_dim=edge_dim, dropout=dropout)
                                    for _ in range(layers)])
        self.norms = nn.ModuleList([nn.LayerNorm(d) for _ in range(layers)])
    def forward(self, z, edge_index, edge_attr):
        # z: node embeddings from the fusion module (features only; never labels of val/test nodes)
        for conv, norm in zip(self.convs, self.norms):
            z = norm(z + torch.relu(conv(z, edge_index, edge_attr)))
        return z
```

Conformal intervals and skill with block bootstrap (Days 66, 76, 86)

```
# src/validation/conformal.py  (Day 76, 86) - CQR with spatially blocked calibration set
import numpy as np

def cqr_offset(y_cal, q_lo_cal, q_hi_cal, alpha=0.1):
    scores = np.maximum(q_lo_cal - y_cal, y_cal - q_hi_cal)
    n = len(scores)
    level = min(1.0, np.ceil((n + 1) * (1 - alpha)) / n)
    return np.quantile(scores, level, method="higher")

def coverage(y, lo, hi):
    return float(np.mean((y >= lo) & (y <= hi)))

def interval_width(lo, hi):
    return float(np.mean(hi - lo))


# src/validation/metrics.py  (Day 66) - skill vs climatology with block bootstrap
import numpy as np

def skill_score(y, pred, clim):
    mse_m = np.mean((y - pred) ** 2)
    mse_c = np.mean((y - clim) ** 2)
    return 1.0 - mse_m / mse_c

def block_bootstrap(metric_fn, blocks, y, pred, clim, n_boot=1000, seed=0):
    rng = np.random.default_rng(seed)
    ub = np.unique(blocks)
    out = []
```

```
    for _ in range(n_boot):
        pick = rng.choice(ub, size=len(ub), replace=True)
        idx = np.concatenate([np.where(blocks == b)[0] for b in pick])
        out.append(metric_fn(y[idx], pred[idx], clim[idx]))
    return np.percentile(out, [2.5, 50, 97.5])
```

Part B. Daily execution plans (Days 1-100)

One block per day, in the order of execution. Blocks may break across pages; the checklist is the acceptance test. Days marked WEEKLY REVIEW (7, 14, ..., 98) also contain a specific technical task plus the review checklist of Section 14.3. Milestone days (10, 25, 40, 55, 65, 80, 90, 100) end a phase and are specified in Section 13. Hours are planned estimates for an experienced researcher working alone; long Earth Engine or training jobs run in the background while documentation and QC proceed.

Phase 1: Research foundation and project setup (Days 1-10)

Literature review, novelty assessment, research questions and hypotheses, study-area definition, dataset inventory, software environment, GitHub repository, directory structure, methodology and initial architecture. **Milestone 1 on Day 10.**

DAY 1 Project charter and scope lock

Phase 1 | 6.0 h planned | Mode: Manual

MAIN OBJECTIVE	Turn the master brief into a one-page charter with operational definitions, constraints and a written list of what will NOT be claimed.
ACTION STEPS	1. Write charter.md: question, unit of analysis candidates, study period 2001-2025, province-wide scope. 2. List 12 non-goals (e.g. no drinking-water quality, migration, or yield claims). 3. Create the project log (docs/LOG.md) and a decision register (docs/DECISIONS.csv: id, date, decision, alternatives, reason). 4. Define success as process criteria (reproducible pipeline, leakage-audited validation); set no accuracy targets. 5. Book fixed daily work blocks; set up Zotero group library and OSF project.
TECHNICAL INSTRUCTIONS	Use the template in Appendix A. The research question is split into three testable sub-questions: (Q1) persistence/recovery are measurable from satellite proxies at village scale; (Q2) models beat seasonal climatology out-of-sample in space and time; (Q3) graph/multimodal structure adds skill beyond tabular baselines. Each is a hypothesis, not a fact.
EXPECTED OUTPUT	charter.md, DECISIONS.csv, LOG.md, Zotero library, OSF stub.
COMPLETION CHECKLIST	☐ Charter lists >= 12 explicit non-goals ☐ Decision register has >= 5 entries ☐ Q1-Q3 each have a falsifiable statement
TIME AND MODE	Core research **2.5** h | Coding / data processing **0.5** h | Documentation **2.0** h | Quality control **1.0** h | Total **6.0** h Mode: Manual (decision: Scope and non-goals.)
DEPENDENCIES	None (start).
FALLBACK STRATEGY	If OSF registration is blocked, store the pre-analysis plan as a time-stamped Git tag and a Zenodo draft.

DAY 2 Literature protocol and first harvest

Phase 1 | 6.0 h planned | Mode: Mixed

MAIN OBJECTIVE	Run a reproducible literature search and collect the first 120+ candidate papers into a tagged library.
ACTION STEPS	1. Write the search protocol (databases, strings, date range 2010-2026, inclusion rules). 2. Run searches in Scopus/Web of Science (or Google Scholar + Semantic Scholar if no access); export RIS/BibTeX. 3. Deduplicate in Zotero; tag by theme (flood, salinity, wetland, drought, poverty, forecasting, GNN, multimodal, uncertainty, spatial CV). 4. Screen titles/abstracts; keep a PRISMA-style count (identified, screened, kept). 5. Add Sindh/Pakistan grey literature (PBS, PDNA 2022, UNOSAT).
TECHNICAL INSTRUCTIONS	Example strings: (Sindh OR Pakistan OR "Indus basin") AND ("remote sensing" OR satellite) AND (flood OR salinity OR drought OR wetland OR poverty); ("graph neural network" OR "graph attention") AND ("remote sensing" OR "satellite") AND (forecast* OR prediction). Script the dedup with pandas on the exported CSV: key = normalised DOI, else title + year.
EXPECTED OUTPUT	search_protocol.md, library of >= 120 screened candidates, PRISMA counts table.

DAY 2 Literature protocol and first harvest

Phase 1 | 6.0 h planned | Mode: Mixed

COMPLETION CHECKLIST	☐ Protocol file committed ☐ >= 120 candidates tagged ☐ Counts reproducible from exports
TIME AND MODE	Core research **3.0** h | Coding / data processing **1.0** h | Documentation **1.0** h | Quality control **1.0** h | Total **6.0** h Mode: Mixed (decision: Inclusion rules.)
DEPENDENCIES	Day 1.
FALLBACK STRATEGY	No database access: use OpenAlex API (free) with the same strings and store the JSON.

DAY 3 Evidence synthesis I: Pakistan and Sindh

Phase 1 | 6.5 h planned | Mode: Manual

MAIN OBJECTIVE	Read and tabulate what has been done in Sindh/Pakistan for floods, salinity, wetlands, drought, poverty and rural environmental monitoring.
ACTION STEPS	1. Read >= 30 papers; fill evidence_table.csv (citation, area, data, method, validation, leakage risk, reported limits). 2. Start from anchor papers: Chen et al. 2024 RSE (2022 floods; confirm details), Agyemang et al. 2023 PLOS ONE (rural poverty Sindh), Sujawal salinity (Agronomy 2019), Tharparkar drought (Remote Sens. 2020). 3. Mark which studies use spatial hold-out validation and which use random splits. 4. Write 600 words: 'What exists in Sindh' with every claim tied to a citation. 5. List unresolved gaps as questions, not conclusions.
TECHNICAL INSTRUCTIONS	Evidence table columns drive the novelty matrix on Day 4. Record validation type explicitly (random split / spatial block / district hold-out / none). Do not copy numbers from abstracts without opening the paper. Verify each DOI resolves (script: requests.head on https://doi.org/<doi>).
EXPECTED OUTPUT	evidence_table.csv (>= 30 rows), sindh_literature_notes.md (600 words).
COMPLETION CHECKLIST	☐ Every row has a resolvable DOI/URL ☐ Validation type filled for all rows ☐ Gap list has >= 6 items
TIME AND MODE	Core research **4.0** h | Coding / data processing **0.5** h | Documentation **1.5** h | Quality control **0.5** h | Total **6.5** h Mode: Manual (decision: Judging study quality.)
DEPENDENCIES	Day 2.
FALLBACK STRATEGY	If full texts are unavailable, use institutional/ILL requests, authors' preprints, or abstract-only rows flagged 'abstract-only'.

DAY 4 Evidence synthesis II: methods and novelty matrix

Phase 1 | 6.5 h planned | Mode: Manual

MAIN OBJECTIVE	Review spatiotemporal forecasting, graph-based remote sensing, multimodal EO, uncertainty and spatial validation; produce the novelty matrix.
ACTION STEPS	1. Read >= 25 methods papers (Transformers, GAT/GATv2, STGCN, Graph WaveNet, GNN-RNN crop yield, PSE-TAE, MOSAIKS, Prithvi, AlphaEarth). 2. Fill methods_matrix.csv: task, inputs, graph?, uncertainty?, spatial-CV?, code available? 3. Search specifically for village/settlement-level environmental forecasting in South Asia; record negative search results with dates. 4. Draft a positioning statement: methodological contribution = leakage-safe village-scale forecasting benchmark + graph/multimodal ablations; regional application second. 5. Define the 'closest prior work' list (<= 8 papers) the manuscript must be compared with.
TECHNICAL INSTRUCTIONS	Never write 'first' unless the search log supports it; phrase as 'to our knowledge, within the searched databases up to <date>'. Record every query and hit count in search_log.csv so a reviewer can repeat it. Treat foundation-model embeddings (AlphaEarth, Prithvi) as competing/complementary baselines.
EXPECTED OUTPUT	methods_matrix.csv, novelty_matrix.md, search_log.csv, positioning.md.

DAY 4 Evidence synthesis II: methods and novelty matrix

Phase 1 | 6.5 h planned | Mode: Manual

COMPLETION CHECKLIST	☐ Novelty matrix has >= 8 rows with citations ☐ search_log.csv has dates and hit counts ☐ No unsupported 'first' claim
TIME AND MODE	Core research **4.0** h | Coding / data processing **0.5** h | Documentation **1.5** h | Quality control **0.5** h | Total **6.5** h Mode: Manual (decision: Positioning wording.)
DEPENDENCIES	Days 2-3.
FALLBACK STRATEGY	If few close works exist, report that as a search result (with log) and strengthen the benchmark framing.

DAY 5 Hypotheses, definitions and pre-analysis plan

Phase 1 | 7.0 h planned | Mode: Manual

MAIN OBJECTIVE	Freeze operational definitions (stress, persistence, recovery), hypotheses H1-H8 and the analysis plan BEFORE seeing any model results.
ACTION STEPS	1. Write definitions v0.9: stress event, persistence, recovery, season, origin, horizon (Section 4 of this PDF is the starting text). 2. Write H1-H8 with direction, test, and what would falsify each. 3. Write the pre-analysis plan: splits, metrics, baselines, ablations, multiple-testing policy, exclusions. 4. Pre-register on OSF (time-stamped) or tag in Git. 5. Specify what may change later (and how it will be logged as a deviation).
TECHNICAL INSTRUCTIONS	Thresholds are parameters with a sensitivity range (e.g. z <= -1.0 with sensitivity -0.75 to -1.5), chosen from training years only. Persistence k = 3 consecutive seasons is a default, not a truth. Test years 2020-2025 and the final spatial test blocks are declared 'locked' in the plan.
EXPECTED OUTPUT	pre_analysis_plan.md (v1.0), hypotheses.md, OSF registration or Git tag 'pap-v1'.
COMPLETION CHECKLIST	☐ PAP lists locked test sets ☐ Each hypothesis has a falsification criterion ☐ Deviation protocol written
TIME AND MODE	Core research **3.5** h | Coding / data processing **0.0** h | Documentation **2.5** h | Quality control **1.0** h | Total **7.0** h Mode: Manual (decision: All definitions.)
DEPENDENCIES	Days 1-4.
FALLBACK STRATEGY	If definitions look infeasible later, amend through the deviation log rather than editing the PAP silently.

DAY 6 Software environment, GEE access and quotas

Phase 1 | 6.0 h planned | Mode: Auto

MAIN OBJECTIVE	Build a reproducible environment and verify Earth Engine and GPU access end to end.
ACTION STEPS	1. Create environment.yml (Python 3.11) with geemap, earthengine-api, rasterio, geopandas, xarray, rioxarray, pandas, numpy, scikit-learn, xgboost, pytorch, torch-geometric, matplotlib, pysal, statsmodels, optuna, mlflow. 2. Register/verify Earth Engine noncommercial access, create a Cloud project, authenticate (ee.Authenticate()), note current quota tier. 3. Test GEE: reduceRegion on a 10 km box in Sindh; export a small table to Drive. 4. Test PyTorch + PyG on CPU/GPU (Colab/Kaggle/local); record versions. 5. Install QGIS LTR; test loading a GeoParquet.
TECHNICAL INSTRUCTIONS	Pin with conda-lock or pip-compile to requirements.lock. Check the Earth Engine quota page and record EECU/concurrent-task limits in docs/compute.md; design all later exports to respect them. Set global seeds (python, numpy, torch) in src/utils/seed.py.
EXPECTED OUTPUT	environment.yml, requirements.lock, docs/compute.md, passing smoke_test.py.

DAY 6 Software environment, GEE access and quotas

Phase 1 | 6.0 h planned | Mode: Auto

COMPLETION CHECKLIST	☐ Fresh env builds from file ☐ GEE test export succeeds ☐ PyG forward pass runs
TIME AND MODE	Core research **0.5** h | Coding / data processing **4.0** h | Documentation **1.0** h | Quality control **0.5** h | Total **6.0** h Mode: Auto
DEPENDENCIES	None.
FALLBACK STRATEGY	PyG install fails: use CPU wheels or fall back to plain PyTorch sparse message passing (Day 75 fallback). No GPU: plan Kaggle/Colab sessions.

DAY 7 Weekly review 1 and repository scaffold

Phase 1 | WEEKLY REVIEW | 6.0 h planned | Mode: Auto

MAIN OBJECTIVE	Review week 1 against its outputs and create the repository skeleton, config system and data-card templates.
ACTION STEPS	1. Run the weekly review checklist (Section 14): outputs present, log updated, risks re-scored. 2. Create the repo tree (Section 12) with README, LICENSE (Apache-2.0 code), CITATION.cff, .gitignore (data/ excluded). 3. Add configs/ (YAML) for paths, GEE assets, seasons, thresholds; load via a single src/config.py. 4. Add pre-commit (black, ruff, nbstripout) and a pytest skeleton. 5. Add data_card.md and model_card.md templates.
TECHNICAL INSTRUCTIONS	Large data stay out of Git: use DVC or a manifest.csv (path, sha256, source, date). Notebooks are for exploration only; every result must come from src/ or scripts/ called by a Makefile target.
EXPECTED OUTPUT	GitHub repo (private), Makefile, configs/, templates, week1_review.md.
COMPLETION CHECKLIST	☐ pre-commit passes ☐ pytest runs (even if trivial) ☐ Week-1 review file lists all outputs
TIME AND MODE	Core research **1.0** h | Coding / data processing **3.0** h | Documentation **1.5** h | Quality control **0.5** h | Total **6.0** h Mode: Auto
DEPENDENCIES	Days 1-6.
FALLBACK STRATEGY	No GitHub: use GitLab/Codeberg or a local bare repo and mirror later.

DAY 8 Dataset inventory and asset verification

Phase 1 | 6.5 h planned | Mode: Auto

MAIN OBJECTIVE	Verify every candidate dataset actually exists with the expected IDs, bands, dates and coverage over Sindh.
ACTION STEPS	1. Write scripts/verify_assets.py: for each GEE ID print first/last date, band names, nominal scale, number of images over a Sindh bbox. 2. Check external sources (PBS, Geofabrik, GeoNames, HDX, Overture, Microsoft footprints, OSM) with HTTP HEAD and file size. 3. Fill dataset_catalogue.csv (Section 6 fields: link, variables, resolution, temporal coverage, limits, GEE ID, epistemic type). 4. Flag missing/deprecated IDs and select alternatives. 5. Record licences for each source.
TECHNICAL INSTRUCTIONS	Example: ee.ImageCollection(id).filterBounds(sindh).aggregate_array('system:time_start') -> min/max. GEE IDs change; treat Section 6 IDs as hypotheses until this script passes. Sentinel-2 SR coverage start over Pakistan, JRC GSW end date (monthly history ends 2021 in v1.4) and S1B end (Dec 2021) must be recorded here.
EXPECTED OUTPUT	dataset_catalogue.csv, assets_report.md, licences.md.

DAY 8 Dataset inventory and asset verification

Phase 1 | 6.5 h planned | Mode: Auto

COMPLETION CHECKLIST	☐ All 17 dataset groups verified or replaced ☐ Date ranges recorded from the API, not memory ☐ Licences listed
TIME AND MODE	Core research **1.0** h | Coding / data processing **3.5** h | Documentation **1.0** h | Quality control **1.0** h | Total **6.5** h Mode: Auto
DEPENDENCIES	Day 6.
FALLBACK STRATEGY	ID missing: search the Earth Engine data catalogue for the successor version; else download via the provider and upload as an asset.

DAY 9 Methodology blueprint and compute budget

Phase 1 | 6.5 h planned | Mode: Mixed

MAIN OBJECTIVE	Write the end-to-end technical blueprint: data flow, tensor schema, experiment registry and compute budget.
ACTION STEPS	1. Draw the pipeline diagram (use Figure 1) and mark each artefact with a file path and schema. 2. Specify the tensor schema (Section 8): x_img, x_ts, x_clim, x_st, y_reg, y_cls, masks. 3. Estimate storage: villages x months x features; chips volume; choose Parquet/Zarr layouts. 4. Estimate GEE tasks: zones x products x years; plan batching (<= 5,000 features per task). 5. Write the experiment registry template (id, config hash, split, seed, metrics).
TECHNICAL INSTRUCTIONS	Rule of thumb: N settlements x 300 months x 60 features x 4 B = about N x 72 kB; 40,000 settlements -> about 2.9 GB (fine). Chips: N x 16 seasons x 8 bands x 32 x 32 x 2 B = N x 262 kB -> about 10 GB for 40,000 (feasible only with subsetting/Zarr chunking). Decide the chip subset policy now (Day 73 uses it).
EXPECTED OUTPUT	blueprint.md, tensor_schema.yaml, compute_budget.md, experiment_registry.csv.
COMPLETION CHECKLIST	☐ Every artefact has path+schema ☐ Compute budget lists fallback if over limit ☐ Schema reviewed against Section 8
TIME AND MODE	Core research **2.0** h | Coding / data processing **2.0** h | Documentation **2.0** h | Quality control **0.5** h | Total **6.5** h Mode: Mixed (decision: Chip subset policy.)
DEPENDENCIES	Days 5, 8.
FALLBACK STRATEGY	If chips look infeasible, plan the Tier-2 fallback (zonal statistics + optional embeddings) from the start.

DAY 10 MILESTONE 1: protocol v1.0 and smoke test

Phase 1 | MILESTONE 1 | 7.0 h planned | Mode: Mixed

MAIN OBJECTIVE	Close Phase 1: freeze the protocol and prove the stack works with a 5-village end-to-end run.
ACTION STEPS	1. Assemble protocol_v1.0.md from charter, PAP, blueprint and definitions. 2. Run smoke pipeline on 5 hand-picked points (Thar, Badin coast, Dadu canal area, Jacobabad, Khairpur): S2 NDVI monthly series -> Parquet -> climatology baseline -> metric. 3. Re-score the risk register; update the manuscript outline (docs/manuscript_outline.md). 4. Tag repo v0.1.0; back up the Zotero library and logs. 5. Self-audit: does every Phase-1 deliverable exist (Section 13, Milestone 1)?
TECHNICAL INSTRUCTIONS	Smoke test checks the leakage guards: climatology must be computed on years < test year; assert in code. This is the first use of the leakage-test pattern reused on Day 66.
EXPECTED OUTPUT	protocol_v1.0.md, smoke_test_report.md, tag v0.1.0, outline.
COMPLETION CHECKLIST	☐ Smoke test passes from a clean clone ☐ Protocol frozen and tagged ☐ Milestone-1 file list 100% present

DAY 10 MILESTONE 1: protocol v1.0 and smoke test

Phase 1 | MILESTONE 1 | 7.0 h planned | Mode: Mixed

TIME AND MODE	Core research **1.5** h | Coding / data processing **2.5** h | Documentation **2.0** h | Quality control **1.0** h | Total **7.0** h Mode: Mixed (decision: Go/no-go on scope.)
DEPENDENCIES	Days 1-9.
FALLBACK STRATEGY	If the smoke test fails on GEE quotas, run with 1 point and extend later; do not skip the leakage assertion.

Phase 2: Village inventory and geospatial database (Days 11-25)

PBS Mouza Census, administrative boundaries, rural settlements, geospatial matching, village identifiers, duplicate resolution, spatial validation, village zones, initial atlas, documented coverage. **Milestone 2 on Day 25.**

DAY 11 PBS Mouza Census 2020: acquire and parse Sindh tables

Phase 2 | 6.5 h planned | Mode: Mixed

MAIN OBJECTIVE	Obtain the PBS Sindh Mouza Census 2020 provincial report and turn its tehsil/district tables into machine-readable files; submit record-level data requests.
ACTION STEPS	1. Download the Sindh provincial report from pbs.gov.pk/mouza-census and the country report; hash the PDFs. 2. Extract Table 1.1 (mouzas by status per taluka), Table 1.2 (settlements per mouza), irrigation, water-table, groundwater taste, disaster tables with pdfplumber/Camelot. 3. Reconcile totals with the published Sindh totals (5,976 dehs total; 5,428 rural populated: rural 5,211 + partly urban 199 + forest populated 18). 4. Send formal requests (PBS Agricultural Census Wing, Sindh Board of Revenue, Survey of Pakistan) for the deh list/boundaries. 5. Write the 'unit of analysis' memo: deh vs settlement.
TECHNICAL INSTRUCTIONS	KEY FACT (verified in the PBS report): the Mouza Census publishes tabulations at tehsil, district and province level; the updated mouza list is the sampling frame and is not published. In Sindh a Deh = Mouza, Taluka = Tehsil, Tapa = Patwar circle; four Karachi districts were not covered. Therefore the 'village inventory' cannot be a simple download; it is built from settlements + (if obtained) deh boundaries, and PBS tables are used as aggregated reference data.
EXPECTED OUTPUT	pbs_sindh_tables/*.csv, totals_reconciliation.csv, data_request_letters/, unit_memo.md.
COMPLETION CHECKLIST	☐ Extracted totals reproduce PBS totals ☐ Request letters sent and logged ☐ Unit memo states how deh/settlement are used
TIME AND MODE	Core research **2.0** h | Coding / data processing **2.5** h | Documentation **1.5** h | Quality control **0.5** h | Total **6.5** h Mode: Mixed (decision: Unit-of-analysis decision.)
DEPENDENCIES	Day 10.
FALLBACK STRATEGY	Parsing fails: key tables manually into CSV for talukas only (about 120 rows each) and double-entry check.

DAY 12 Administrative boundaries and crosswalks

Phase 2 | 6.5 h planned | Mode: Mixed

MAIN OBJECTIVE	Assemble and harmonise province, district, taluka and union-council boundaries and crosswalk them to PBS names.
ACTION STEPS	1. Download candidates: OCHA COD-AB Pakistan (HDX), geoBoundaries, GADM, OSM admin boundaries. 2. Reproject to EPSG:4326 (storage) and EPSG:32642 UTM 42N (metric work); fix invalid geometries (make_valid). 3. Build taluka crosswalk: PBS tehsil names (Day 11) <-> boundary names with rapidfuzz token_set_ratio and manual review. 4. Handle district changes (new districts since 2017) with a district_history table. 5. Overlay check: area per taluka vs PBS 'area of mouza' sums where available.

DAY 12 Administrative boundaries and crosswalks

Phase 2 | 6.5 h planned | Mode: Mixed

TECHNICAL INSTRUCTIONS	Use UTM 42N for buffers and distances; document that the province lies within 66-72E. Crosswalk acceptance rule: score >= 92 auto; 80-92 review; < 80 manual. Store the final tree in data/village_inventory/admin/ as GeoParquet with columns pcode, name_en, name_pbs, parent, valid_from, valid_to.
EXPECTED OUTPUT	admin_boundaries.gpkg, taluka_crosswalk.csv, district_history.csv.
COMPLETION CHECKLIST	☐ No invalid geometries ☐ 100% PBS talukas matched or flagged ☐ District history table covers 2017-2025
TIME AND MODE	Core research **1.0** h | Coding / data processing **3.5** h | Documentation **1.0** h | Quality control **1.0** h | Total **6.5** h Mode: Mixed (decision: Review of ambiguous matches.)
DEPENDENCIES	Day 11.
FALLBACK STRATEGY	Boundary mismatch: use the most recent OCHA/OSM layer as the spine and record unmatched PBS rows as 'unmapped'.

DAY 13 Settlement evidence layers: footprints, OSM, GeoNames

Phase 2 | 6.5 h planned | Mode: Auto

MAIN OBJECTIVE	Ingest every free settlement-evidence layer for Sindh and quantify each against PBS counts.
ACTION STEPS	1. Download Geofabrik Pakistan PBF; extract place=village/hamlet/isolated_dwelling, landuse=residential, building=*. 2. Download GeoNames PK; filter feature class P (populated places) within Sindh. 3. Get building footprints: Microsoft Global ML Footprints, Overture buildings, Google Open Buildings (verify Pakistan coverage); store as GeoParquet. 4. Load GHSL built-up/settlement model and WorldPop for cross-checks. 5. Count evidence per taluka and compare with PBS settlement totals.
TECHNICAL INSTRUCTIONS	Use osmium/pyosmium or ogr2ogr for extraction; DuckDB spatial for fast joins on tens of millions of footprints. Record per-source completeness: buildings per km2 by taluka; flag talukas with < 1 building/km2 outside deserts as suspicious. Respect licences (ODbL attribution for OSM).
EXPECTED OUTPUT	data/village_inventory/raw/{osm,geonames,footprints}.parquet, source_coverage_by_taluka.csv.
COMPLETION CHECKLIST	☐ Each source loaded for all talukas ☐ Counts per taluka tabulated vs PBS ☐ Licence notes saved
TIME AND MODE	Core research **1.0** h | Coding / data processing **4.0** h | Documentation **1.0** h | Quality control **0.5** h | Total **6.5** h Mode: Auto
DEPENDENCIES	Days 8, 12.
FALLBACK STRATEGY	Footprints too heavy: process by taluka tiles; or use GHSL built-up connected components as the fallback settlement source.

DAY 14 Weekly review 2 and pilot settlement clustering

Phase 2 | WEEKLY REVIEW | 6.5 h planned | Mode: Mixed

MAIN OBJECTIVE	Review week 2 and test settlement-polygon construction on three contrasting pilot talukas.
ACTION STEPS	1. Run weekly checklist; update risks (coordinate quality, name inconsistency). 2. Pick pilots: desert (Mithi/Tharparkar), canal plain (Mirpurkhas or Sanghar), delta (Sujawal/Badin). 3. Cluster footprints: buffer 40 m, dissolve, drop clusters < 5 buildings; compare DBSCAN (eps 60-120 m) and morphological closing. 4. Compare cluster counts with PBS settlement counts and with OSM place nodes. 5. Pick parameters by visual check of 30 clusters per pilot.

DAY 14 Weekly review 2 and pilot settlement clustering

Phase 2 | WEEKLY REVIEW | 6.5 h planned | Mode: Mixed

TECHNICAL INSTRUCTIONS	Parameters are scientific decisions: eps and min_buildings change counts strongly in dispersed Thar settlements. Report sensitivity (eps 60/90/120 m; min 3/5/10). Use shapely unary_union + geopandas dissolve; sklearn DBSCAN on UTM coordinates.
EXPECTED OUTPUT	pilot_clusters.gpkg, parameter_sensitivity.csv, week2_review.md.
COMPLETION CHECKLIST	☐ Sensitivity table for 3 pilots ☐ Visual QC notes for 90 clusters ☐ Chosen parameters justified
TIME AND MODE	Core research **1.5** h | Coding / data processing **3.0** h | Documentation **1.0** h | Quality control **1.0** h | Total **6.5** h Mode: Mixed (decision: Cluster parameters.)
DEPENDENCIES	Day 13.
FALLBACK STRATEGY	If clusters are unstable, use two-tier definition (hamlet vs village) and carry both into the inventory.

DAY 15 Province-wide settlement polygons

Phase 2 | 7.0 h planned | Mode: Auto

MAIN OBJECTIVE	Generate settlement polygons for all of Sindh with attributes and per-taluka QC.
ACTION STEPS	1. Run the chosen clustering per taluka tile (parallel, chunked). 2. Attach attributes: n_buildings, built area, centroid, equivalent radius, elevation, nearest road distance. 3. Remove urban cores using GHSL SMOD/PBS urban talukas and city polygons; keep peri-urban villages flagged. 4. Write topology checks: overlaps, slivers, multi-part geometries. 5. Export settlements_v0.parquet.
TECHNICAL INSTRUCTIONS	Mask urban areas with a documented rule (GHS-SMOD class >= urban centre, or OSM place=city/town polygons). Keep a flag 'peri_urban'. Use tiled processing with joblib; write each taluka to disk to survive failures.
EXPECTED OUTPUT	settlements_v0.parquet (+ per-taluka logs), qc_topology.csv.
COMPLETION CHECKLIST	☐ All talukas processed ☐ 0 invalid geometries ☐ Settlement counts per taluka reported
TIME AND MODE	Core research **0.5** h | Coding / data processing **4.5** h | Documentation **1.0** h | Quality control **1.0** h | Total **7.0** h Mode: Auto
DEPENDENCIES	Day 14.
FALLBACK STRATEGY	Memory limits: shrink tile size; use DuckDB spatial or Dask-GeoPandas.

DAY 16 Names: attach, normalise and transliterate

Phase 2 | 7.0 h planned | Mode: Mixed

MAIN OBJECTIVE	Attach names from OSM/GeoNames/PBS lists to settlement polygons using a documented normalisation pipeline.
ACTION STEPS	1. Spatial join place names to nearest polygon within 300 m (or contained). 2. Normalise: lowercase, unify Goth/Gawn/Basti/Deh/Mohalla/Village tokens, remove honorific noise, strip punctuation. 3. Create phonetic keys and token sets suited to Sindhi/Urdu transliteration variants. 4. Store all name variants in names_alt.parquet with source and language. 5. Report % polygons named, by source and district.
TECHNICAL INSTRUCTIONS	Normalisation is deliberately conservative: never merge two names unless distance and admin context agree. Use rapidfuzz (token_set_ratio) plus a custom variant dictionary (e.g. 'Muhammad/Mohammad/Muhammed', 'Khan/Khaan'). Keep raw names untouched.

DAY 16 Names: attach, normalise and transliterate

Phase 2 | 7.0 h planned | Mode: Mixed

EXPECTED OUTPUT	settlements_v0_named.parquet, names_alt.parquet, naming_coverage.csv.
COMPLETION CHECKLIST	☐ Raw names preserved ☐ >= 1 normalised key per named polygon ☐ Naming coverage table by district
TIME AND MODE	Core research **1.5** h | Coding / data processing **3.5** h | Documentation **1.0** h | Quality control **1.0** h | Total **7.0** h Mode: Mixed (decision: Variant dictionary entries.)
DEPENDENCIES	Day 15.
FALLBACK STRATEGY	Few names available: keep unnamed settlements (ID-only) — the model does not need names.

DAY 17 Deh boundaries and assignment

Phase 2 | 6.5 h planned | Mode: Mixed

MAIN OBJECTIVE	Obtain deh (mouza) polygons if available and assign settlements to dehs; otherwise create documented proxies.
ACTION STEPS	1. Check responses to Day-11 requests; search Sindh Board of Revenue/LRMIS, Survey of Pakistan, PBS GIS, HDX, OSM for deh polygons. 2. If polygons exist: point-in-polygon assignment, handle boundary cases by largest overlap. 3. If not: record 'deh = unknown' and assign settlements to taluka/UC only; use Voronoi influence zones as spatial partition. 4. Quantify assignment coverage and cross-check settlement counts per deh against PBS Table 1.2 ranges at taluka level. 5. Document legal/licensing limits.
TECHNICAL INSTRUCTIONS	A deh is a revenue unit, not a social village; do not equate counts. If deh polygons are obtained, the resource zone is clipped by deh boundary where it improves ecological meaning; otherwise unclipped. Either way, ALL later steps run on settlements.
EXPECTED OUTPUT	deh_assignment.parquet or deh_unavailable.md, coverage_table.csv.
COMPLETION CHECKLIST	☐ Assignment rule documented ☐ Coverage % computed per district ☐ Fallback path documented if no deh data
TIME AND MODE	Core research **2.0** h | Coding / data processing **2.5** h | Documentation **1.5** h | Quality control **0.5** h | Total **6.5** h Mode: Mixed (decision: Rule for boundary cases.)
DEPENDENCIES	Days 11, 12, 15.
FALLBACK STRATEGY	No deh polygons: Voronoi partition within taluka/UC; PBS aggregates remain taluka-level references.

DAY 18 Village identifier system and registry

Phase 2 | 7.0 h planned | Mode: Auto

MAIN OBJECTIVE	Create persistent, name-independent identifiers and a registry with crosswalks.
ACTION STEPS	1. Define VID = 'SD' + district code + taluka code + H3 index (res 10) of centroid + 2-char disambiguator. 2. Create village_registry.parquet: vid, centroid, polygon hash, quality class, source ids, name keys, deh code if known. 3. Write crosswalk tables (OSM id, GeoNames id, PBS codes, footprint cluster id). 4. Add registry versioning (v1.0, change log) and a collision test. 5. Unit-test: IDs stable across reruns.
TECHNICAL INSTRUCTIONS	H3 (h3-py) gives deterministic spatial keys; a hash of rounded centroid (1e-5 deg) breaks ties. IDs never encode names. Registry updates create new versions and keep retired IDs with a 'superseded_by' field.
EXPECTED OUTPUT	village_registry.parquet, crosswalks/*.csv, test_ids.py.

DAY 18 Village identifier system and registry

Phase 2 | 7.0 h planned | Mode: Auto

COMPLETION CHECKLIST	☐ 0 duplicate VIDs ☐ IDs identical after rerun ☐ Crosswalks load without orphans
TIME AND MODE	Core research **0.5** h | Coding / data processing **4.0** h | Documentation **1.5** h | Quality control **1.0** h | Total **7.0** h Mode: Auto
DEPENDENCIES	Days 15-17.
FALLBACK STRATEGY	H3 not installable: use geohash (level 9) + counter.

DAY 19 Duplicate detection and resolution

Phase 2 | 6.5 h planned | Mode: Mixed

MAIN OBJECTIVE	Resolve duplicated names and duplicated detections of the same settlement.
ACTION STEPS	1. Find same-name groups within taluka/UC and across Sindh. 2. Detect same-settlement duplicates: centroid distance < 150 m and name similarity >= 0.85. 3. Create a manual review queue for 300 ambiguous pairs; decide merge/keep. 4. Apply merges; log every merge with before/after IDs. 5. Produce duplicate_report.md with counts per district.
TECHNICAL INSTRUCTIONS	Distinguish three cases: (a) same name, different places (keep; disambiguate by VID); (b) same place, several detections (merge); (c) near neighbours with similar names (keep). Report rates; do not drop villages merely for sharing a name.
EXPECTED OUTPUT	merge_log.csv, duplicate_report.md, registry v0.5.
COMPLETION CHECKLIST	☐ Merge log complete ☐ 300 pairs reviewed ☐ Post-merge uniqueness tests pass
TIME AND MODE	Core research **2.0** h | Coding / data processing **2.5** h | Documentation **1.0** h | Quality control **1.0** h | Total **6.5** h Mode: Mixed (decision: Merge decisions.)
DEPENDENCIES	Day 18.
FALLBACK STRATEGY	If review is too slow, review a stratified 150 and fix thresholds by precision on that sample.

DAY 20 Missing coordinates and quality classes

Phase 2 | 6.5 h planned | Mode: Mixed

MAIN OBJECTIVE	Handle records without coordinates and assign a coordinate quality class to every village.
ACTION STEPS	1. Collect non-georeferenced records (PBS lists, other tables, names without points). 2. Geocode by name within taluka against settlement names (probabilistic, top-3 candidates with scores). 3. Define classes: Q1 direct coordinates/footprints; Q2 name+admin match; Q3 admin-only; Q4 unmatched. 4. Assign classes and flag training eligibility (Q1-Q2 only). 5. Summarise class shares per district.
TECHNICAL INSTRUCTIONS	Matching score = 0.5*name_sim + 0.3*admin_agreement + 0.2*distance_prior; threshold tuned on 200 hand-verified pairs. Q3/Q4 stay in the inventory but never enter model training. Report the share excluded; this bounds the geographic coverage claim.
EXPECTED OUTPUT	quality_class column in registry, unmatched_records.csv, quality_summary.csv.
COMPLETION CHECKLIST	☐ Every VID has a class ☐ Threshold tuned on verified pairs ☐ Excluded share reported by district

DAY 20 Missing coordinates and quality classes

Phase 2 | 6.5 h planned | Mode: Mixed

TIME AND MODE	Core research **1.5** h | Coding / data processing **3.0** h | Documentation **1.0** h | Quality control **1.0** h | Total **6.5** h Mode: Mixed (decision: Class thresholds.)
DEPENDENCIES	Days 16, 19.
FALLBACK STRATEGY	Poor matching: raise the threshold and accept lower coverage; never force matches.

DAY 21 Weekly review 3 and validation sample design

Phase 2 | WEEKLY REVIEW | 6.0 h planned | Mode: Manual

MAIN OBJECTIVE	Review week 3 and design the independent spatial validation of the inventory.
ACTION STEPS	1. Run weekly checklist; update risk scores (coordinate quality risk). 2. Draw a stratified sample of 600 settlements (by district, size class, source agreement, biome) plus 150 random 1-km cells to estimate omission. 3. Prepare QGIS projects with Sentinel-2 composite and a high-resolution basemap used for visual interpretation only. 4. Write the interpretation protocol (correct/over-split/merged/false/ missing) and a second-rater subset (100). 5. Pre-compute metric formulas (Section 5.7).
TECHNICAL INSTRUCTIONS	Sampling frame includes the 'missing' strata: random cells containing buildings but no detected settlement. Use Wilson intervals for proportions. Note basemap terms of use: interpretation is allowed, redistribution of derived imagery is not.
EXPECTED OUTPUT	validation_sample.gpkg, interpretation_protocol.md, qgis_validation.qgz, week3_review.md.
COMPLETION CHECKLIST	☐ 600+150 units sampled with seed logged ☐ Protocol has class definitions ☐ 100-unit second-rater subset marked
TIME AND MODE	Core research **2.0** h | Coding / data processing **2.0** h | Documentation **1.5** h | Quality control **0.5** h | Total **6.0** h Mode: Manual (decision: Protocol choices.)
DEPENDENCIES	Days 15-20.
FALLBACK STRATEGY	Basemap unavailable: use Sentinel-2 10 m only and accept coarser omission/commission estimates.

DAY 22 Execute inventory validation

Phase 2 | 6.0 h planned | Mode: Manual

MAIN OBJECTIVE	Perform the manual validation and compute detection and matching accuracy with uncertainty.
ACTION STEPS	1. Interpret 600 settlement samples and 150 cells in QGIS (about 5 hours). 2. Second rater labels 100 units; compute Cohen's kappa. 3. Compute precision (commission), recall (omission), split/merge rates, name-match accuracy. 4. Compute Wilson 95% CIs overall and by district group. 5. Record systematic errors (e.g. dune-field hamlets, flooded areas, solar farms).
TECHNICAL INSTRUCTIONS	Report metrics as measured, with CIs; no predetermined target. Precision = verified true settlements / sampled detections; recall = detected settlements in cells / (detected + missed). Keep raw interpretation CSV in outputs/tables for the supplement.
EXPECTED OUTPUT	inventory_validation.csv, kappa.txt, error_taxonomy.md.
COMPLETION CHECKLIST	☐ All 750 units labelled ☐ CIs computed ☐ Error taxonomy with >= 5 error types
TIME AND MODE	Core research **1.0** h | Coding / data processing **1.0** h | Documentation **1.0** h | Quality control **3.0** h | Total **6.0** h Mode: Manual (decision: Visual labelling.)

DAY 22 Execute inventory validation

Phase 2 | 6.0 h planned | Mode: Manual

DEPENDENCIES	Day 21.
FALLBACK STRATEGY	If time runs short, finish the 300-unit core sample and flag the remainder as 'planned'.

DAY 23 Village zones: core, resource and connectivity

Phase 2 | 7.0 h planned | Mode: Auto

MAIN OBJECTIVE	Implement and test the three-zone geometry for every village.
ACTION STEPS	1. Implement core C_i = S_i buffered ~100 m (sensitivity 50-200 m). 2. Implement resource zone R_i = (Voronoi cell of settlement centroid within taluka) ∩ buffer(S_i, r_max) minus all cores; default r_max = 2 km, desert 3-5 km. 3. Mask permanent water/urban from R_i; keep area attributes. 4. Build connectivity zone K_i = corridors along roads/canals/drains within 5 km buffer toward k=8 nearest neighbours. 5. Run geometry QC: area distribution, overlaps, empty zones.
TECHNICAL INSTRUCTIONS	Use shapely 2.x vectorised ops on UTM 42N; cap Voronoi cells with taluka polygons; store zones as three layers (core, resource, connectivity) keyed by VID. Zone choices are scientific decisions: resource-zone radius will be justified by the Day-44 sensitivity analysis, not asserted.
EXPECTED OUTPUT	zones_core.parquet, zones_resource.parquet, zones_connectivity.parquet, zone_qc.csv.
COMPLETION CHECKLIST	☐ Zones exist for every Q1-Q2 village ☐ No resource zone overlaps a foreign core ☐ Area statistics reported
TIME AND MODE	Core research **1.0** h | Coding / data processing **4.0** h | Documentation **1.0** h | Quality control **1.0** h | Total **7.0** h Mode: Auto
DEPENDENCIES	Days 15, 18.
FALLBACK STRATEGY	Memory/time: compute zones per taluka; if Voronoi is slow, use rasterised nearest-seed allocation at 30 m.

DAY 24 Village atlas v0.1 and coverage report

Phase 2 | 7.0 h planned | Mode: Auto

MAIN OBJECTIVE	Generate the first atlas and a province-wide coverage report.
ACTION STEPS	1. Create province map of settlements by quality class and density. 2. Create district summary table: PBS dehs, PBS settlements (if available), detected settlements, Q1-Q4 shares, validation precision/recall. 3. Generate 20 sample atlas sheets (settlement + zones on Sentinel-2). 4. Build an interactive HTML map (folium/lonboard) for QC. 5. Write coverage.md with the exact definition of 'coverage'.
TECHNICAL INSTRUCTIONS	Define: geographic coverage = share of PBS-reported rural populated area (or population) within a taluka represented by Q1-Q2 villages; matching accuracy = validated name+location agreement (Day 22). PBS counts are used for comparison, not as ground truth.
EXPECTED OUTPUT	atlas_v0.1.pdf (sample), coverage_report.md, coverage_by_district.csv, interactive_map.html.
COMPLETION CHECKLIST	☐ Coverage table for all districts ☐ Definitions written ☐ 20 atlas sheets visually checked
TIME AND MODE	Core research **1.5** h | Coding / data processing **3.0** h | Documentation **1.5** h | Quality control **1.0** h | Total **7.0** h Mode: Auto
DEPENDENCIES	Days 18-23.
FALLBACK STRATEGY	Atlas rendering slow: restrict to 10 pilot districts.

DAY 25 MILESTONE 2: inventory freeze v1.0

Phase 2 | MILESTONE 2 | 7.0 h planned | Mode: Mixed

MAIN OBJECTIVE	Close Phase 2 with a frozen, documented village inventory and decision memo.
ACTION STEPS	1. Run all inventory tests; fix remaining defects. 2. Freeze registry v1.0 (GeoParquet + GPKG), data dictionary and licence statement. 3. Write the decision memo: final analysis unit, counts per quality class, excluded areas. 4. Archive a draft Zenodo record (embargoed); tag v0.2.0. 5. Update risk register and manuscript outline (Methods section 1).
TECHNICAL INSTRUCTIONS	From here the inventory is immutable; later improvements create v1.1 with a change log. State clearly the final N (measured, not assumed) and per-district coverage.
EXPECTED OUTPUT	village_inventory_v1.0 (+ zones), data_dictionary.md, decision_memo_inventory.md, tag v0.2.0.
COMPLETION CHECKLIST	☐ Registry passes all unit tests ☐ N and coverage reported from files ☐ Milestone-2 file list complete
TIME AND MODE	Core research **1.5** h | Coding / data processing **2.0** h | Documentation **2.0** h | Quality control **1.5** h | Total **7.0** h Mode: Mixed (decision: Go/no-go on inventory.)
DEPENDENCIES	Days 11-24.
FALLBACK STRATEGY	If validation shows poor recall in some districts, label them 'low confidence' and train models only where confidence is adequate.

Phase 3: Satellite data acquisition and processing (Days 26-40)

Sentinel-1, Sentinel-2 cloud masking, Landsat, Dynamic World, MODIS, rainfall, temperature, surface water, DEM and terrain, harmonised environmental dataset. **Milestone 3 on Day 40.**

DAY 26 GEE zonal-statistics framework

Phase 3 | 7.0 h planned | Mode: Auto

MAIN OBJECTIVE	Build a reusable, fault-tolerant Earth Engine pipeline that turns any image collection into village-zone time series.
ACTION STEPS	1. Upload zone polygons as Earth Engine assets per district (simplify to 5 m tolerance; keep VID). 2. Write src/preprocessing/gee_zonal.py: batch features (<= 5,000 per task), multi-band yearly stacks, reduceRegions with tileScale, retry/backoff. 3. Write a task manifest (task id, product, year, chunk, status) and a monitor script. 4. Test end to end on one pilot taluka for one year of Sentinel-2. 5. Define output schema: long Parquet (vid, zone, date_start, date_end, product, band, stat, value, n_valid).
TECHNICAL INSTRUCTIONS	Prefer yearly multi-band images (12 months x bands) reduced once per chunk over one task per month: it cuts task counts by about 12x. Always export n_valid (pixel count of valid observations) with each mean. Use ee.Reducer.mean().combine(ee.Reducer.count(), sharedInputs=True). Respect quotas from Day 6.
EXPECTED OUTPUT	gee_zonal.py, task_manifest.csv, pilot export Parquet.
COMPLETION CHECKLIST	☐ Pilot export reproduces a local rasterio check within tolerance ☐ Retry logic tested by forced failure ☐ Manifest tracks all tasks
TIME AND MODE	Core research **0.5** h | Coding / data processing **4.5** h | Documentation **1.0** h | Quality control **1.0** h | Total **7.0** h Mode: Auto
DEPENDENCIES	Days 6, 23-25.
FALLBACK STRATEGY	Asset upload too large: split by district; or avoid assets by sending geometries in chunks from Python (slower).

DAY 27 Sentinel-2 preprocessing and cloud masking

Phase 3 | 7.0 h planned | Mode: Mixed

MAIN OBJECTIVE	Create analysis-ready monthly Sentinel-2 composites with explicit cloud and quality handling.
ACTION STEPS	1. Select COPERNICUS/S2_SR_HARMONIZED; verify coverage start over Sindh (Day 8 report). 2. Link GOOGLE/CLOUD_SCORE_PLUS/V1/S2_HARMONIZED (band cs_cdf); mask below a tuned threshold (start 0.6). 3. Build monthly median composites (and 16-day for crop phenology) with valid-observation counts. 4. Compute NDVI, EVI, NDWI(green-NIR), MNDWI, NDMI per image before compositing. 5. Visual QC on 10 chips (Thar dust, Indus floodplain, delta).
TECHNICAL INSTRUCTIONS	Scale reflectance by 1/10000. NDVI = (B8-B4)/(B8+B4); EVI = 2.5(B8-B4)/(B8+6B4-7.5B2+1); MNDWI = (B3-B11)/(B3+B11). Bright desert sand and salt crust can be flagged as cloud: tune the threshold per ecoregion on a visual sample and store the value in configs/.
EXPECTED OUTPUT	s2_monthly_composite.py, s2_qc_chips.png, threshold choice recorded.
COMPLETION CHECKLIST	☐ Valid-observation count exported ☐ Threshold justified on a sample ☐ No negative-reflectance artefacts
TIME AND MODE	Core research **1.0** h | Coding / data processing **4.0** h | Documentation **1.0** h | Quality control **1.0** h | Total **7.0** h Mode: Mixed (decision: Threshold choice.)
DEPENDENCIES	Day 26.
FALLBACK STRATEGY	Cloud Score+ unavailable: use COPERNICUS/S2_CLOUD_PROBABILITY (s2cloudless, prob < 40) + SCL mask.

DAY 28 Weekly review 4 and Sentinel-2 province run (batch 1)

Phase 3 | WEEKLY REVIEW | 7.0 h planned | Mode: Auto

MAIN OBJECTIVE	Review week 4, then launch the province-wide Sentinel-2 extraction in batches.
ACTION STEPS	1. Run weekly checklist; confirm quota consumption so far. 2. Launch S2 jobs for 2018-2025 by district groups; record task IDs. 3. While running, compute S2 coverage statistics for finished tasks (valid fraction by month). 4. Spot-check 20 villages against QGIS zonal statistics. 5. Write week4_review.md.
TECHNICAL INSTRUCTIONS	Run jobs overnight; poll status with ee.data.getTaskList. Do not re-run succeeded tasks. Record EECU time per task to calibrate later batches. Valid fraction in July-September is expected to be low (monsoon cloud); keep it as an explicit covariate, never fill silently.
EXPECTED OUTPUT	s2_extract_batch1/*.parquet, coverage_stats.csv, week4_review.md.
COMPLETION CHECKLIST	☐ Batch-1 tasks complete or queued ☐ Spot checks within tolerance ☐ Quota log updated
TIME AND MODE	Core research **0.5** h | Coding / data processing **4.0** h | Documentation **1.0** h | Quality control **1.5** h | Total **7.0** h Mode: Auto
DEPENDENCIES	Days 26-27.
FALLBACK STRATEGY	Quota hit: reduce to the 2019-2025 window first, then extend; or restrict to the resource zone only.

DAY 29 Sentinel-2 province run (batch 2) and QC

Phase 3 | 7.0 h planned | Mode: Auto

MAIN OBJECTIVE	Finish Sentinel-2 extraction and quality-control the full table.

DAY 29 Sentinel-2 province run (batch 2) and QC

Phase 3 | 7.0 h planned | Mode: Auto

ACTION STEPS	1. Complete remaining districts/years; re-run failed tasks. 2. Merge Parquets; check duplicates, missing zone-months and units. 3. Plot valid-observation heatmaps by month x district. 4. Flag zone-months with n_valid below the minimum (e.g. < 10 pixels or < 30% of zone). 5. Version the dataset (s2_v1).
TECHNICAL INSTRUCTIONS	Minimum-coverage rule is part of the data model: value = NaN when coverage is insufficient; mask array carries that information into the model. Keep both the unfilled and (later) causally filled series.
EXPECTED OUTPUT	s2_zonal_v1.parquet, qc_heatmaps.png.
COMPLETION CHECKLIST	☐ Row counts match expected zones x months ☐ Masks stored ☐ Dataset hash logged
TIME AND MODE	Core research **0.5** h | Coding / data processing **4.0** h | Documentation **1.0** h | Quality control **1.5** h | Total **7.0** h Mode: Auto
DEPENDENCIES	Day 28.
FALLBACK STRATEGY	Missing years: carry as NaN; Landsat and MODIS provide the long record.

DAY 30 Landsat Collection 2 harmonisation

Phase 3 | 7.0 h planned | Mode: Mixed

MAIN OBJECTIVE	Build consistent 30 m Landsat series (TM/ETM+/OLI) for 2000-2025.
ACTION STEPS	1. Merge LANDSAT/LT05, LE07, LC08, LC09 C02 T1_L2 collections. 2. Apply scale factors (0.0000275, -0.2 offset) and QA_PIXEL masks (cloud, shadow, cirrus, dilated cloud). 3. Harmonise ETM+/TM to OLI (or vice versa) with published linear coefficients (Roy et al. 2016) and verify with overlap statistics. 4. Build monthly and seasonal (Rabi, Kharif) composites; handle the ETM+ SLC-off stripes post-May 2003. 5. Compute NDVI, EVI, MNDWI, NDMI, and surface temperature (ST_B10) per composite.
TECHNICAL INSTRUCTIONS	Seasons: Rabi = Nov-Apr, Kharif = May-Oct (documented as a modelling convention; Sindh crop calendars vary by crop and canal command). ETM+ strips are masked, not interpolated. Record per-composite sensor mix as a covariate.
EXPECTED OUTPUT	landsat_composite.py, sensor_mix_table.csv, coefficient_check.png.
COMPLETION CHECKLIST	☐ Overlap regression R^2 and bias reported ☐ SLC-off pixels masked ☐ Season definition in configs
TIME AND MODE	Core research **1.0** h | Coding / data processing **4.0** h | Documentation **1.0** h | Quality control **1.0** h | Total **7.0** h Mode: Mixed (decision: Season definition.)
DEPENDENCIES	Day 26.
FALLBACK STRATEGY	Harmonisation unstable: use NDVI only from OLI-era plus MODIS for pre-2013; or analyse sensor-specific series with sensor dummies.

DAY 31 Landsat extraction and cross-sensor calibration

Phase 3 | 7.0 h planned | Mode: Mixed

MAIN OBJECTIVE	Extract Landsat zonal series and quantify agreement with Sentinel-2 and MODIS.

DAY 31 Landsat extraction and cross-sensor calibration

Phase 3 | 7.0 h planned | Mode: Mixed

ACTION STEPS	1. Run Landsat exports (2000-2025) by district group. 2. Compare Landsat NDVI with S2 NDVI on 2018-2025 overlap at zone level (OLS, Deming, bias by ecoregion). 3. Compare with MODIS MOD13Q1 NDVI at the resource-zone scale. 4. Decide primary long-term vegetation series and the role of S2. 5. Log the decision with evidence.
TECHNICAL INSTRUCTIONS	Primary design choice (to be confirmed here with evidence): long record = MODIS NDVI/EVI (250 m, 2000-) for the forecasting backbone; Landsat 30 m for cross-checks and chips; Sentinel-2 10 m for 2017+ chips and spatial features. Report slope, intercept, RMSE of cross-sensor fits; do not splice series without a stated adjustment.
EXPECTED OUTPUT	landsat_zonal_v1.parquet, crosssensor_report.md, DECISIONS entry.
COMPLETION CHECKLIST	☐ Overlap statistics by ecoregion ☐ Primary series chosen and justified ☐ Landsat table QC'd
TIME AND MODE	Core research **1.5** h | Coding / data processing **3.5** h | Documentation **1.0** h | Quality control **1.0** h | Total **7.0** h Mode: Mixed (decision: Primary series choice.)
DEPENDENCIES	Days 29, 30.
FALLBACK STRATEGY	Landsat export too heavy: restrict Landsat to seasonal composites and resource zones.

DAY 32 Sentinel-1 SAR processing

Phase 3 | 7.0 h planned | Mode: Mixed

MAIN OBJECTIVE	Create consistent Sentinel-1 backscatter and water-sensitive layers for 2015-2025.
ACTION STEPS	1. Filter COPERNICUS/S1_GRD: IW, VV+VH, one orbit direction per area (avoid mixing geometries). 2. Convert to dB, apply multi-temporal or focal-median speckle reduction, optional slope normalisation near the Kirthar hills. 3. Compute monthly median VV, VH, VH-VV and monthly minimum VH (water sensitivity). 4. Define inundation detection: VH threshold per scene via Otsu on a regional histogram plus change from the dry-season baseline. 5. Document S1B end (Dec 2021) and the resulting sampling-rate change.
TECHNICAL INSTRUCTIONS	GEE S1_GRD is already thermal-noise corrected, calibrated and terrain-corrected, but not speckle filtered. Sampling drops after 2021 (single satellite): include revisit count as covariate; never compare raw flood durations across the break without normalising by revisits. Flat Indus plain minimises layover/shadow issues except near hills.
EXPECTED OUTPUT	s1_monthly_composite.py, inundation_prototype.png, s1_revisit_table.csv.
COMPLETION CHECKLIST	☐ Revisit count stored ☐ Water prototype visually checked on the 2022 flood ☐ Orbit policy documented
TIME AND MODE	Core research **1.0** h | Coding / data processing **4.0** h | Documentation **1.0** h | Quality control **1.0** h | Total **7.0** h Mode: Mixed (decision: Threshold approach.)
DEPENDENCIES	Day 26.
FALLBACK STRATEGY	Orbit mixing unavoidable: add orbit as covariate and use relative (change) metrics only.

DAY 33 Dynamic World integration

Phase 3 | 7.0 h planned | Mode: Auto

MAIN OBJECTIVE	Extract land-cover probability series from Dynamic World and define their permitted use.

DAY 33 Dynamic World integration

Phase 3 | 7.0 h planned | Mode: Auto

ACTION STEPS	1. Select GOOGLE/DYNAMICWORLD/V1; verify start date over Sindh. 2. Create monthly mean probabilities for water, crops, trees, grass, shrub_and_scrub, built, bare, flooded_vegetation. 3. Zonal statistics for core and resource zones; keep n_valid. 4. Compare 'crops' probability with NDVI seasonality and PBS crop presence (taluka). 5. Write the use policy: DW is a proxy classifier, not ground truth.
TECHNICAL INSTRUCTIONS	Use probability means, not hard labels, to avoid classification flicker. Known confusion: salt crusts, dust, fallow and bright sand may be labelled bare/built; flooded vegetation and rice paddies overlap water. Document these in the data card.
EXPECTED OUTPUT	dw_zonal_v1.parquet, dw_policy.md.
COMPLETION CHECKLIST	☐ Probabilities within [0,1] ☐ Seasonality plausible vs NDVI ☐ Policy written
TIME AND MODE	Core research **1.0** h | Coding / data processing **3.5** h | Documentation **1.0** h | Quality control **1.5** h | Total **7.0** h Mode: Auto
DEPENDENCIES	Day 26.
FALLBACK STRATEGY	Coverage gaps: build crop/water fractions directly from S2 thresholds.

DAY 34 MODIS vegetation, LST and ET products

Phase 3 | 7.0 h planned | Mode: Auto

MAIN OBJECTIVE	Extract MODIS NDVI/EVI, LST, and optional ET/GPP time series with correct QA and timing.
ACTION STEPS	1. MODIS/061/MOD13Q1: NDVI, EVI, SummaryQA; MODIS/061/MOD11A2 and MYD11A2: LST_Day_1km, LST_Night_1km, QC; optional MOD16A2GF ET and MOD17A2HGF GPP. 2. Apply scale factors and QA masks; compute availability date = composite END date. 3. Zonal means over the resource zone only (cores are sub-pixel). 4. Build monthly aggregates with count of good composites. 5. Check LST seasonality against ERA5-Land skin temperature.
TECHNICAL INSTRUCTIONS	Time stamps: MODIS 16-day/8-day composites carry the start date in system:time_start; the information is only available at the END of the window. Gap-filled products (GF) use temporal information that may include future observations: use only for descriptive analysis, not as forecasting inputs unless re-derived causally.
EXPECTED OUTPUT	modis_zonal_v1.parquet, availability_lag_table.csv.
COMPLETION CHECKLIST	☐ Availability date column present ☐ GF products flagged 'non-causal' ☐ LST day/night plausible ranges
TIME AND MODE	Core research **1.0** h | Coding / data processing **3.5** h | Documentation **1.0** h | Quality control **1.5** h | Total **7.0** h Mode: Auto
DEPENDENCIES	Day 26.
FALLBACK STRATEGY	QA too restrictive: use SummaryQA 0-1 and report coverage.

DAY 35 Weekly review 5 and CHIRPS rainfall

Phase 3 | WEEKLY REVIEW | 7.0 h planned | Mode: Auto

MAIN OBJECTIVE	Review week 5 and build the rainfall series.

DAY 35 Weekly review 5 and CHIRPS rainfall

Phase 3 | WEEKLY REVIEW | 7.0 h planned | Mode: Auto

ACTION STEPS	1. Run weekly checklist; confirm batch statuses. 2. UCSB-CHG/CHIRPS/DAILY: zonal monthly totals, rainy days (> 1 mm), max 1-day, max 5-day, longest dry spell. 3. Compute SPI-1, 3, 6, 12 per village with gamma fits on the 1981-2010 baseline (handle zeros with mixed distribution). 4. Compare monthly CHIRPS against any available PMD gauge records (request if needed). 5. Write week5_review.md.
TECHNICAL INSTRUCTIONS	Gamma fit by maximum likelihood on non-zero months; probability of zero treated as q; transform with the standard normal inverse. Baseline 1981-2010 is fixed ahead of evaluation to avoid using test-period information. CHIRPS is satellite+station blended; in the arid Thar it is a modelled proxy with coarse 0.05 degree resolution.
EXPECTED OUTPUT	chirps_zonal_v1.parquet, spi_params.parquet, gauge_comparison.md.
COMPLETION CHECKLIST	☐ SPI series have mean about 0 and SD about 1 on baseline ☐ Zero-rain handling documented ☐ Review file complete
TIME AND MODE	Core research **1.0** h | Coding / data processing **3.5** h | Documentation **1.0** h | Quality control **1.5** h | Total **7.0** h Mode: Auto
DEPENDENCIES	Day 26.
FALLBACK STRATEGY	No gauges: compare CHIRPS with ERA5-Land precipitation only and state the limitation.

DAY 36 ERA5-Land climate variables and SPEI

Phase 3 | 7.0 h planned | Mode: Auto

MAIN OBJECTIVE	Extract reanalysis temperature, soil moisture, PET and compute SPEI.
ACTION STEPS	1. ECMWF/ERA5_LAND/MONTHLY_AGGR: temperature_2m, total_precipitation_sum, potential_evaporation_sum, volumetric_soil_water_layer_1-3, skin_temperature, surface_runoff_sum. 2. Zonal means (9 km grid, many villages share a cell: store the cell id). 3. Compute climatic water balance D = P - PET and SPEI (log-logistic) for scales 3, 6, 12. 4. Compare ERA5-Land and CHIRPS rainfall at taluka level. 5. Anomaly baselines recorded.
TECHNICAL INSTRUCTIONS	ERA5-Land potential evaporation is reported with negative sign convention in some products: verify and convert. SPEI parameters fit on 1981-2010. Many villages share a coarse pixel: spatial variance of climate covariates is small; do not over-interpret village-level differences.
EXPECTED OUTPUT	era5_zonal_v1.parquet, spei_params.parquet.
COMPLETION CHECKLIST	☐ PET sign verified ☐ SPEI mean about 0 on baseline ☐ Cell-id column stored
TIME AND MODE	Core research **1.0** h | Coding / data processing **3.5** h | Documentation **1.0** h | Quality control **1.5** h | Total **7.0** h Mode: Auto
DEPENDENCIES	Day 35.
FALLBACK STRATEGY	ERA5-Land PET problematic: use Hargreaves PET from t2m range if needed.

DAY 37 Surface water: JRC GSW and S1/S2 extension

Phase 3 | 7.0 h planned | Mode: Mixed

MAIN OBJECTIVE	Create a consistent surface-water series and quantify the gap after 2021.

DAY 37 Surface water: JRC GSW and S1/S2 extension

Phase 3 | 7.0 h planned | Mode: Mixed

ACTION STEPS	1. JRC/GSW1_4 MonthlyHistory (1984-2021), GlobalSurfaceWater (occurrence, recurrence, seasonality): zonal water frequency per month. 2. Build the S1/S2-based water fraction series (2016-2025) from Days 27, 32. 3. Compare GSW and S1/S2 water on 2016-2021 at zone-month level (confusion matrix, correlation). 4. Define a single 'water_fraction' variable with a source flag and adjustment if needed. 5. Document the GSW end date and what replaces it.
TECHNICAL INSTRUCTIONS	GSW monthly water has gaps (clouds, no-data): keep 'valid' vs 'water' counts. Wet-season cloud lowers recall during floods. Because GSW stops in 2021, 2022+ rely on S1/S2; the model will use only the S1/S2 variant for forecasting targets to avoid a source break inside the test period.
EXPECTED OUTPUT	water_zonal_v1.parquet, gsw_vs_s1_agreement.csv.
COMPLETION CHECKLIST	☐ Agreement metrics reported ☐ Source flag in table ☐ Post-2021 policy written
TIME AND MODE	Core research **1.5** h | Coding / data processing **3.0** h | Documentation **1.0** h | Quality control **1.5** h | Total **7.0** h Mode: Mixed (decision: Source-break policy.)
DEPENDENCIES	Days 27, 32.
FALLBACK STRATEGY	Poor agreement: restrict target construction to the S1/S2 era (2016-2025) and use GSW only as a static covariate.

DAY 38 DEM, terrain and hydrology covariates

Phase 3 | 7.0 h planned | Mode: Auto

MAIN OBJECTIVE	Derive static terrain and drainage covariates for all villages.
ACTION STEPS	1. Load USGS/SRTMGL1_003 and COPERNICUS/DEM/GLO30; compare in a 30-village sample. 2. Compute elevation, slope, TWI, curvature; use MERIT/Hydro (elv, upa, hnd) for upstream area and HAND. 3. Compute distances to Indus main stem, canals (OSM), drains, coast. 4. Create static feature table with units and vertical-accuracy notes. 5. Check HAND plausibility in the flat floodplain.
TECHNICAL INSTRUCTIONS	Flat deltaic terrain makes DEM vertical error (metres) comparable to relief: treat HAND/TWI as coarse covariates, not flood-depth predictors. Use the same DEM across all villages (record which). Distances computed in UTM 42N.
EXPECTED OUTPUT	static_terrain_v1.parquet, dem_comparison.md.
COMPLETION CHECKLIST	☐ No NaN for Q1-Q2 villages ☐ Units documented ☐ DEM choice justified
TIME AND MODE	Core research **1.0** h | Coding / data processing **3.5** h | Documentation **1.0** h | Quality control **1.5** h | Total **7.0** h Mode: Auto
DEPENDENCIES	Days 23, 26.
FALLBACK STRATEGY	HAND unreliable: use distance to river/canal and elevation above local minimum in a 5 km window.

DAY 39 Exposure, accessibility and settlement context layers

Phase 3 | 7.0 h planned | Mode: Auto

MAIN OBJECTIVE	Add slow-varying context covariates: population, built-up, accessibility, optional night lights.

DAY 39 Exposure, accessibility and settlement context layers

Phase 3 | 7.0 h planned | Mode: Auto

ACTION STEPS	1. JRC/GHSL/P2023A GHS_POP, GHS_BUILT_S, GHS_SMOD: zonal sums/means by epoch (1990-2020; treat 2025/2030 as projections). 2. WorldPop 100 m population (2000-2020): zonal sums; tag as modelled estimate. 3. Oxford MAP accessibility to cities 2015; VIIRS annual night lights (optional). 4. OSM road density and nearest paved road by VID. 5. Create the 'context' table with epistemic type labels.
TECHNICAL INSTRUCTIONS	Population layers are modelled disaggregations of census data, not observations. They are used as exposure/size context, never as outcomes. Interpolate epochs causally (last-available), not by linear interpolation using future epochs.
EXPECTED OUTPUT	context_v1.parquet, epistemic_labels.csv.
COMPLETION CHECKLIST	☐ Epistemic type column complete ☐ No future-epoch interpolation ☐ Missing values summarised
TIME AND MODE	Core research **1.0** h | Coding / data processing **3.5** h | Documentation **1.0** h | Quality control **1.5** h | Total **7.0** h Mode: Auto
DEPENDENCIES	Days 13, 23.
FALLBACK STRATEGY	WorldPop missing for some years: use GHSL only.

DAY 40 MILESTONE 3: harmonised environmental dataset v0.9

Phase 3 | MILESTONE 3 | 7.0 h planned | Mode: Mixed

MAIN OBJECTIVE	Close Phase 3: assemble all extractions into one aligned store and write the data cards.
ACTION STEPS	1. Align all series to the monthly grid with explicit availability dates; build the Zarr/Parquet store (vid x month x variable). 2. Generate completeness matrices per variable x region x year. 3. Write unit tests for alignment, no future timestamps, and units. 4. Write data cards for each source (variable, scale, proxy/observation/model, limits). 5. Tag v0.3.0 and update the manuscript Methods draft (data section).
TECHNICAL INSTRUCTIONS	Store a mask tensor alongside values. 'available_at' = last date information could have been known; the loader will enforce available_at <= origin. A test will assert that no feature column has available_at > month end.
EXPECTED OUTPUT	env_store_v0.9 (Zarr/Parquet), completeness_report.pdf, data_cards/*.md, tag v0.3.0.
COMPLETION CHECKLIST	☐ Alignment tests pass ☐ Completeness matrices produced ☐ Milestone-3 file list complete
TIME AND MODE	Core research **1.0** h | Coding / data processing **2.5** h | Documentation **2.0** h | Quality control **1.5** h | Total **7.0** h Mode: Mixed (decision: Go/no-go on dataset.)
DEPENDENCIES	Days 26-39.
FALLBACK STRATEGY	If one product fails (e.g. S1), proceed without it and register the loss as an ablation condition.

Phase 4: Environmental indicator engineering (Days 41-55)

NDVI, EVI, NDWI, water persistence, LST and rainfall anomalies, productivity anomalies, floods, agricultural proxies, settlement change, stress and recovery indicators; formulas and limits in Section 7. **Milestone 4 on Day 55.**

DAY 41 Indicator catalogue and data model

Phase 4 | 7.0 h planned | Mode: Mixed

MAIN OBJECTIVE	Specify every indicator formally before computing it: formula, inputs, units, normalisation, limits.
ACTION STEPS	1. Create indicators_spec.yaml for the 12 indicator families (Section 7). 2. For each: input sensor(s), spatial zone, temporal step, valid range, normalisation, known biases. 3. Implement the indicator registry in src/feature_engineering/registry.py. 4. Implement causal gap handling (last-observation-carried-forward with age counter; no look-ahead). 5. Test on 5 villages.
TECHNICAL INSTRUCTIONS	No composite 'stress score': every indicator remains an individual variable with its own definition. Gap handling stores 'age_since_last_obs' as a feature so the model can learn reliability. Smoothing, if any, must be causal (past-only windows).
EXPECTED OUTPUT	indicators_spec.yaml, registry.py, tests.
COMPLETION CHECKLIST	☐ Spec covers all 12 families ☐ Causal-fill unit test passes ☐ Registry reproduces 5-village outputs
TIME AND MODE	Core research **2.0** h | Coding / data processing **3.0** h | Documentation **1.5** h | Quality control **0.5** h | Total **7.0** h Mode: Mixed (decision: Indicator definitions.)
DEPENDENCIES	Day 40.
FALLBACK STRATEGY	Spec too ambitious: drop agricultural proxies beyond cropping intensity.

DAY 42 Weekly review 6 and spectral indices

Phase 4 | WEEKLY REVIEW | 7.0 h planned | Mode: Auto

MAIN OBJECTIVE	Review week 6 and compute vegetation and water indices consistently across sensors.
ACTION STEPS	1. Run weekly checklist. 2. Compute NDVI, EVI, NDWI (McFeeters), MNDWI (Xu), NDMI, SAVI for Landsat, S2 and MODIS zone series. 3. Verify sensor band mappings (S2 B8/B4/B2/B3/B11; Landsat OLI B5/B4/B2/B3/B6; MODIS bands). 4. Cross-sensor consistency table by ecoregion. 5. Record saturation and soil-background issues.
TECHNICAL INSTRUCTIONS	NDVI = (NIR-Red)/(NIR+Red); EVI = 2.5 (NIR-Red)/(NIR + 6 Red - 7.5 Blue + 1); NDWI = (Green-NIR)/(Green+NIR); MNDWI = (Green-SWIR1)/(Green+SWIR1); NDMI = (NIR-SWIR1)/(NIR+SWIR1). NDVI saturates in dense canopy and is affected by soil brightness in sparse Thar vegetation: SAVI/EVI provide sensitivity checks.
EXPECTED OUTPUT	indices_v1.parquet, cross_sensor_indices.csv, week6_review.md.
COMPLETION CHECKLIST	☐ Band maps documented ☐ Consistency table complete ☐ Range checks pass
TIME AND MODE	Core research **1.0** h | Coding / data processing **3.5** h | Documentation **1.0** h | Quality control **1.5** h | Total **7.0** h Mode: Auto
DEPENDENCIES	Days 29-31, 41.
FALLBACK STRATEGY	Index inconsistent across sensors: use sensor-specific indices with sensor covariates and never mix within one training series without adjustment.

DAY 43 Seasonal normalisation framework

Phase 4 | 7.0 h planned | Mode: Mixed

MAIN OBJECTIVE	Implement and compare normalisation methods that produce anomalies without leakage.
ACTION STEPS	1. Compute village x calendar-month climatology from TRAINING years only (2001-2017). 2. Implement three anomalies: z-score (mean/SD), robust z (median/MAD), and percent-of-median; plus VCI (min-max). 3. Compare stability and distribution shape (skew, kurtosis) per ecoregion. 4. Check trend contamination: detrended vs non-detrended climatology. 5. Select the default with a written rationale.
TECHNICAL INSTRUCTIONS	z = (x - mu_{v,m}) / sigma_{v,m}. Robust z = (x - median)/(1.4826 MAD). Short baselines (17 years) give noisy sigma for villages with low variability: apply shrinkage of sigma toward the regional pooled value (empirical Bayes, weight w = n/(n+k), k = 5). Document non-stationarity: a fixed baseline treats long-term change as anomaly.
EXPECTED OUTPUT	anomaly.py, normalisation_comparison.md.
COMPLETION CHECKLIST	☐ Climatology uses only years <= 2017 ☐ Shrinkage implemented and tested ☐ Default chosen with evidence
TIME AND MODE	Core research **2.0** h | Coding / data processing **3.0** h | Documentation **1.0** h | Quality control **1.0** h | Total **7.0** h Mode: Mixed (decision: Choice of default.)
DEPENDENCIES	Day 42.
FALLBACK STRATEGY	Short-baseline instability: pool by ecoregion x month; report village-level anomalies with wider caution.

DAY 44 Resource-zone sensitivity and vegetation productivity

Phase 4 | 7.0 h planned | Mode: Mixed

MAIN OBJECTIVE	Test resource-zone radius choices and build seasonal productivity indicators.
ACTION STEPS	1. Compute NDVI series for r_max = 1, 2, 3, 5 km; evaluate variance explained by rainfall and cloud cover share. 2. Select r_max per ecoregion (or one global value) from the sensitivity table. 3. Compute integrated seasonal NDVI (area above baseline), seasonal peak, and amplitude per season. 4. Compute phenology: start/end of season by threshold (20-50% of amplitude), season length. 5. Anomaly-normalise each metric (Day 43).
TECHNICAL INSTRUCTIONS	Productivity proxy P_{v,s} = sum over season of max(NDVI_t - NDVI_base, 0) * dt, NDVI_base = village-specific dry-season 10th percentile in training years. It is a vegetation-greenness proxy; it is NOT agricultural yield. Report sensitivity of conclusions to r_max.
EXPECTED OUTPUT	productivity_v1.parquet, rmax_sensitivity.csv.
COMPLETION CHECKLIST	☐ r_max chosen from data ☐ Phenology plausible on 20 samples ☐ Anomalies computed with train-only baseline
TIME AND MODE	Core research **2.0** h | Coding / data processing **3.0** h | Documentation **1.0** h | Quality control **1.0** h | Total **7.0** h Mode: Mixed (decision: r_max selection.)
DEPENDENCIES	Days 23, 42, 43.
FALLBACK STRATEGY	Radius ambiguous: keep 2 km (3-5 km in desert) and report sensitivity as a limitation.

DAY 45 Rainfall anomaly indicators

Phase 4 | 7.0 h planned | Mode: Auto

MAIN OBJECTIVE	Finalise rainfall and drought-forcing indicators.

DAY 45 Rainfall anomaly indicators

Phase 4 | 7.0 h planned | Mode: Auto

ACTION STEPS	1. Monthly and seasonal rainfall anomalies (z, percent of normal). 2. SPI-3/6/12 and SPEI-3/6/12 from Day 35-36 verified. 3. Dry-spell length, monsoon onset proxy, R95p, RX1day, RX5day. 4. Lagged correlation between SPI-3 and vegetation anomalies by ecoregion (0-6 month lags). 5. Flag low-skill rainfall products in hyper-arid areas.
TECHNICAL INSTRUCTIONS	SPI_k = Phi^-1(F_gamma(P_k; a, b)) with mixed zero handling. Lagged correlations are descriptive; do not read them as causal. Report where CHIRPS and ERA5-Land disagree strongly (taluka-level) as uncertainty in the forcing.
EXPECTED OUTPUT	rain_indicators_v1.parquet, lag_corr_by_region.csv.
COMPLETION CHECKLIST	☐ Baseline SPI moments ok ☐ Lag table for all ecoregions ☐ Disagreement map saved
TIME AND MODE	Core research **1.5** h | Coding / data processing **3.0** h | Documentation **1.0** h | Quality control **1.5** h | Total **7.0** h Mode: Auto
DEPENDENCIES	Days 35-36.
FALLBACK STRATEGY	Rainfall products disagree: use both as separate inputs and include disagreement as a feature.

DAY 46 Land-surface temperature anomalies

Phase 4 | 7.0 h planned | Mode: Auto

MAIN OBJECTIVE	Build LST and heat-related anomaly indicators with explicit caveats.
ACTION STEPS	1. MODIS day/night LST monthly means and valid counts; diurnal range. 2. LST anomalies vs training climatology (z and robust z). 3. Heat-stress event proxy: number of 8-day composites with LST_day z >= +1 within a season. 4. ERA5-Land t2m anomalies for comparison. 5. Document LST limitations (clear-sky bias, emissivity, mixed pixels).
TECHNICAL INSTRUCTIONS	LST is skin temperature at the satellite overpass under clear sky; it is not air temperature and not human heat exposure. Salt flats and sand have emissivity issues. Landsat ST (30 m) is optional for the resource zone.
EXPECTED OUTPUT	lst_indicators_v1.parquet, lst_caveats.md.
COMPLETION CHECKLIST	☐ Valid-count column present ☐ z-scores finite ☐ Caveats documented
TIME AND MODE	Core research **1.0** h | Coding / data processing **3.5** h | Documentation **1.0** h | Quality control **1.5** h | Total **7.0** h Mode: Auto
DEPENDENCIES	Day 34, 43.
FALLBACK STRATEGY	Gaps in night LST: use day LST only.

DAY 47 Surface-water persistence and wetland dynamics

Phase 4 | 7.0 h planned | Mode: Mixed

MAIN OBJECTIVE	Compute water persistence and seasonality indicators, with checks on known wetlands.
ACTION STEPS	1. P_water = fraction of months with water in the resource zone per year; also max monthly water fraction. 2. Classify water regimes: permanent, seasonal, ephemeral, flood-only using occurrence/seasonality (JRC) and S1/S2. 3. Check known features (Manchar, Keenjhar, Haleji, Indus delta creeks, Indus main stem). 4. Compute water-anomaly series with training-year baselines. 5. Write limitations (turbidity, cloud in monsoon, mixed pixels).

DAY 47 Surface-water persistence and wetland dynamics

Phase 4 | 7.0 h planned | Mode: Mixed

TECHNICAL INSTRUCTIONS	Persistence formula: Pers_{v,y} = (1/12) * sum_m 1[w_{v,m} >= tau], tau = 0.05 of zone area (sensitivity 0.02-0.10). Rice paddies and canal irrigation produce seasonal water that is not flooding: treat 'water' as a surface signal and classify cause only through additional logic (Day 48).
EXPECTED OUTPUT	water_indicators_v1.parquet, wetland_checks.png.
COMPLETION CHECKLIST	☐ Known wetlands show expected persistence ☐ tau sensitivity table ☐ Anomaly series causal
TIME AND MODE	Core research **1.5** h | Coding / data processing **3.0** h | Documentation **1.0** h | Quality control **1.5** h | Total **7.0** h Mode: Mixed (decision: tau threshold.)
DEPENDENCIES	Day 37.
FALLBACK STRATEGY	Water signal too noisy: use a persistence class (3 levels) instead of continuous values.

DAY 48 Flood frequency and persistence

Phase 4 | 7.0 h planned | Mode: Mixed

MAIN OBJECTIVE	Create flood indicators from Sentinel-1/Optical data and validate them against independent event products.
ACTION STEPS	1. Event-based inundation: change detection against a dry-season baseline; mask permanent water and irrigated rice via cropping calendar and HAND. 2. Per village-year: flood frequency (events), flood duration (revisit-normalised), max inundated fraction in core and resource zones. 3. Retrospective pre-2015 events (2010, 2011) from Landsat/MODIS only as low-confidence layers. 4. Compare 2020 and 2022 inundation with published maps (Chen et al. 2024; UNOSAT/Copernicus EMS where available). 5. Quantify agreement (F1, IoU) on a sample.
TECHNICAL INSTRUCTIONS	Known false positives: wind-roughened water, wet sand, rice paddies, shadowed terrain; false negatives: flooded vegetation, urban double-bounce. Report agreement honestly; treat pre-2015 flood layers as sparse and cloud-limited. Duration measured in 'days between observations', normalised by revisit gaps.
EXPECTED OUTPUT	flood_indicators_v1.parquet, flood_validation.csv.
COMPLETION CHECKLIST	☐ 2022 event agreement metrics reported ☐ Revisit normalisation implemented ☐ Pre-2015 flagged low-confidence
TIME AND MODE	Core research **1.5** h | Coding / data processing **3.5** h | Documentation **1.0** h | Quality control **1.0** h | Total **7.0** h Mode: Mixed (decision: Event delineation.)
DEPENDENCIES	Days 32, 37, 47.
FALLBACK STRATEGY	No external flood layer: validate by visual interpretation of 100 village-year samples.

DAY 49 Weekly review 7 and agricultural activity proxies

Phase 4 | WEEKLY REVIEW | 7.0 h planned | Mode: Mixed

MAIN OBJECTIVE	Review week 7 and build agricultural-activity proxies without claiming yield.
ACTION STEPS	1. Run weekly checklist. 2. Cropping intensity proxy: count of NDVI peaks per year (peak prominence rule) in the resource zone. 3. Fallow-fraction proxy: share of Rabi/Kharif seasons with peak NDVI below a threshold. 4. Dry-season greenness as an irrigated-area proxy; S1 VH seasonal dynamics as a cross-check. 5. Compare taluka-level aggregates with PBS Table 2.1 crop presence shares (wheat, cotton, rice, sugarcane, maize).

DAY 49 Weekly review 7 and agricultural activity proxies

Phase 4 | WEEKLY REVIEW | 7.0 h planned | Mode: Mixed

TECHNICAL INSTRUCTIONS	scipy.signal.find_peaks on smoothed (causal) NDVI with prominence >= 0.1 (sensitivity 0.05-0.2). These indicators describe greenness cycles; they do not identify crops or yield. PBS crop tables are counts of mouzas reporting a crop, aggregated by taluka: use rank correlations only.
EXPECTED OUTPUT	agri_proxies_v1.parquet, pbs_crop_comparison.csv, week7_review.md.
COMPLETION CHECKLIST	☐ Prominence sensitivity table ☐ PBS comparison Spearman rho reported ☐ Limitations noted
TIME AND MODE	Core research **1.5** h | Coding / data processing **3.0** h | Documentation **1.0** h | Quality control **1.5** h | Total **7.0** h Mode: Mixed (decision: Peak rules.)
DEPENDENCIES	Days 42-44.
FALLBACK STRATEGY	Cloud limits Kharif peaks: use S1 as the Kharif cross-check and flag cloud-limited seasons.

DAY 50 Settlement expansion and contraction signals

Phase 4 | 7.0 h planned | Mode: Mixed

MAIN OBJECTIVE	Quantify built-up change around villages with noise controls.
ACTION STEPS	1. Built-up fraction in core zone from GHSL epochs (1990-2020) and Dynamic World built probability (2016+). 2. Compute net change and change rate with a minimum-detectable-change threshold from stable-reference zones. 3. Cross-check with building footprint counts where multi-year footprints exist. 4. Classify: expansion, stable, contraction, uncertain. 5. Write the 'what this is not' paragraph: no migration inference.
TECHNICAL INSTRUCTIONS	Net change = A_core(t2) - A_core(t1) over core area; noise floor = 2 SD of change in unchanged control polygons (urban-free desert samples). Contraction can reflect flooding, demolition, classifier noise or abandonment: satellite built-up change is NOT evidence of migration or displacement.
EXPECTED OUTPUT	settlement_change_v1.parquet, noise_floor.csv.
COMPLETION CHECKLIST	☐ Noise floor computed ☐ Classes assigned ☐ Disclaimer in data card
TIME AND MODE	Core research **1.5** h | Coding / data processing **3.0** h | Documentation **1.0** h | Quality control **1.5** h | Total **7.0** h Mode: Mixed (decision: Noise-floor policy.)
DEPENDENCIES	Days 33, 39.
FALLBACK STRATEGY	Signal weak: keep as static context (built fraction) rather than a dynamic indicator.

DAY 51 Environmental stress indicators (event-based)

Phase 4 | 7.0 h planned | Mode: Mixed

MAIN OBJECTIVE	Define and compute separate stress events per indicator, without any composite score.
ACTION STEPS	1. Vegetation stress: z_NDVI <= -1.0 (training-year thresholds) for the season. 2. Thermal stress: z_LST_day >= +1.0 for >= 2 composites in season. 3. Water-deficit stress: SPI-3 <= -1 or SPEI-3 <= -1. 4. Flood stress: inundated fraction >= 5% of core or >= 20% of resource zone (sensitivity). 5. Report stress frequency and co-occurrence as descriptive tables only.
TECHNICAL INSTRUCTIONS	E_{v,s}^{(k)} = 1[z^{(k)}_{v,s} crosses threshold theta_k], k indexes the indicator. Thresholds are parameters with sensitivity ranges frozen from training years only. Co-occurrence matrices are descriptive; do not weight or sum events into an index.

DAY 51 Environmental stress indicators (event-based)

Phase 4 | 7.0 h planned | Mode: Mixed

EXPECTED OUTPUT	stress_events_v1.parquet, threshold_sensitivity.csv.
COMPLETION CHECKLIST	☐ Four event types computed ☐ Sensitivity grid saved ☐ No composite variable exists
TIME AND MODE	Core research **2.0** h | Coding / data processing **3.0** h | Documentation **1.0** h | Quality control **1.0** h | Total **7.0** h Mode: Mixed (decision: Threshold freeze.)
DEPENDENCIES	Days 43-48.
FALLBACK STRATEGY	Thresholds produce degenerate frequencies: widen to percentile-based events (e.g. lowest quintile) and record the change.

DAY 52 Environmental recovery indicators

Phase 4 | 7.0 h planned | Mode: Mixed

MAIN OBJECTIVE	Define recovery and resilience metrics from stress spells.
ACTION STEPS	1. Identify stress spells (run-lengths of stress events) per indicator. 2. After each spell: recovery time to z >= -0.5 within m seasons, recovery rate (slope), depth (min z), area under the deficit curve. 3. Handle right-censoring (no recovery observed by the end of the record). 4. Compute AR(1) and variance of anomalies in rolling windows as resilience metrics (hypothesis-level). 5. Classify outcomes: recovered, partial, not recovered, censored.
TECHNICAL INSTRUCTIONS	Recovery time T_r = min{ k >= 1 : z_{s+k} >= -0.5 } after the last stressed season s. Recovery after a drought could be rainfall-driven rather than land-condition recovery; the indicator is conditional on climate and must be interpreted with rainfall anomalies. Critical-slowing-down metrics (rising AR(1)) are exploratory hypotheses (Scheffer et al. 2009; Dakos et al. 2012), not established predictors here.
EXPECTED OUTPUT	recovery_indicators_v1.parquet, spell_table.parquet.
COMPLETION CHECKLIST	☐ Censoring flag stored ☐ Spell table matches stress events ☐ Class counts reported
TIME AND MODE	Core research **2.0** h | Coding / data processing **3.0** h | Documentation **1.0** h | Quality control **1.0** h | Total **7.0** h Mode: Mixed (decision: Recovery thresholds.)
DEPENDENCIES	Day 51.
FALLBACK STRATEGY	Few spells in some regions: pool by ecoregion for statistics, keep village labels where defined.

DAY 53 Indicator validation I: internal and cross-sensor

Phase 4 | 7.0 h planned | Mode: Mixed

MAIN OBJECTIVE	Quantify reliability of indicators before using them as inputs or labels.
ACTION STEPS	1. Cross-sensor agreement for NDVI, water, built-up. 2. Temporal stability: repeat-pass noise on invariant targets (desert dunes, deep water). 3. Missingness analysis by month/region; effect of valid-observation thresholds. 4. Sensitivity to r_max, thresholds, normalisation (Day 43-51). 5. Write indicator_validation_I.md.
TECHNICAL INSTRUCTIONS	Use pseudo-invariant features (bare dune areas, large permanent lakes) to estimate measurement noise: SD of detrended anomalies. Report indicators with noise > signal as 'unreliable' and exclude them from labels.
EXPECTED OUTPUT	validation_I.md, noise_table.csv.

DAY 53 Indicator validation I: internal and cross-sensor

Phase 4 | 7.0 h planned | Mode: Mixed

COMPLETION CHECKLIST	☐ Noise estimates for >= 6 indicators ☐ Unreliable indicators listed ☐ Sensitivity summary complete
TIME AND MODE	Core research **1.5** h | Coding / data processing **3.0** h | Documentation **1.0** h | Quality control **1.5** h | Total **7.0** h Mode: Mixed (decision: Exclusion of unreliable indicators.)
DEPENDENCIES	Days 42-52.
FALLBACK STRATEGY	Too few invariants: use stable-reference polygons by ecoregion.

DAY 54 Indicator validation II: external reference data

Phase 4 | 7.0 h planned | Mode: Mixed

MAIN OBJECTIVE	Compare indicators with independent reference observations at the scale where they exist.
ACTION STEPS	1. PBS Mouza Census Sindh tehsil tables: share of mouzas reporting drought/flood in the 5 years before Jan 2020; brackish groundwater share; irrigation source; water-table depth. 2. Aggregate indicators to taluka and compute Spearman rho (e.g. drought share vs vegetation-stress frequency 2015-2019). 3. Compare 2022 flood exposure with Post-Disaster Needs Assessment district damage. 4. Visual validation sample: 300 village-seasons labelled by an independent reader. 5. Write limitations (ecological fallacy, reporting bias, one time point).
TECHNICAL INSTRUCTIONS	PBS tables are respondent-based and aggregated: they test rank agreement across talukas, not village-level accuracy. Use bootstrapped CIs over talukas; with about 100+ talukas, expect modest power. Report all comparisons including those that fail.
EXPECTED OUTPUT	validation_II.csv, reference_comparison.md.
COMPLETION CHECKLIST	☐ Rho with CIs for >= 5 comparisons ☐ Failed comparisons reported ☐ Visual sample labelled
TIME AND MODE	Core research **2.0** h | Coding / data processing **2.0** h | Documentation **1.5** h | Quality control **1.5** h | Total **7.0** h Mode: Mixed (decision: Interpretation.)
DEPENDENCIES	Days 11, 48-52.
FALLBACK STRATEGY	Reference tables unusable: rely on visual validation and published event maps; state the gap.

DAY 55 MILESTONE 4: indicator dataset v1.0 and handbook

Phase 4 | MILESTONE 4 | 7.0 h planned | Mode: Mixed

MAIN OBJECTIVE	Close Phase 4: freeze indicators, thresholds and the indicator handbook.
ACTION STEPS	1. Freeze indicator definitions and thresholds in configs/indicators_v1.yaml. 2. Export the feature store v1.0 (Zarr/Parquet) and schema docs. 3. Compile the indicator handbook (formulas, interpretation, limits) and validation reports. 4. Update manuscript Methods (indicators) and risk register; tag v0.4.0. 5. Peer-check: another reader reproduces three indicators from raw tables.
TECHNICAL INSTRUCTIONS	After this milestone, thresholds cannot be tuned on test-period data. Any change is a logged deviation. Keep the pre-registered lists of primary vs exploratory indicators.
EXPECTED OUTPUT	feature_store_v1.0, indicator_handbook.pdf, indicators_v1.yaml, tag v0.4.0.
COMPLETION CHECKLIST	☐ Reproduction check passes for 3 indicators ☐ Config frozen with hash ☐ Milestone-4 file list complete

DAY 55 MILESTONE 4: indicator dataset v1.0 and handbook

Phase 4 | MILESTONE 4 | 7.0 h planned | Mode: Mixed

TIME AND MODE	Core research **1.5** h | Coding / data processing **2.0** h | Documentation **2.0** h | Quality control **1.5** h | Total **7.0** h Mode: Mixed (decision: Freeze.)
DEPENDENCIES	Days 41-54.
FALLBACK STRATEGY	If some indicators remain unreliable, drop them from the model and list them as exploratory.

Phase 5: Environmental trajectory analysis (Days 56-65)

Trends, seasonal decomposition, persistent stress, recovery trajectories, change-points, clustering, spatial autocorrelation, regional comparison, forecasting targets and training/testing datasets. **Milestone 5 on Day 65.**

DAY 56 Weekly review 8 and exploratory analysis with ecological regions

Phase 5 | WEEKLY REVIEW | 7.0 h planned | Mode: Mixed

MAIN OBJECTIVE	Review week 8, explore distributions, and define ecological regions used for stratification and transfer tests.
ACTION STEPS	1. Run weekly checklist. 2. Summarise each indicator by district and month (distributions, missingness). 3. Build static covariate matrix (aridity, mean rain, elevation, HAND, distance to river/coast, irrigated fraction). 4. Cluster villages into 5-7 data-driven ecoregions (k-means/GMM) and compare to named regions (Thar desert, Kirthar hills, canal plains, riverine/kacho, delta/coast). 5. Choose k with silhouette, gap and stability (bootstrap ARI).
TECHNICAL INSTRUCTIONS	Ecoregions are a stratification device, not a scientific taxonomy. Use only static, non-target covariates (to avoid circularity). Keep a name-based region label for interpretability. Save region ids in the registry.
EXPECTED OUTPUT	ecoregions_v1.parquet, eda_summary.pdf, week8_review.md.
COMPLETION CHECKLIST	☐ k chosen with stability evidence ☐ Ecoregion map produced ☐ Named-region cross-tab saved
TIME AND MODE	Core research **2.0** h | Coding / data processing **3.0** h | Documentation **1.0** h | Quality control **1.0** h | Total **7.0** h Mode: Mixed (decision: Choice of k.)
DEPENDENCIES	Day 55.
FALLBACK STRATEGY	Unstable clusters: use named agro-ecological zones from literature and assign by taluka.

DAY 57 Long-term trend analysis

Phase 5 | 7.0 h planned | Mode: Auto

MAIN OBJECTIVE	Estimate trends per village and indicator with proper autocorrelation and multiplicity control.
ACTION STEPS	1. Deseasonalise annual/seasonal series (training and descriptive use only). 2. Mann-Kendall with Hamed-Rao autocorrelation correction; Theil-Sen slope per village. 3. Control false discoveries (Benjamini-Hochberg) and account for spatial dependence (cluster-level FDR by ecoregion). 4. Compare vegetation trends with rainfall trends to avoid misattribution. 5. Create trend maps for NDVI anomaly, LST, water persistence.
TECHNICAL INSTRUCTIONS	S = sum_{i<j} sign(x_j - x_i); slope = median_{i<j} (x_j - x_i)/(j - i). Report effect sizes with CIs, not just significance. Trends over 25 years may reflect sensor changes; cross-check with MODIS vs Landsat series.
EXPECTED OUTPUT	trends_v1.parquet, trend_maps.png.

DAY 57 Long-term trend analysis

Phase 5 | 7.0 h planned | Mode: Auto

COMPLETION CHECKLIST	☐ FDR applied ☐ Sensor-artefact check done ☐ Effect sizes reported
TIME AND MODE	Core research **2.0** h | Coding / data processing **3.0** h | Documentation **1.0** h | Quality control **1.0** h | Total **7.0** h Mode: Auto
DEPENDENCIES	Day 55.
FALLBACK STRATEGY	Too many insignificant trends: report regional medians with bootstrap CIs.

DAY 58 Seasonal decomposition

Phase 5 | 7.0 h planned | Mode: Auto

MAIN OBJECTIVE	Decompose series to quantify seasonality and residual dynamics; separate descriptive from forecasting use.
ACTION STEPS	1. Apply STL/MSTL (statsmodels) to monthly series of NDVI, LST, water fraction per village. 2. Compute seasonal strength F_s and trend strength F_t. 3. Fit harmonic regression as a causal alternative (expanding-window fit). 4. Compare residual autocorrelation structure by ecoregion. 5. Document: full-sample STL is descriptive only (uses future data).
TECHNICAL INSTRUCTIONS	F_s = max(0, 1 - Var(R)/Var(S+R)). For model inputs, only causal (expanding-window or past-only) decomposition is allowed. Missing months are masked, not interpolated.
EXPECTED OUTPUT	decomposition_stats.parquet, causal_decomp.py.
COMPLETION CHECKLIST	☐ Strength metrics for all villages ☐ Causal variant tested for no look-ahead ☐ Plots for 10 villages
TIME AND MODE	Core research **1.5** h | Coding / data processing **3.5** h | Documentation **1.0** h | Quality control **1.0** h | Total **7.0** h Mode: Auto
DEPENDENCIES	Day 55.
FALLBACK STRATEGY	STL fails on sparse series: use monthly climatology + AR(1) residual summaries.

DAY 59 Persistent stress detection

Phase 5 | 7.0 h planned | Mode: Mixed

MAIN OBJECTIVE	Detect persistent stress with explicit rules and statistical alternatives.
ACTION STEPS	1. Compute run-lengths of stress events per indicator. 2. Persistent stress: >= k consecutive seasons (k=3; sensitivity 2-4). 3. Compare to a two-state hidden Markov model (statsmodels MarkovRegression or hmmlearn). 4. Estimate P(stress at s+1 | stress at s) by ecoregion and compare with the climatological expectation. 5. Map persistent-stress villages and their duration.
TECHNICAL INSTRUCTIONS	Persistence must be compared against what chance (and climate autocorrelation) would produce: simulate surrogate series preserving seasonality and autocorrelation (phase-randomised) and compute the expected frequency of runs. Report the ratio observed/expected.
EXPECTED OUTPUT	persistence_v1.parquet, markov_vs_rule.csv, persistence_maps.png.
COMPLETION CHECKLIST	☐ Surrogate test completed ☐ k-sensitivity table ☐ HMM comparison reported
TIME AND MODE	Core research **2.0** h | Coding / data processing **3.0** h | Documentation **1.0** h | Quality control **1.0** h | Total **7.0** h Mode: Mixed (decision: k choice.)

DAY 59 Persistent stress detection

Phase 5 | 7.0 h planned | Mode: Mixed

DEPENDENCIES	Days 51-52.
FALLBACK STRATEGY	HMM unstable: use rule-based only and note the limitation.

DAY 60 Recovery trajectory identification

Phase 5 | 7.0 h planned | Mode: Mixed

MAIN OBJECTIVE	Characterise recovery after stress using survival analysis and trajectory classes.
ACTION STEPS	1. Build spell-level dataset: duration, depth, preceding rainfall, region, irrigation proxy. 2. Kaplan-Meier curves of time-to-recovery with censoring; log-rank by ecoregion. 3. Cox proportional hazards (lifelines) for covariate effects; check assumptions. 4. Define trajectory classes: rapid, slow, partial, none. 5. Create labels for the recovery target (Day 64-65).
TECHNICAL INSTRUCTIONS	h(t|x) = h_0(t) exp(beta^T x). Covariates are descriptive; recovery is not causal evidence for any intervention. Spells close to the end of record are censored; do not treat them as non-recovery.
EXPECTED OUTPUT	recovery_survival.csv, km_curves.png, trajectory_classes.parquet.
COMPLETION CHECKLIST	☐ Censoring handled ☐ PH assumption checked ☐ Class counts by region
TIME AND MODE	Core research **2.0** h | Coding / data processing **3.0** h | Documentation **1.0** h | Quality control **1.0** h | Total **7.0** h Mode: Mixed (decision: Class definitions.)
DEPENDENCIES	Day 59.
FALLBACK STRATEGY	lifelines unavailable: use statsmodels PHReg or a discrete-time logistic hazard.

DAY 61 Change-point detection

Phase 5 | 7.0 h planned | Mode: Auto

MAIN OBJECTIVE	Detect abrupt shifts in environmental series with false-positive control.
ACTION STEPS	1. Deseasonalise (descriptive) and apply PELT/BinSeg (ruptures) with penalty tuned by BIC. 2. Run Pettitt test as an independent check. 3. Run BFAST-like approach (seasonal+trend breakpoints) where feasible (R via rpy2 or Python port). 4. Map break timing; compare with 2010, 2011, 2014, 2020 and 2022 events. 5. Evaluate detection stability under penalty and noise perturbations.
TECHNICAL INSTRUCTIONS	Penalty beta ~ 2*sigma^2*log(n) as default (BIC), with sensitivity x0.5 to x2. Breaks near the series ends are unreliable (minimum segment size 12 months). Breaks may reflect sensor changes (e.g. Landsat 7 -> 8, S1B end): flag breaks coinciding with sensor events.
EXPECTED OUTPUT	changepoints_v1.parquet, break_timing_map.png.
COMPLETION CHECKLIST	☐ Sensor-event overlap flagged ☐ Stability metric reported ☐ Break counts per region
TIME AND MODE	Core research **1.5** h | Coding / data processing **3.5** h | Documentation **1.0** h | Quality control **1.0** h | Total **7.0** h Mode: Auto
DEPENDENCIES	Day 58.
FALLBACK STRATEGY	Too many breaks: increase penalty or use annual aggregates only.

DAY 62 Environmental clustering and typology

Phase 5 | 7.0 h planned | Mode: Mixed

MAIN OBJECTIVE	Create an interpretable typology of village trajectories.
ACTION STEPS	1. Features per village: trend, seasonal amplitude, interannual SD, stress frequency, persistence ratio, recovery time, rainfall sensitivity, flood frequency. 2. Standardise; reduce with PCA; cluster with GMM/HDBSCAN and with DTW k-means (tslearn) on annual anomaly trajectories. 3. Select k by BIC/silhouette and bootstrap stability (ARI). 4. Profile clusters and name them cautiously. 5. Map clusters; compare with ecoregions.
TECHNICAL INSTRUCTIONS	Clusters are descriptive types, not ground truth: report stability and avoid causal names (e.g. 'degraded'). Use clusters only for stratified evaluation and interpretation, never as a model input if they were built using test-period data.
EXPECTED OUTPUT	typology_v1.parquet, cluster_profiles.csv.
COMPLETION CHECKLIST	☐ Stability ARI reported ☐ Test-period data not used for cluster fitting used in models ☐ Maps produced
TIME AND MODE	Core research **2.0** h | Coding / data processing **3.0** h | Documentation **1.0** h | Quality control **1.0** h | Total **7.0** h Mode: Mixed (decision: Naming and k.)
DEPENDENCIES	Days 56-61.
FALLBACK STRATEGY	Poor stability: reduce to 3-4 broad types.

DAY 63 Weekly review 9 and spatial autocorrelation

Phase 5 | WEEKLY REVIEW | 7.0 h planned | Mode: Mixed

MAIN OBJECTIVE	Review week 9 and quantify spatial dependence to design the validation blocks.
ACTION STEPS	1. Run weekly checklist. 2. Global Moran's I and Geary's C for indicators and for baseline residuals (esda) using KNN (k=8) and distance-band weights. 3. Local Moran's I (LISA) and Getis-Ord Gi* hotspots with FDR. 4. Empirical semivariograms (skgstat) of baseline residuals: estimate range. 5. Decide block size and buffer from the range estimates.
TECHNICAL INSTRUCTIONS	I = (N/W) * sum_ij w_ij (x_i - xbar)(x_j - xbar) / sum_i (x_i - xbar)^2. The semivariogram range of climatology residuals sets minimum block size and buffer: blocks smaller than the range leak information between training and test. Record range per target and use the maximum (conservative).
EXPECTED OUTPUT	moran_results.csv, variograms.png, block_design.md.
COMPLETION CHECKLIST	☐ Range estimated per target ☐ Block size/buffer decision logged ☐ Hotspot maps saved
TIME AND MODE	Core research **2.0** h | Coding / data processing **3.0** h | Documentation **1.0** h | Quality control **1.0** h | Total **7.0** h Mode: Mixed (decision: Block size.)
DEPENDENCIES	Day 62.
FALLBACK STRATEGY	Variogram unreliable: use a fixed 50 km block with 10 km buffer and test sensitivity (25/100 km).

DAY 64 Regional comparisons and target definition

Phase 5 | 7.0 h planned | Mode: Manual

MAIN OBJECTIVE	Compare regions and districts statistically and finalise the forecasting targets.

DAY 64 Regional comparisons and target definition

Phase 5 | 7.0 h planned | Mode: Manual

ACTION STEPS	1. Kruskal-Wallis and effect sizes (epsilon^2) across ecoregions; Dunn post-hoc with Holm correction. 2. District summaries for Tharparkar, Umerkot, Badin, Sujawal, Dadu, Jamshoro, Sanghar, Mirpurkhas, Khairpur, Jacobabad and others. 3. Mixed-effects model (district random intercept) to partition variance (ICC). 4. Define targets: T1 vegetation anomaly (h=3,6,12 months), T2 LST anomaly, T3 water-fraction anomaly, T4 flood-event next monsoon, T5 persistence of stress, T6 recovery within 2 seasons (given stress). 5. Check that target windows begin strictly after origin; check class balance.
TECHNICAL INSTRUCTIONS	Targets are computed from indicators of the label window; inputs end at t0. For T5/T6 only villages currently in stress enter the conditional samples. Expect T4 to be rare (few flood years): plan AUPRC. Do not define a target that is mechanically a function of an input (e.g. predicting a 3-month mean that includes input months).
EXPECTED OUTPUT	targets_definition.md, class_balance.csv, icc_table.csv.
COMPLETION CHECKLIST	☐ Windows disjoint test passes ☐ Class balance reported ☐ Variance partition table saved
TIME AND MODE	Core research **2.5** h | Coding / data processing **2.5** h | Documentation **1.5** h | Quality control **0.5** h | Total **7.0** h Mode: Manual (decision: Target definitions.)
DEPENDENCIES	Days 55, 59, 60.
FALLBACK STRATEGY	Target too rare: merge T4 into a 'wet-season inundation anomaly' regression.

DAY 65 MILESTONE 5: forecasting dataset v1.0 and splits

Phase 5 | MILESTONE 5 | 7.0 h planned | Mode: Mixed

MAIN OBJECTIVE	Create leakage-audited training/validation/test datasets and freeze the split manifest.
ACTION STEPS	1. Build samples (vid, origin, horizon) with features at <= origin and labels after origin. 2. Create hexagonal blocks (H3) sized from Day 63; assign folds stratified by ecoregion; add buffers. 3. Define splits: train origins 2001-2017, validation 2018-2019, test 2020-2025; spatial test blocks held out in all stages. 4. Run the leakage audit: automated tests (time, space, normalisation, climatology, filling). 5. Export datasets, manifests and data card; tag v0.5.0.
TECHNICAL INSTRUCTIONS	Locked spatial test blocks (about 15% of H3 blocks, stratified by ecoregion) are removed first; CV folds and LODO use only the remaining blocks. Spatial blocks and temporal splits are combined: the final test is future years AND unseen blocks (strict) plus separate future-years-only and unseen-blocks-only evaluations (to isolate the two generalisation types). Normalisers fit on training samples only. The split manifest is hashed; any change is a deviation.
EXPECTED OUTPUT	forecast_dataset_v1.0, split_manifest.parquet, leakage_audit.md, tag v0.5.0.
COMPLETION CHECKLIST	☐ Leakage tests all pass ☐ Split manifest hashed ☐ Milestone-5 file list complete
TIME AND MODE	Core research **1.5** h | Coding / data processing **3.0** h | Documentation **1.5** h | Quality control **1.0** h | Total **7.0** h Mode: Mixed (decision: Final splits.)
DEPENDENCIES	Days 55-64.
FALLBACK STRATEGY	Dataset too large: subsample training villages by stratified block sampling; never subsample test blocks.

Phase 6: Advanced AI development (Days 66-80)

Eight model classes, graph, encoders, fusion, uncertainty, tuning and model selection. Simpler models are retained if they perform as well. **Milestone 6 on Day 80.**

DAY 66 Evaluation harness and data loaders

Phase 6 | 7.0 h planned | Mode: Auto

MAIN OBJECTIVE	Build the PyTorch data pipeline and metric library with automated leakage tests.
ACTION STEPS	1. Implement VistaDataset (tabular/series), optional chip loader, collate with masks. 2. Implement train-only normalisation and mask-aware losses. 3. Implement metrics: MAE, RMSE, R2, skill score, F1, AUROC, AUPRC, CRPS, coverage, ECE. 4. Write tests: block disjointness, origin ordering, label windows, determinism. 5. Set up MLflow (local) and the experiment registry.
TECHNICAL INSTRUCTIONS	Input shapes: x_ts [B, L=24, F_ts], x_clim [B, L, F_c], x_st [B, F_s], masks [B, L], y_reg [B, H=3, K=3], y_cls [B, 3]. Use persistent workers and pin_memory; keep seeds in the config; log config hash with every run.
EXPECTED OUTPUT	src/models/data.py, src/validation/metrics.py, tests/test_leakage.py.
COMPLETION CHECKLIST	☐ All tests pass ☐ Loader deterministic given seed ☐ Metrics verified against sklearn on toy data
TIME AND MODE	Core research **0.5** h | Coding / data processing **4.5** h | Documentation **1.0** h | Quality control **1.0** h | Total **7.0** h Mode: Auto
DEPENDENCIES	Day 65.
FALLBACK STRATEGY	Chips too heavy: skip chip loader; keep zonal-statistics inputs (Tier 2).

DAY 67 Baselines 1-2: climatology, persistence, autoregression

Phase 6 | 7.0 h planned | Mode: Auto

MAIN OBJECTIVE	Implement and score the simplest baselines.
ACTION STEPS	1. Seasonal climatology: predicted anomaly 0 (z space) and climatological probabilities for classes. 2. Persistence: last observed anomaly (and exponentially damped persistence). 3. AR/ARX: pooled AR(p) with village-month fixed effects, ridge with lags 1-12 and SPI. 4. Compute metrics on validation partitions only; compute skill vs climatology. 5. Write results_v0.md.
TECHNICAL INSTRUCTIONS	Skill score SS = 1 - MSE_model / MSE_climatology (ref). For classification, use prevalence as the reference (AUPRC baseline = prevalence). These baselines establish how hard the problem is; good baselines can be hard to beat at long horizons.
EXPECTED OUTPUT	baseline_results_v0.csv, results_v0.md.
COMPLETION CHECKLIST	☐ Climatology uses train years only ☐ Skill scores computed ☐ Horizon-wise table complete
TIME AND MODE	Core research **1.0** h | Coding / data processing **4.0** h | Documentation **1.0** h | Quality control **1.0** h | Total **7.0** h Mode: Auto
DEPENDENCIES	Days 65-66.
FALLBACK STRATEGY	AR too heavy per village: fit per ecoregion with village offsets.

DAY 68 Baselines 3-4: Random Forest and XGBoost

Phase 6 | 7.0 h planned | Mode: Auto

MAIN OBJECTIVE	Train tabular tree models with proper blocked validation and explainability.

DAY 68 Baselines 3-4: Random Forest and XGBoost

Phase 6 | 7.0 h planned | Mode: Auto

ACTION STEPS	1. Feature set: lags 1-12 of anomalies, rolling means/SD (causal), SPI/SPEI, season, static covariates, ecoregion. 2. Tune with Optuna (40 trials) using spatially blocked inner validation. 3. Class weights/scale_pos_weight for imbalanced targets. 4. Compute SHAP on a sample; check importance stability across folds. 5. Results table v1.
TECHNICAL INSTRUCTIONS	Neighbour-averaged features are allowed only if they use neighbours' INPUT features at <= origin; never neighbour labels. Fit on training blocks; early stopping on validation blocks, never on test.
EXPECTED OUTPUT	rf_xgb_runs/, shap_summary.png, results_v1.csv.
COMPLETION CHECKLIST	☐ Blocked tuning logs saved ☐ No neighbour-label features ☐ SHAP stability checked
TIME AND MODE	Core research **1.0** h | Coding / data processing **4.0** h | Documentation **1.0** h | Quality control **1.0** h | Total **7.0** h Mode: Auto
DEPENDENCIES	Day 67.
FALLBACK STRATEGY	Compute-limited: use LightGBM or reduce trials to 20.

DAY 69 LSTM forecaster

Phase 6 | 7.0 h planned | Mode: Auto

MAIN OBJECTIVE	Train a sequence model and compare with tree baselines.
ACTION STEPS	1. Architecture: static embedding + 2-layer LSTM (hidden 128) over 24 months, masked inputs; heads for T1-T6. 2. Losses: Gaussian NLL for regression, BCE with pos_weight for classification; multi-task weights fixed at 1 initially. 3. Training: AdamW (lr 1e-3, wd 1e-4), cosine schedule, batch 512, early stopping on validation blocks, mixed precision. 4. 3 seeds; log curves; check calibration quickly. 5. Results table v2.
TECHNICAL INSTRUCTIONS	L(theta) = sum_h sum_k [0.5 log sigma^2 + (y - mu)^2 / (2 sigma^2)] + lambda * BCE. Handle missing inputs with a learned mask embedding and 'age since last observation' features. Gradient clipping at 1.0.
EXPECTED OUTPUT	lstm_runs/, curves.png, results_v2.csv.
COMPLETION CHECKLIST	☐ 3 seeds finished ☐ Training stable (no NaN) ☐ Validation-only reporting
TIME AND MODE	Core research **0.5** h | Coding / data processing **4.5** h | Documentation **1.0** h | Quality control **1.0** h | Total **7.0** h Mode: Auto
DEPENDENCIES	Days 66-68.
FALLBACK STRATEGY	GPU unavailable: reduce hidden size to 64, batch 256, train on subset; or use GRU.

DAY 70 Weekly review 10 and Temporal Transformer

Phase 6 | WEEKLY REVIEW | 7.0 h planned | Mode: Auto

MAIN OBJECTIVE	Review week 10 and implement the Temporal Transformer encoder.

DAY 70 Weekly review 10 and Temporal Transformer

Phase 6 | WEEKLY REVIEW | 7.0 h planned | Mode: Auto

ACTION STEPS	1. Run weekly checklist; check experiment registry integrity. 2. Implement encoder: linear input projection, month-of-year embedding + learned positional encoding, 3 layers, 4 heads, d_model 128, ff 256, dropout 0.1, key-padding mask. 3. Pooling: attention pooling over time -> z_i (128). 4. Train with the same heads, losses and protocol as the LSTM. 5. Compare with LSTM and XGBoost (validation).
TECHNICAL INSTRUCTIONS	Attention: Attn(Q,K,V) = softmax(QK^T/sqrt(d_k) + M) V. With only 24 time steps and 17-25 years of data the Transformer may not beat the LSTM or XGBoost; report whichever is better and keep the simpler model as the default fallback.
EXPECTED OUTPUT	transformer_runs/, results_v3.csv, week10_review.md.
COMPLETION CHECKLIST	☐ Padding mask verified by test ☐ 3 seeds finished ☐ Comparison table complete
TIME AND MODE	Core research **0.5** h | Coding / data processing **4.5** h | Documentation **1.0** h | Quality control **1.0** h | Total **7.0** h Mode: Auto
DEPENDENCIES	Day 69.
FALLBACK STRATEGY	Overfitting: smaller d_model (64), fewer layers (2), stronger dropout (0.3).

DAY 71 Village graph construction

Phase 6 | 7.0 h planned | Mode: Mixed

MAIN OBJECTIVE	Build the multi-relational village graph from proximity, roads, drainage and irrigation data, with reliability tests.
ACTION STEPS	1. Proximity edges: kNN (k=8) within 15 km (UTM 42N), weights exp(-d/d0). 2. Road edges: OSM road network (osmnx/networkx); snap villages to nodes; connect if network distance <= 10 km. 3. Drainage edges: MERIT Hydro flow direction/upstream area and OSM waterways; directed upstream->downstream links within a catchment. 4. Irrigation edges: OSM canals and any Sindh irrigation GIS obtained; connect villages sharing a canal reach within 5 km. 5. Assess reliability per edge type (OSM completeness audit on 20 samples); mask unreliable types.
TECHNICAL INSTRUCTIONS	Edge features: distance, edge type one-hot, elevation difference, flow direction flag. Use PyG Data(x, edge_index, edge_attr, edge_type). If irrigation data are incomplete, DO NOT impute connectivity; keep the type but mark coverage and include/exclude as an ablation. Edges are built from static geography, not from outcomes.
EXPECTED OUTPUT	graph_v1.pt, edge_reliability.csv, graph_stats.csv.
COMPLETION CHECKLIST	☐ Graph connected statistics reported ☐ Reliability audit done ☐ No outcome-derived edges
TIME AND MODE	Core research **1.0** h | Coding / data processing **4.5** h | Documentation **1.0** h | Quality control **0.5** h | Total **7.0** h Mode: Mixed (decision: Reliability threshold.)
DEPENDENCIES	Days 23, 38, 65.
FALLBACK STRATEGY	Irrigation/drainage data poor: use proximity + road graph (Tier 2 fallback).

DAY 72 Graph neural network baseline

Phase 6 | 7.0 h planned | Mode: Auto

MAIN OBJECTIVE	Train GraphSAGE/GCN/GAT baselines with inductive spatial hold-out.

DAY 72 Graph neural network baseline

Phase 6 | 7.0 h planned | Mode: Auto

ACTION STEPS	1. Node features = causal summaries (e.g. last-12-month anomalies + statics). 2. Models: GraphSAGE, GCN, GATv2 (2 layers) with NeighborLoader/Cluster-GCN mini-batches. 3. Inductive protocol: test nodes' features visible; no labels of test (or validation) nodes used. 4. Compare against XGBoost with neighbour-averaged inputs (same information). 5. Record results v4.
TECHNICAL INSTRUCTIONS	Message passing: h_i^(l+1) = sigma(sum_{j in N(i)} alpha_ij W h_j^(l)). To test if graph structure matters, use shuffled-edge and distance-only controls. A GNN that equals the neighbour-averaged XGBoost is evidence that graph attention adds little.
EXPECTED OUTPUT	gnn_runs/, results_v4.csv, edge_shuffle_control.csv.
COMPLETION CHECKLIST	☐ Edge-shuffle control run ☐ Test labels never in message passing ☐ Seeds >= 3
TIME AND MODE	Core research **0.5** h | Coding / data processing **4.5** h | Documentation **1.0** h | Quality control **1.0** h | Total **7.0** h Mode: Auto
DEPENDENCIES	Days 70-71.
FALLBACK STRATEGY	PyG mini-batching unstable: precompute neighbour-aggregated features and use MLP (graph-free approximation).

DAY 73 Spatial feature encoder and chip pipeline

Phase 6 | 7.0 h planned | Mode: Mixed

MAIN OBJECTIVE	Decide and build the spatial encoder: CNN, ViT-tiny, or zonal-statistics (+ optional embeddings).
ACTION STEPS	1. Define chip subset policy (Day 9): e.g. stratified 12,000 villages; 32x32 px seasonal composites at 10 m (S2) or 30 m (Landsat) centred on the settlement. 2. Export chips from GEE (sampleRectangles or image export to Cloud Storage as TFRecord/GeoTIFF) into Zarr. 3. Implement ResNet-lite (4 blocks) and ViT-tiny (patch 4); weight-share over time steps. 4. Alternative Tier-2 encoder: MLP over zonal statistics and histograms; optional AlphaEarth annual embeddings lagged by one year. 5. Run a small pilot: encoder-only prediction of NDVI anomaly.
TECHNICAL INSTRUCTIONS	Chip tensor: [B, T_s=8-16, C=8, 32, 32]. Augmentation: flips/rot90 only (no radiometric changes that mimic anomalies). AlphaEarth annual embeddings incorporate data across the calendar year: use only embeddings of years strictly before the forecast origin year to avoid leakage.
EXPECTED OUTPUT	chips.zarr (subset), spatial_encoder.py, pilot_results.md.
COMPLETION CHECKLIST	☐ Chip counts and sizes logged ☐ Encoder forward pass tested ☐ Leakage rule for embeddings in code
TIME AND MODE	Core research **1.0** h | Coding / data processing **4.5** h | Documentation **1.0** h | Quality control **0.5** h | Total **7.0** h Mode: Mixed (decision: Chip subset.)
DEPENDENCIES	Days 27-31, 66.
FALLBACK STRATEGY	Chips infeasible: Tier 2 (zonal statistics only) becomes the primary spatial encoder.

DAY 74 Climate encoder and cross-modal attention fusion

Phase 6 | 7.0 h planned | Mode: Auto

MAIN OBJECTIVE	Implement the climate branch and fusion module with ablation switches.

DAY 74 Climate encoder and cross-modal attention fusion

Phase 6 | 7.0 h planned | Mode: Auto

ACTION STEPS	1. Climate encoder: 1D-CNN (kernel 3-5) then 2-layer Transformer on CHIRPS/ERA5-Land sequences [B, L, F_c] -> [B, L, 128]. 2. Cross-modal attention: queries = temporal tokens, keys/values = concatenated spatial and climate tokens; heads=4. 3. Gated residual: z = g * Attn + (1 - g) * h_ts, g = sigmoid(W[h_ts; Attn]). 4. Unit tests for shapes and mask propagation. 5. Add config flags: use_spatial, use_climate, fusion in {concat, cross_attn}.
TECHNICAL INSTRUCTIONS	Fusion: CA(Q=H_t, K=V=[H_s; H_c]) = softmax(Q W_Q (K W_K)^T / sqrt(d)) K W_V. The concat fusion is the ablation control; if it matches cross-attention, prefer it. Document parameter counts.
EXPECTED OUTPUT	fusion.py, test_fusion.py, ablation_flags.yaml.
COMPLETION CHECKLIST	☐ Shape tests pass ☐ Flags switch components off cleanly ☐ Parameter counts logged
TIME AND MODE	Core research **0.5** h | Coding / data processing **4.5** h | Documentation **1.0** h | Quality control **1.0** h | Total **7.0** h Mode: Auto
DEPENDENCIES	Days 69-70, 73.
FALLBACK STRATEGY	Cross-attention unstable: replace by concatenation + MLP.

DAY 75 Village Graph Attention Network integration

Phase 6 | 7.0 h planned | Mode: Auto

MAIN OBJECTIVE	Integrate the graph attention module into the multimodal network, with end-to-end and two-stage training modes.
ACTION STEPS	1. Implement relational GATv2: separate attention per edge type or edge-type embeddings in attention; 2 layers x 4 heads, residual + LayerNorm. 2. Mode A: end-to-end with cluster mini-batches; Mode B: freeze fused embeddings z_i (precomputed) then train the GAT head. 3. Neighbour sampling honours the inductive protocol (no label leakage). 4. Verify memory on 2-3 pilot districts. 5. Compare Mode A and B on validation blocks.
TECHNICAL INSTRUCTIONS	alpha_ij = softmax_j(a^T LeakyReLU(W [h_i || h_j || e_ij])) (GATv2 uses a dynamic attention ordering). Two-stage training reduces memory and instability and is the fallback if end-to-end training fails.
EXPECTED OUTPUT	gat_module.py, graph_batching.py, modeA_vs_B.csv.
COMPLETION CHECKLIST	☐ Both modes run ☐ Memory profile logged ☐ Leakage test passes with graph
TIME AND MODE	Core research **0.5** h | Coding / data processing **4.5** h | Documentation **1.0** h | Quality control **1.0** h | Total **7.0** h Mode: Auto
DEPENDENCIES	Days 71-74.
FALLBACK STRATEGY	GAT too slow: use GraphSAGE mean aggregator or precomputed neighbour pooling.

DAY 76 Multivariate heads and uncertainty

Phase 6 | 7.0 h planned | Mode: Auto

MAIN OBJECTIVE	Implement regression/classification heads and a layered uncertainty system.

DAY 76 Multivariate heads and uncertainty

Phase 6 | 7.0 h planned | Mode: Auto

ACTION STEPS	1. Regression heads: mu and log sigma^2 for T1-T3 at h=3,6,12. 2. Classification heads for T4-T6 with focal or weighted BCE. 3. Deep ensembles (M=5) for epistemic uncertainty; optional MC dropout. 4. Conformal prediction: split conformal / CQR using spatially blocked calibration blocks. 5. Test uncertainty API on a toy dataset.
TECHNICAL INSTRUCTIONS	Predictive variance: sigma^2_tot = mean_m sigma_m^2 + var_m(mu_m). CQR intervals: [q_lo - Q, q_hi + Q] with Q the (1-alpha)(1+1/n) empirical quantile of conformity scores on calibration blocks. Conformal guarantees assume exchangeability, which spatial/temporal shift weakens; report empirical coverage under shift.
EXPECTED OUTPUT	uncertainty.py, conformal.py, test_uncertainty.py.
COMPLETION CHECKLIST	☐ Coverage test on toy data near nominal ☐ Ensemble inference script runs ☐ API documented
TIME AND MODE	Core research **0.5** h | Coding / data processing **4.5** h | Documentation **1.0** h | Quality control **1.0** h | Total **7.0** h Mode: Auto
DEPENDENCIES	Days 69-75.
FALLBACK STRATEGY	Ensembles too costly: use 3 members or snapshot ensembles.

DAY 77 Weekly review 11 and full model assembly

Phase 6 | WEEKLY REVIEW | 7.0 h planned | Mode: Auto

MAIN OBJECTIVE	Review week 11 and assemble VISTA-Net end to end with sanity checks.
ACTION STEPS	1. Run weekly checklist. 2. Assemble: spatial encoder + temporal Transformer + climate encoder + cross-modal fusion + GAT + heads. 3. Overfit-tiny-batch test (64 samples -> loss near 0); gradient flow check per module. 4. Profile speed/memory; set mixed precision and checkpoints. 5. Run the first end-to-end training on a 3-district pilot.
TECHNICAL INSTRUCTIONS	If the tiny-batch test fails, debug before any tuning. Parameter budget target: < 5 M parameters for the fused model to match data volume. Log module-wise gradient norms.
EXPECTED OUTPUT	vista_net.py, sanity_report.md, pilot_run/.
COMPLETION CHECKLIST	☐ Tiny-batch overfit passes ☐ All modules receive gradients ☐ Pilot trains without NaN
TIME AND MODE	Core research **0.5** h | Coding / data processing **4.5** h | Documentation **1.0** h | Quality control **1.0** h | Total **7.0** h Mode: Auto
DEPENDENCIES	Days 73-76.
FALLBACK STRATEGY	Debug loop too long: freeze modules and add them one at a time (progressive assembly).

DAY 78 Hyper-parameter selection

Phase 6 | 7.0 h planned | Mode: Auto

MAIN OBJECTIVE	Select hyper-parameters with a fixed budget and blocked validation only.
ACTION STEPS	1. Define search spaces per model class (lr, wd, hidden, heads, layers, dropout, k, loss weights). 2. Run Optuna with ASHA/successive halving; objective = validation-block loss (not test). 3. Cap trials: 30 per neural model, 40 per tree model. 4. Record the best configs and the budget used. 5. Lock hyper-parameters in configs/best_*.yaml.

DAY 78 Hyper-parameter selection

Phase 6 | 7.0 h planned | Mode: Auto

TECHNICAL INSTRUCTIONS	Report the full search log (all trials). Compare models with equal tuning effort to avoid inflating the proposed model. Test sets remain untouched until Day 83.
EXPECTED OUTPUT	optuna_studies/, best_configs/, tuning_report.md.
COMPLETION CHECKLIST	☐ Equal-effort rule documented ☐ Search logs saved ☐ Best configs hashed
TIME AND MODE	Core research **0.5** h | Coding / data processing **4.5** h | Documentation **1.0** h | Quality control **1.0** h | Total **7.0** h Mode: Auto
DEPENDENCIES	Days 68-77.
FALLBACK STRATEGY	Compute limited: tune only learning rate, hidden size, dropout and keep others fixed.

DAY 79 Full training runs on primary split

Phase 6 | 7.0 h planned | Mode: Auto

MAIN OBJECTIVE	Train all eight model classes with final configs and seeds on the primary split and prepare the CV job queue.
ACTION STEPS	1. Train: climatology, persistence/AR, RF, XGBoost, LSTM, Transformer, GNN, VISTA-Net with 3 seeds each. 2. Save checkpoints, predictions and logs with config hashes. 3. Prepare the job queue for spatial-CV folds (Day 81) and LODO runs (Day 82). 4. Run sanity checks (prediction ranges, calibration plots). 5. Update the experiment registry.
TECHNICAL INSTRUCTIONS	Do not evaluate on test partitions here. Use validation-block predictions only. Schedule long runs on Kaggle/Colab/local GPU with checkpoint resume. Keep a table of wall-clock time per model for the compute-cost discussion.
EXPECTED OUTPUT	predictions_val/*.parquet, checkpoints/, runtime_table.csv, job_queue.yaml.
COMPLETION CHECKLIST	☐ 8 model classes x 3 seeds trained ☐ Registry complete ☐ Job queue ready
TIME AND MODE	Core research **0.5** h | Coding / data processing **4.5** h | Documentation **1.0** h | Quality control **1.0** h | Total **7.0** h Mode: Auto
DEPENDENCIES	Day 78.
FALLBACK STRATEGY	Time shortage: reduce seeds to 2 for the baselines and keep 3 for the proposed model.

DAY 80 MILESTONE 6: model zoo frozen

Phase 6 | MILESTONE 6 | 7.0 h planned | Mode: Manual

MAIN OBJECTIVE	Close Phase 6: freeze the final candidates and choose the reporting architecture using the one-standard-error rule.
ACTION STEPS	1. Compare model classes on validation blocks (mean +/- SE over seeds/folds). 2. Apply the one-standard-error rule: choose the simplest model within 1 SE of the best. 3. Write model cards (inputs, training data, known limits). 4. Freeze code and configs (tag v0.6.0); archive checkpoints. 5. Document fallback tier chosen (Tier 1/2/3, Section 10).
TECHNICAL INSTRUCTIONS	Do not force VISTA-Net if simpler models are equivalent: the finding 'graph/multimodal fusion does not add skill beyond X' is a valid and publishable result when supported by ablations. After this milestone, only bug fixes (logged) are allowed.
EXPECTED OUTPUT	model_cards/, frozen_configs/, tag v0.6.0, model_selection_memo.md.

DAY 80 MILESTONE 6: model zoo frozen

Phase 6 | MILESTONE 6 | 7.0 h planned | Mode: Manual

COMPLETION CHECKLIST	☐ Selection rule applied and documented ☐ Model cards complete ☐ Milestone-6 file list complete
TIME AND MODE	Core research **1.5** h | Coding / data processing **2.0** h | Documentation **2.0** h | Quality control **1.5** h | Total **7.0** h Mode: Manual (decision: Architecture decision.)
DEPENDENCIES	Days 66-79.
FALLBACK STRATEGY	If VISTA-Net fails to train reliably, ship Tier-2 (tabular + temporal + optional GAT) as the main model.

Phase 7: Validation and scientific experiments (Days 81-90)

Spatial blocks, district hold-out, temporal and extreme-event tests, ablations, error analysis, calibration, OOD detection, regional comparison, independent reference validation. **Milestone 7 on Day 90.**

DAY 81 Spatially blocked cross-validation

Phase 7 | 7.0 h planned | Mode: Auto

MAIN OBJECTIVE	Run all models through 5-fold spatially blocked cross-validation with buffers.
ACTION STEPS	1. Load the H3 blocks and fold assignment from the manifest; verify buffers. 2. Run all model classes on 5 folds (fixed hyper-parameters). 3. Collect out-of-fold predictions; compute metrics per fold and pooled. 4. Block-bootstrap (resample blocks) for 95% CIs. 5. Plot random-split vs blocked results for a subset to demonstrate leakage inflation.
TECHNICAL INSTRUCTIONS	Report both naive random-split and blocked scores for XGBoost only as a diagnostic of spatial leakage; the paper's headline results use blocked/held-out designs. Never select models on the held-out folds of this run.
EXPECTED OUTPUT	cv_blocked_results.parquet, cv_summary.csv, leakage_inflation_plot.png.
COMPLETION CHECKLIST	☐ All folds complete ☐ Block-bootstrap CIs present ☐ Leakage-inflation figure produced
TIME AND MODE	Core research **1.0** h | Coding / data processing **4.0** h | Documentation **1.0** h | Quality control **1.0** h | Total **7.0** h Mode: Auto
DEPENDENCIES	Days 65, 79-80.
FALLBACK STRATEGY	Compute short: run 3 folds for neural models and 5 for baselines, state it.

DAY 82 District-held-out and ecological-region transfer

Phase 7 | 7.0 h planned | Mode: Auto

MAIN OBJECTIVE	Test generalisation to entire unseen districts and ecological regions.
ACTION STEPS	1. Leave-one-district-out (LODO) for >= 10 districts including Tharparkar, Badin, Sujawal, Dadu, Jacobabad, Khairpur. 2. Ecological-region transfer: train on region A, test on B (matrix of region pairs). 3. Compute metrics and skill vs climatology per held-out unit. 4. Summarise heterogeneity (range, IQR) and its covariates. 5. Document failures honestly.
TECHNICAL INSTRUCTIONS	A district hold-out removes the district's villages and a buffer from training. LODO is scored on non-locked blocks only; the locked test blocks stay unseen until Day 83. Report the whole distribution of held-out performance, not only the mean; worst-district performance matters for deployment.
EXPECTED OUTPUT	lodo_results.csv, region_transfer_matrix.csv, transfer_heatmap.png.

DAY 82 District-held-out and ecological-region transfer

Phase 7 | 7.0 h planned | Mode: Auto

COMPLETION CHECKLIST	☐ LODO for >= 10 districts ☐ Transfer matrix complete ☐ Worst-case districts listed
TIME AND MODE	Core research **1.0** h | Coding / data processing **4.0** h | Documentation **1.0** h | Quality control **1.0** h | Total **7.0** h Mode: Auto
DEPENDENCIES	Day 81.
FALLBACK STRATEGY	Compute limit: LODO for baselines + final model only; others in supplement 'planned'.

DAY 83 Temporal holdout and extreme-event testing

Phase 7 | 7.0 h planned | Mode: Auto

MAIN OBJECTIVE	Evaluate on the locked test years and on event-withheld experiments.
ACTION STEPS	1. Evaluate final models ONCE on test origins 2020-2025 (unseen blocks AND unseen years; plus years-only and blocks-only variants). 2. Rolling-origin evaluation (expanding window) as a robustness check on validation years. 3. Event-withheld retraining: remove 2010-2011 (flood), 2013-2014 (Thar drought) from training and test on those periods. 4. Stratify by extremeness (|SPI| >= 1.5, flood years) and compare skill. 5. Record all results, including negative ones.
TECHNICAL INSTRUCTIONS	This is the single permitted look at the locked test set; log the date, config hashes and results in the registry. Retrospective forecasting caveat: reanalysis/climate products are produced in hindsight; features are therefore 'pseudo-prospective' unless restricted to information available at the origin. Prospective evaluation (real-time forecasts) is a future extension.
EXPECTED OUTPUT	test_results_locked.csv, event_withheld.csv, extreme_stratified.csv.
COMPLETION CHECKLIST	☐ Test set evaluated exactly once ☐ Event-withheld experiments run ☐ Extreme vs normal skill table
TIME AND MODE	Core research **1.0** h | Coding / data processing **4.0** h | Documentation **1.0** h | Quality control **1.0** h | Total **7.0** h Mode: Auto
DEPENDENCIES	Days 80-82.
FALLBACK STRATEGY	If runs fail, rerun with identical configs and log the deviation; never tune against the test result.

DAY 84 Weekly review 12 and ablation studies

Phase 7 | WEEKLY REVIEW | 7.0 h planned | Mode: Auto

MAIN OBJECTIVE	Review week 12 and run the pre-registered ablation suite.
ACTION STEPS	1. Run weekly checklist. 2. Ablate: spatial encoder, climate encoder, cross-attention (-> concat), GAT (-> MLP), each edge type, uncertainty head. 3. Modality ablations: remove Sentinel-1, Dynamic World, DEM/static, MODIS LST. 4. Paired block-bootstrap comparisons with Holm correction. 5. Summarise which components matter and where.
TECHNICAL INSTRUCTIONS	Ablation effect = metric(full) - metric(ablated), CI by block bootstrap over the same blocks. Interpret small differences cautiously; multiple comparisons inflate false positives. Present a ranked component contribution table by target and horizon.
EXPECTED OUTPUT	ablation_results.csv, ablation_forest_plot.png, week12_review.md.
COMPLETION CHECKLIST	☐ All pre-registered ablations run ☐ Holm-adjusted p/CI reported ☐ Component ranking by target

DAY 84 Weekly review 12 and ablation studies

Phase 7 | WEEKLY REVIEW | 7.0 h planned | Mode: Auto

TIME AND MODE	Core research **1.0** h | Coding / data processing **4.0** h | Documentation **1.0** h | Quality control **1.0** h | Total **7.0** h Mode: Auto
DEPENDENCIES	Days 80-83.
FALLBACK STRATEGY	Too many ablations: prioritise graph, fusion, uncertainty, S1, climate; list the rest as 'not run'.

DAY 85 Error analysis

Phase 7 | 7.0 h planned | Mode: Mixed

MAIN OBJECTIVE	Diagnose where and why the models fail.
ACTION STEPS	1. Map residuals; compute Moran's I of residuals. 2. Error vs valid-observation fraction, cloud cover, land cover, settlement size, distance to river, region. 3. Horizon-wise error growth; compare with climatology. 4. Failure gallery: 30 worst villages with imagery and series. 5. Confusion matrices and precision-recall analysis for classification targets.
TECHNICAL INSTRUCTIONS	If residuals remain spatially autocorrelated, spatial structure is not captured (or targets contain unresolved region-wide shocks). Distinguish measurement error (cloud, sensor) from model error: stratify by n_valid.
EXPECTED OUTPUT	error_analysis.pdf, residual_maps.png, failure_gallery.pdf.
COMPLETION CHECKLIST	☐ Residual Moran's I reported ☐ Failure gallery of 30 villages ☐ Error vs n_valid analysis
TIME AND MODE	Core research **2.5** h | Coding / data processing **2.5** h | Documentation **1.0** h | Quality control **1.0** h | Total **7.0** h Mode: Mixed (decision: Interpretation.)
DEPENDENCIES	Day 83.
FALLBACK STRATEGY	Imagery gallery heavy: use thumbnails from S2 composites only.

DAY 86 Uncertainty calibration

Phase 7 | 7.0 h planned | Mode: Auto

MAIN OBJECTIVE	Evaluate and, if needed, recalibrate predictive uncertainty under spatial and temporal shift.
ACTION STEPS	1. Reliability diagrams and ECE for classification. 2. PICP (coverage), MPIW (width), CRPS for regression; compare ensemble, NLL and conformal intervals. 3. Error by uncertainty decile; sparsification curves. 4. Conditional coverage by region, year, season; coverage under spatial hold-out. 5. Recalibration (temperature scaling, isotonic) on calibration blocks only.
TECHNICAL INSTRUCTIONS	PICP = (1/n) sum 1[y in interval]; ECE = sum_b (n_b/n)|acc_b - conf_b|. Use proper scoring rules (Gneiting and Raftery 2007). Under distribution shift coverage often falls below nominal: report the shortfall, do not hide it.
EXPECTED OUTPUT	calibration_results.csv, reliability_plots.png, coverage_by_region.csv.
COMPLETION CHECKLIST	☐ Coverage by region reported ☐ Recalibration uses calibration blocks only ☐ Sparsification curves saved
TIME AND MODE	Core research **1.5** h | Coding / data processing **3.0** h | Documentation **1.0** h | Quality control **1.5** h | Total **7.0** h Mode: Auto
DEPENDENCIES	Days 76, 83.

DAY 86 Uncertainty calibration

Phase 7 | 7.0 h planned | Mode: Auto

FALLBACK STRATEGY	If calibration is poor, report uncalibrated and calibrated results side by side and restrict the claim.

DAY 87 Out-of-distribution detection and area of applicability

Phase 7 | 7.0 h planned | Mode: Mixed

MAIN OBJECTIVE	Assess whether the model can flag where it should not be trusted.
ACTION STEPS	1. Compute OOD scores: Mahalanobis distance in embedding space, dissimilarity index (Meyer and Pebesma), ensemble variance, input-missingness. 2. Test whether OOD scores predict high error (Spearman, AUROC of 'high error'). 3. Evaluate on held-out ecoregions and event years. 4. Produce the Area of Applicability map for Sindh. 5. Decide the masking rule for the final maps.
TECHNICAL INSTRUCTIONS	DI_i = d_i / dbar_train, with d_i the minimum standardised, importance-weighted distance to training samples. Threshold from the training cross-validation DI distribution (e.g. 3rd quartile + 1.5 IQR). Reports must show masked villages explicitly.
EXPECTED OUTPUT	ood_results.csv, aoa_map.png, masking_rule.md.
COMPLETION CHECKLIST	☐ AUROC for error detection reported ☐ AoA map produced ☐ Masking rule defined
TIME AND MODE	Core research **1.5** h | Coding / data processing **3.0** h | Documentation **1.0** h | Quality control **1.5** h | Total **7.0** h Mode: Mixed (decision: Masking rule.)
DEPENDENCIES	Days 83-86.
FALLBACK STRATEGY	OOD scores uninformative: rely on ensemble variance and valid-observation masks.

DAY 88 Comparison across ecological regions

Phase 7 | 7.0 h planned | Mode: Mixed

MAIN OBJECTIVE	Compare performance by ecoregion, target and horizon, including regions where skill is absent.
ACTION STEPS	1. Table: metric by ecoregion x target x horizon with CIs. 2. Skill vs climatology and vs best baseline by region. 3. Identify regions where model skill is not distinguishable from climatology. 4. Relate regional skill to predictability drivers (rain variability, cloud, irrigation). 5. Prepare the regional summary figure.
TECHNICAL INSTRUCTIONS	Express differences as effect sizes with CIs. In irrigated canal commands, vegetation dynamics are partly management-driven and may be less rainfall-predictable; in the Thar, rain-fed dynamics may be more predictable but coarse rainfall inputs limit skill. These are hypotheses to be tested, not assumptions.
EXPECTED OUTPUT	regional_performance.csv, regional_skill_figure.png.
COMPLETION CHECKLIST	☐ All regions covered ☐ No-skill regions flagged ☐ CIs present
TIME AND MODE	Core research **2.0** h | Coding / data processing **2.5** h | Documentation **1.5** h | Quality control **1.0** h | Total **7.0** h Mode: Mixed (decision: Interpretation.)
DEPENDENCIES	Days 82-87.
FALLBACK STRATEGY	Region sizes too small: merge adjacent regions and note it.

DAY 89 Independent validation against reference observations

Phase 7 | 7.0 h planned | Mode: Mixed

MAIN OBJECTIVE	Validate predictions and indicators against independent data not used for training.
ACTION STEPS	1. PBS Mouza Census (taluka): rank agreement between predicted/observed stress frequencies and reported disaster shares; brackish groundwater share vs salinity-sensitive indicators (hypothesis only). 2. 2022 floods: probabilities issued at origin before monsoon 2022 vs UNOSAT/Copernicus extents and district damage (PDNA). 3. Independent visual validation: 300 village-seasons labelled blind to predictions. 4. Gauge sanity checks for rainfall (PMD) if available. 5. Summarise strengths and limits per source.
TECHNICAL INSTRUCTIONS	Report as agreement statistics (Spearman, AUROC, F1) with CIs, and note the unit mismatch (aggregated vs village). No claim is made about drinking-water quality, migration or yield from these comparisons.
EXPECTED OUTPUT	independent_validation.csv, flood2022_check.png, validation_summary.md.
COMPLETION CHECKLIST	☐ Blind labelling protocol followed ☐ All comparisons reported ☐ Limits per source documented
TIME AND MODE	Core research **2.0** h | Coding / data processing **2.5** h | Documentation **1.0** h | Quality control **1.5** h | Total **7.0** h Mode: Mixed (decision: Interpretation.)
DEPENDENCIES	Days 54, 83.
FALLBACK STRATEGY	No reference data: strengthen visual validation and cite published event maps for plausibility.

DAY 90 MILESTONE 7: validation package and claims audit

Phase 7 | MILESTONE 7 | 7.0 h planned | Mode: Manual

MAIN OBJECTIVE	Close Phase 7: freeze results and decide which claims the evidence supports.
ACTION STEPS	1. Compile all result tables and figure source data into outputs/. 2. Run the leakage audit checklist (Section 11) and sign it. 3. Write the claims-evidence matrix: each manuscript claim linked to a table/figure and uncertainty. 4. Document negative and null results. 5. Tag v0.7.0; update risk register.
TECHNICAL INSTRUCTIONS	Allowed claim language depends on evidence: 'outperforms climatology in X regions at horizons Y' vs 'comparable to'. Prohibited: unsupported causal language, 'first' without search evidence, accuracy claims from random splits.
EXPECTED OUTPUT	results_package_v1/, leakage_audit_signed.md, claims_matrix.csv, tag v0.7.0.
COMPLETION CHECKLIST	☐ Every claim has evidence ☐ Audit checklist signed ☐ Milestone-7 file list complete
TIME AND MODE	Core research **2.0** h | Coding / data processing **1.5** h | Documentation **2.0** h | Quality control **1.5** h | Total **7.0** h Mode: Manual (decision: Claims decisions.)
DEPENDENCIES	Days 81-89.
FALLBACK STRATEGY	If key results are weak, reframe the manuscript as a benchmark/negative-results paper; do not extend experiments indefinitely.

Phase 8: Final maps, manuscript and research release (Days 91-100)

Provincial and village maps, uncertainty maps, figures and tables, reproducibility checks, manuscript sections, supplements, GitHub release, submission files. **Milestone 8 on Day 100.**

DAY 91 Weekly review 13 and province-wide inference

Phase 8 | WEEKLY REVIEW | 7.0 h planned | Mode: Auto

MAIN OBJECTIVE	Review week 13 and generate province-wide predictions with uncertainty.
ACTION STEPS	1. Run weekly checklist. 2. Run final models for all eligible villages at the latest available origin (with ensemble and conformal intervals). 3. Apply AoA masking and quality-class filters. 4. Save predictions Parquet + GeoParquet; verify counts against the inventory. 5. Compute province summaries by district.
TECHNICAL INSTRUCTIONS	Predictions for villages outside the area of applicability are suppressed or flagged. Use only information up to the latest complete month; store model version and data cutoff in metadata.
EXPECTED OUTPUT	predictions_province_v1.parquet, district_summary.csv, week13_review.md.
COMPLETION CHECKLIST	☐ All eligible villages predicted ☐ Masking applied ☐ Metadata complete
TIME AND MODE	Core research **0.5** h | Coding / data processing **4.0** h | Documentation **1.0** h | Quality control **1.5** h | Total **7.0** h Mode: Auto
DEPENDENCIES	Days 80, 87.
FALLBACK STRATEGY	Inference slow: batch by district and cache embeddings.

DAY 92 Provincial environmental maps

Phase 8 | 7.0 h planned | Mode: Auto

MAIN OBJECTIVE	Produce publication-quality provincial maps of indicators and trajectories.
ACTION STEPS	1. Maps: stress frequency by indicator, persistence ratio, recovery time, flood frequency, water persistence, trend maps, typology. 2. Cartography: EPSG:32642 or an appropriate projection, scale bar, north arrow, colour-blind-safe palettes. 3. Use hexagon aggregation for readability while keeping village-level data available. 4. Export vector PDFs and 300-dpi PNGs. 5. Write captions with data sources and caveats.
TECHNICAL INSTRUCTIONS	Choose diverging palettes for anomalies and sequential palettes for frequencies; avoid red-green. Always include the sample size per aggregation cell. QGIS print layouts or matplotlib + geopandas are both acceptable; script them (src/visualization/).
EXPECTED OUTPUT	outputs/maps/prov_*.pdf (8-10 maps), captions.md.
COMPLETION CHECKLIST	☐ Each map has legend/scale/source ☐ Colour-blind-safe check done ☐ Captions state caveats
TIME AND MODE	Core research **0.5** h | Coding / data processing **4.0** h | Documentation **1.5** h | Quality control **1.0** h | Total **7.0** h Mode: Auto
DEPENDENCIES	Days 55-65, 91.
FALLBACK STRATEGY	Rendering slow: aggregate to taluka/hex for the province and show village-level only in zooms.

DAY 93 Village-level prediction and uncertainty maps

Phase 8 | 7.0 h planned | Mode: Auto

MAIN OBJECTIVE	Create zoomed village-level maps with uncertainty and applicability overlays.
ACTION STEPS	1. Select 6 contrasting districts (e.g. Tharparkar, Badin, Sujawal, Dadu, Jacobabad, Khairpur). 2. Maps: predicted vegetation anomaly, stress persistence probability, interval width, AoA mask. 3. Create atlas v1.0 pages for example villages. 4. Export an interactive map for the supplement (static HTML). 5. Check privacy/ethics (settlement-level only; no household data).
TECHNICAL INSTRUCTIONS	Show uncertainty beside every prediction map. Use transparency or hatching for masked villages. Document that outputs are environmental-stress forecasts based on proxies and not assessments of household welfare.
EXPECTED OUTPUT	maps/district_*.pdf, atlas_v1.0.pdf, interactive_map.html.
COMPLETION CHECKLIST	☐ Uncertainty shown with every prediction ☐ 6 districts covered ☐ Ethics note included
TIME AND MODE	Core research **0.5** h | Coding / data processing **4.0** h | Documentation **1.5** h | Quality control **1.0** h | Total **7.0** h Mode: Auto
DEPENDENCIES	Day 92.
FALLBACK STRATEGY	Interactive map too large: provide district-level files only.

DAY 94 Scientific figures and tables

Phase 8 | 7.0 h planned | Mode: Auto

MAIN OBJECTIVE	Generate the final manuscript figures and tables from source data with consistent style.
ACTION STEPS	1. Fix the list: study area, workflow, inventory validation, indicator examples, trajectory typology, architecture, main results, ablations, calibration, regional performance. 2. Use one matplotlib style, vector output, 300-dpi rasters, consistent fonts and colours. 3. Generate tables: dataset catalogue, indicator definitions, model comparison, ablation, validation summary. 4. Write captions and alt-text; export source-data CSVs. 5. Internal review of every figure for clarity and honesty (axes, CIs).
TECHNICAL INSTRUCTIONS	Every figure must be reproducible with one command (make figures). Do not truncate axes to exaggerate differences; show uncertainty everywhere.
EXPECTED OUTPUT	outputs/figures/*.pdf, outputs/tables/*.csv, source_data/, captions.md.
COMPLETION CHECKLIST	☐ make figures runs from clean outputs ☐ Captions complete ☐ No figure without uncertainty where applicable
TIME AND MODE	Core research **0.5** h | Coding / data processing **3.5** h | Documentation **1.5** h | Quality control **1.5** h | Total **7.0** h Mode: Auto
DEPENDENCIES	Days 90-93.
FALLBACK STRATEGY	Time short: prioritise 6 main figures; move the rest to supplement.

DAY 95 Reproducibility checks and release engineering

Phase 8 | 7.0 h planned | Mode: Auto

MAIN OBJECTIVE	Prove the pipeline reproduces from a clean machine and prepare the public release.

DAY 95 Reproducibility checks and release engineering

Phase 8 | 7.0 h planned | Mode: Auto

ACTION STEPS	1. Fresh clone -> environment build -> run 'make reproduce-small' (pilot district) end to end. 2. Add Dockerfile/Apptainer or lock file; GitHub Actions for tests/lint. 3. Record dataset hashes and Earth Engine export manifests; note unavoidable non-reproducibility (asset updates). 4. Choose licences (code Apache-2.0; derived data CC BY 4.0 subject to upstream terms, e.g. ODbL/OSM). 5. Create a Zenodo release draft with DOI reservation.
TECHNICAL INSTRUCTIONS	Reproducibility levels: (1) re-run analysis from released feature store; (2) re-extract from Earth Engine; (3) re-train. Level 1 must be guaranteed; Level 2 depends on asset versions; state this in the Data and Code Availability statement.
EXPECTED OUTPUT	Dockerfile, CI workflow, reproduce_report.md, Zenodo draft.
COMPLETION CHECKLIST	☐ Clean-clone reproduction passes ☐ Licences compatible ☐ Hashes recorded
TIME AND MODE	Core research **0.5** h | Coding / data processing **4.0** h | Documentation **1.5** h | Quality control **1.0** h | Total **7.0** h Mode: Auto
DEPENDENCIES	Days 55, 80, 94.
FALLBACK STRATEGY	Docker not possible: provide conda-lock and a step-by-step REPRODUCE.md.

DAY 96 Manuscript I: Methods and supplementary methods

Phase 8 | 7.0 h planned | Mode: Manual

MAIN OBJECTIVE	Write the complete Methods section and Supplement S1 from the documentation.
ACTION STEPS	1. Write: study area, village inventory, datasets, indicators, trajectory analysis, models, validation, uncertainty. 2. Write Supplement S1: detailed parameters, thresholds, GEE IDs, versions. 3. Insert equations and algorithm boxes. 4. Check each statement against the code/configs. 5. Compile references (verified DOIs).
TECHNICAL INSTRUCTIONS	Write what was done, not what was planned. Mark deviations from the pre-analysis plan explicitly. Keep the Methods reproducible enough that a reader can rebuild each step.
EXPECTED OUTPUT	manuscript/methods.md (about 4,000-5,000 words), supplement_S1.md.
COMPLETION CHECKLIST	☐ All deviations listed ☐ Every parameter traceable to config ☐ Citations resolved
TIME AND MODE	Core research **1.0** h | Coding / data processing **0.5** h | Documentation **4.5** h | Quality control **1.0** h | Total **7.0** h Mode: Manual (decision: Wording of limitations.)
DEPENDENCIES	Days 55, 80, 95.
FALLBACK STRATEGY	Time short: write Methods in bullet-sentence form first and polish later.

DAY 97 Manuscript II: Results and abstract draft

Phase 8 | 7.0 h planned | Mode: Manual

MAIN OBJECTIVE	Write the Results section from the claims matrix and a first abstract.

DAY 97 Manuscript II: Results and abstract draft

Phase 8 | 7.0 h planned | Mode: Manual

ACTION STEPS	1. Write results in order: inventory validation, indicators, trajectories, baselines, proposed model, ablations, regional/spatial/temporal generalisation, uncertainty, independent validation. 2. State uncertainty with every result (CIs). 3. Include negative/null findings. 4. Draft a 250-word structured abstract. 5. Run a numbers check: every number in text matches a table/figure.
TECHNICAL INSTRUCTIONS	No adjectives that exceed the evidence ('excellent', 'robust') unless backed by effect sizes and CIs. Numbers must be generated by scripts (insert via templated values) to avoid transcription errors.
EXPECTED OUTPUT	manuscript/results.md, abstract_v1.md, numbers_check.csv.
COMPLETION CHECKLIST	☐ Numbers check passes ☐ Negative results included ☐ Abstract <= 250 words
TIME AND MODE	Core research **1.0** h | Coding / data processing **0.5** h | Documentation **4.5** h | Quality control **1.0** h | Total **7.0** h Mode: Manual (decision: Interpretation.)
DEPENDENCIES	Day 90, 96.
FALLBACK STRATEGY	If results are weak, emphasise the benchmark and diagnostic findings.

DAY 98 Weekly review 14 and manuscript III: Introduction and literature positioning

Phase 8 | WEEKLY REVIEW | 7.0 h planned | Mode: Manual

MAIN OBJECTIVE	Review week 14 and write the Introduction and related-work positioning.
ACTION STEPS	1. Run weekly checklist. 2. Write the Introduction: problem, gap, objective, contributions (as methodological contributions). 3. Write related work using the novelty matrix (Pakistan/Sindh, flood/salinity/wetland/drought/poverty, forecasting, GNN-RS, multimodal, uncertainty, spatial CV). 4. Check every 'novelty' sentence against the search log; use 'to our knowledge, within the searched sources'. 5. Verify all DOIs and fix reference formatting.
TECHNICAL INSTRUCTIONS	Contributions stated as: (i) leakage-audited, village-scale benchmark for environmental stress/recovery forecasting in Sindh; (ii) systematic test of graph and multimodal components against strong baselines; (iii) uncertainty and applicability analysis under spatial/temporal shift; (iv) open code and inventory. Do not claim first-of-kind.
EXPECTED OUTPUT	manuscript/introduction.md, related_work.md, references.bib (verified).
COMPLETION CHECKLIST	☐ All DOIs resolve ☐ Novelty sentences tied to search log ☐ Contributions list approved
TIME AND MODE	Core research **1.0** h | Coding / data processing **0.5** h | Documentation **4.5** h | Quality control **1.0** h | Total **7.0** h Mode: Manual (decision: Positioning.)
DEPENDENCIES	Days 4, 90, 97.
FALLBACK STRATEGY	Reference management slow: keep BibTeX from Zotero and run a DOI-check script.

DAY 99 Manuscript IV: Discussion, limitations, availability and ethics

Phase 8 | 7.0 h planned | Mode: Manual

MAIN OBJECTIVE	Write the Discussion, Limitations, Conclusion, Data/Code availability and Ethics statements.

DAY 99 Manuscript IV: Discussion, limitations, availability and ethics

Phase 8 | 7.0 h planned | Mode: Manual

ACTION STEPS	1. Discussion: what worked, what did not, why (hypotheses), practical use. 2. Limitations: proxies vs ground truth, retrospective forecasting, product changes, cloud, GRACE use, coordinate quality, spatial shift, short record, causal limits. 3. Future work: prospective forecasting, field validation, higher-resolution data, formal causal analysis. 4. Data and Code availability; ethics (no household data, no individual identification, potential misuse). 5. Prepare Supplement S2 (extra results) and S3 (validation protocols).
TECHNICAL INSTRUCTIONS	State explicitly that results are not measures of drinking-water quality, migration, welfare or yield. State what is retrospective and what would be needed for prospective operational use. Discuss responsible use: model outputs must not be used alone for aid targeting.
EXPECTED OUTPUT	manuscript/discussion.md, limitations.md, availability.md, supplements S2-S3.
COMPLETION CHECKLIST	☐ Limitations list >= 10 items ☐ Ethics statement written ☐ Future work separated from claims
TIME AND MODE	Core research **1.0** h | Coding / data processing **0.5** h | Documentation **4.5** h | Quality control **1.0** h | Total **7.0** h Mode: Manual (decision: Interpretive claims.)
DEPENDENCIES	Days 96-98.
FALLBACK STRATEGY	Short on time: finalise limitations first; shorten discussion.

DAY 100 MILESTONE 8: submission package and release

Phase 8 | MILESTONE 8 | 6.5 h planned | Mode: Manual

MAIN OBJECTIVE	Close the project: abstract, submission files, GitHub v1.0 release, and a candid retrospective.
ACTION STEPS	1. Finalise abstract, title, keywords, highlights, graphical abstract; format for the chosen journal. 2. Select journal after checking current scope/guidelines (e.g. Remote Sensing of Environment, ISPRS JPRS, Int. J. Applied Earth Observation and Geoinformation); prepare cover letter, CRediT statement, data availability. 3. Release GitHub v1.0 and Zenodo DOI; check README. 4. Internal red-team review: read as a hostile reviewer; list weaknesses and responses. 5. Retrospective: what was completed vs deferred (Section 17); plan the next 30 days.
TECHNICAL INSTRUCTIONS	Submission does not guarantee acceptance; plan for revision and possible rejection with a cascade of target journals. Archive everything needed to respond to reviewers (registry, logs, tests).
EXPECTED OUTPUT	submission/ (manuscript PDF/DOCX, figures, supplements, cover letter), GitHub v1.0, Zenodo DOI, retrospective.md.
COMPLETION CHECKLIST	☐ Journal checklist satisfied ☐ Release reproducible ☐ Milestone-8 file list complete
TIME AND MODE	Core research **1.5** h | Coding / data processing **1.0** h | Documentation **3.0** h | Quality control **1.0** h | Total **6.5** h Mode: Manual (decision: Journal choice.)
DEPENDENCIES	Days 1-99.
FALLBACK STRATEGY	If not submission-ready, submit to a preprint server (EarthArXiv) and complete the manuscript in the following weeks.

13. Milestones and required deliverables

Each milestone lists the files, datasets, scripts, figures, models and documents that must exist (file names are targets; folders as in Section 12.2). A milestone is passed only when every acceptance criterion is met; otherwise the shortfall is recorded and the next phase begins with the repair as its first task.

Milestone 1 - Day 10: Research foundation	**Protocol v1.0 frozen; stack proven**
Required artefacts	• charter.md, DECISIONS.csv, LOG.md • search_protocol.md, search_log.csv, evidence_table.csv (>=30 rows), methods_matrix.csv, novelty_matrix.md • pre_analysis_plan.md v1.0 + OSF/Git tag • environment.yml, requirements.lock, repo tree, Makefile, configs/ • dataset_catalogue.csv verified (assets_report.md), licences.md • blueprint.md, tensor_schema.yaml, compute_budget.md • smoke_test_report.md (5 villages), tag v0.1.0
Acceptance	☐ Hypotheses H1-H8 with falsification criteria ☐ Smoke test passes from clean clone ☐ All 17 dataset groups verified or replaced

Milestone 2 - Day 25: Village inventory	**Frozen inventory v1.0 with measured coverage**
Required artefacts	• admin_boundaries.gpkg, taluka_crosswalk.csv, district_history.csv • pbs_sindh_tables/*.csv + totals_reconciliation.csv • settlements_v1 polygons, village_registry.parquet, crosswalks, merge_log.csv • zones_core / zones_resource / zones_connectivity (GeoParquet) • inventory_validation.csv (precision, recall, kappa, CIs), coverage_by_district.csv • atlas_v0.1.pdf, interactive_map.html, data_dictionary.md, decision_memo_inventory.md, tag v0.2.0
Acceptance	☐ N villages reported from files (not assumed) ☐ Q1-Q4 shares by district ☐ Validation with CIs complete

Milestone 3 - Day 40: Harmonised dataset	**Aligned environmental store v0.9**
Required artefacts	• gee_zonal.py + task manifests • Zonal tables: Sentinel-2, Landsat, Sentinel-1, Dynamic World, MODIS, CHIRPS, ERA5-Land, water, terrain, context • env_store_v0.9 (Zarr/Parquet with masks, available_at) • completeness_report.pdf, cross-sensor reports, data_cards/*.md • alignment and no-future-timestamp tests, tag v0.3.0
Acceptance	☐ Completeness matrices exist ☐ Alignment tests pass ☐ Primary vegetation series chosen with evidence

Milestone 4 - Day 55: Indicators	**Indicator dataset v1.0 + handbook**
Required artefacts	• indicators_spec.yaml (frozen config with hash) • feature_store_v1.0 (all indicator families) • validation_I.md, validation_II.csv, threshold_sensitivity.csv • stress_events_v1, recovery_indicators_v1, spell_table • indicator_handbook.pdf; manuscript Methods (indicators) draft; tag v0.4.0
Acceptance	☐ Reproduction of 3 indicators by a second reader ☐ Unreliable indicators listed/excluded ☐ No composite variable

Milestone 5 - Day 65: Trajectories and datasets	**Leakage-audited forecasting dataset v1.0**
Required artefacts	• trends, decomposition, persistence, change-point and typology tables • ecoregions_v1, moran_results.csv, variograms.png, block_design.md • targets_definition.md, class_balance.csv, icc_table.csv • forecast_dataset_v1.0, split_manifest.parquet (hashed), leakage_audit.md • tag v0.5.0
Acceptance	☐ Leakage tests pass ☐ Block size justified by variogram range ☐ Class balance reported

Milestone 6 - Day 80: Model zoo	Frozen candidate models
Required artefacts	• data.py, metrics.py, baselines, lstm.py, transformer.py, gnn.py, vista_net.py, uncertainty.py, conformal.py • graph_v1.pt, edge_reliability.csv • tuning_report.md, best_configs/, runtime_table.csv • predictions_val/*.parquet, checkpoints/, model_cards/ • model_selection_memo.md (1-SE rule, tier decision), tag v0.6.0
Acceptance	☐ Tiny-batch overfit and gradient tests pass ☐ Equal-effort tuning documented ☐ Selected tier stated

Milestone 7 - Day 90: Validation	Results frozen; claims audit complete
Required artefacts	• cv_blocked_results, lodo_results, region_transfer_matrix, test_results_locked, event_withheld, extreme_stratified • ablation_results + forest plot • error_analysis.pdf, calibration_results, ood_results, aoa_map.png • regional_performance.csv, independent_validation.csv • leakage_audit_signed.md, claims_matrix.csv, results_package_v1/, tag v0.7.0
Acceptance	☐ Test set touched once (logged) ☐ Every claim linked to evidence ☐ Negative results documented

Milestone 8 - Day 100: Release and submission	Reproducible release and manuscript package
Required artefacts	• predictions_province_v1, maps (provincial + 6 districts), atlas_v1.0.pdf, interactive map • figures + tables + source data; make figures • Dockerfile/lock, CI, REPRODUCE.md, GitHub v1.0, Zenodo DOI • manuscript (all sections), supplements S1-S3, references.bib (verified) • submission/: cover letter, highlights, graphical abstract, CRediT, data availability; retrospective.md
Acceptance	☐ Clean-clone reproduction passes ☐ Numbers check passes ☐ Journal checklist satisfied

14. Daily productivity system and 100-day calendar

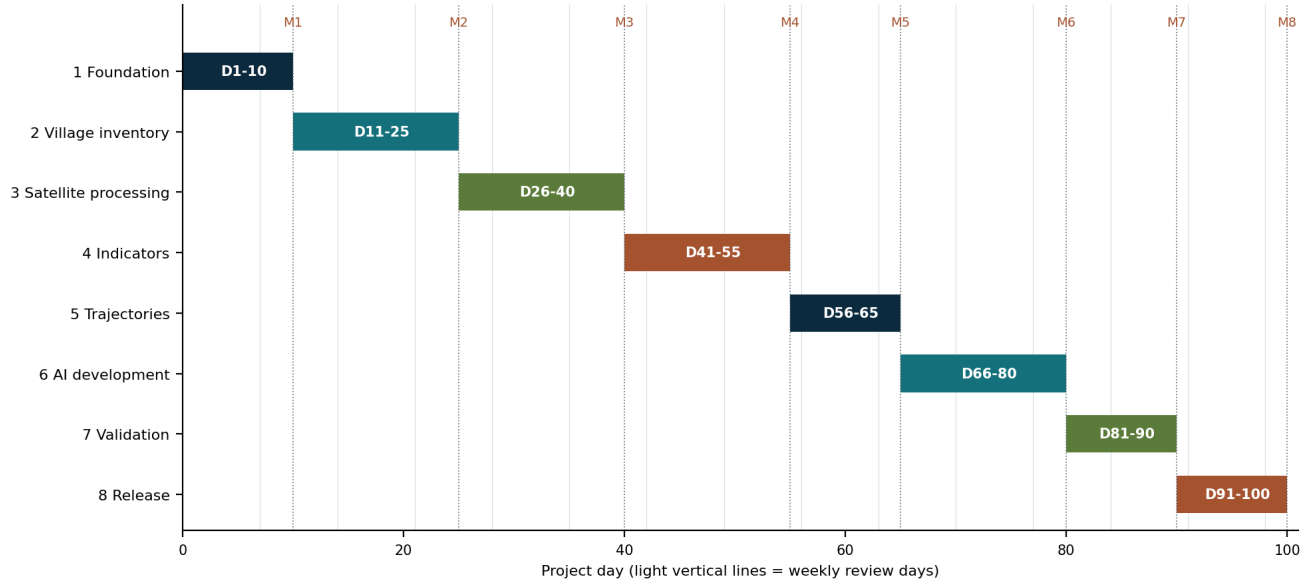


Figure 6. 100-day schedule by phase; light vertical lines mark weekly review days (7, 14, ..., 98).

14.1 Weekly rhythm

Weeks	Days	Focus
1	1-7	Charter, literature, hypotheses, environment, repository (review Day 7)
2	8-14	Dataset verification, blueprint, Milestone 1, PBS tables, boundaries, settlements (review Day 14)
3	15-21	Settlement polygons, names, deh, IDs, duplicates, missing coordinates (review Day 21)
4	22-28	Inventory validation, zones, atlas, Milestone 2, GEE framework, Sentinel-2 (review Day 28)
5	29-35	Sentinel-2/Landsat extraction, S1, Dynamic World, MODIS, CHIRPS (review Day 35)
6	36-42	ERA5-Land, water, DEM, context, Milestone 3, indicator spec, spectral indices (review Day 42)
7	43-49	Normalisation, productivity, rainfall, LST, water, floods, agriculture (review Day 49)
8	50-56	Settlement change, stress, recovery, validation, Milestone 4, EDA (review Day 56)
9	57-63	Trends, STL, persistence, recovery, change-points, clusters, autocorrelation (review Day 63)
10	64-70	Targets, Milestone 5, harness, baselines, LSTM, Transformer (review Day 70)
11	71-77	Graph, GNN, encoders, fusion, GAT, uncertainty, assembly (review Day 77)
12	78-84	Hyper-parameters, training, Milestone 6, CV, LODO, temporal tests, ablations (review Day 84)
13	85-91	Error, calibration, OOD, regions, independent validation, Milestone 7, inference (review Day 91)
14	92-98	Maps, figures, reproducibility, Methods, Results, Introduction (review Day 98)
15	99-100	Discussion/limitations, Milestone 8: submission package

14.2 Daily time blocks (5-7 focused hours)

Block	Typical time	Content
Plan	20-30 min	Read the day page; confirm dependencies; write the 3 outputs you will leave behind in LOG.md
Core research	per day page (R)	Scientific decisions, reading, parameter choices, interpretation (manual)
Coding / data processing	per day page (C)	Scripts, GEE jobs, model runs; start long jobs early, write documentation while they run
Documentation	per day page (D)	LOG.md, DECISIONS.csv, data/model cards, manuscript notes
Quality control	per day page (Q)	Tests, checklist, visual inspection, commit and tag; update experiment registry

Across the 100 daily plans the planned workload is 688 h: research 136 h, coding/data 310 h, documentation 134 h, quality control 109 h (mean 6.9 h/day). Days are marked Auto (45), Mixed (41) or Manual (14).

14.3 Weekly review checklist (every 7th day)

- All outputs and checklist items of the past 6 days exist (list missing items and schedule repair).
- LOG.md and DECISIONS.csv updated; any deviation from the pre-analysis plan recorded.
- Tests pass; repository tagged; dataset manifest and hashes updated.
- Risk register re-scored (likelihood x impact); triggers checked.
- Quota and compute check (Earth Engine tasks, GPU hours, storage).
- Measurable progress: share of checklist items completed, phase milestone distance, open decisions count.
- Plan the next 6 days; apply the cut list if more than one day behind.

14.4 Automation versus manual scientific decisions

Can be automated (scripts, scheduled jobs)	Requires manual scientific judgement
Downloads, parsing, boundary reprojection, clustering runs, ID generation, GEE exports and merges, indicator computation, tests, model training and tuning, figure generation, DOI checks.	Unit of analysis; cluster parameters; duplicate decisions; zone radius; cloud thresholds; primary sensor series; stress/recovery thresholds; target definitions; block size; model/tier selection; interpretation of failures; claims and wording; journal choice.

Manual decision days: 1, 3, 4, 5, 21, 22, 64, 80, 90, 96, 97, 98, 99, 100. Mixed days contain a named decision in the day page.

14.5 Slippage policy and cut list

All 100 days are individually specified, so there is no hidden buffer. If you fall more than one day behind, apply this cut list in order: (1) optional embeddings and WorldPop; (2) irrigation edges; (3) lower-priority ablations; (4) leave-one-district-out restricted to the baselines plus final model; (5) chips and the CNN/ViT encoder (adopt Tier 2). Protected items that are never cut: inventory validation, leakage tests and split manifest, baselines, spatial/temporal validation, uncertainty reporting and the manuscript Methods/Limitations.

15. Risk register and mitigation

Likelihood (L) and impact (I) are scored on Day 10 and re-scored in every weekly review.

ID	Risk	L	I	Early warning	Mitigation	Alternative method
R1	Missing village coordinates / record-level PBS list unavailable	High	High	Match rate below threshold; no response to requests by Day 20	Build inventory from footprints/OSM/GeoNames; quality classes Q1-Q4; train on Q1-Q2 only; use PBS taluka tables as aggregates	Settlement-only analysis without deh linkage; state coverage loss
R2	Inconsistent or duplicated village names (transliteration)	High	Med	Many ambiguous pairs in review queue	Name-independent VIDs; conservative merge rules; manual review of ambiguous pairs; variant dictionary	Operate on settlement IDs; names are metadata
R3	Cloud contamination (monsoon) and dust	High	Med	n_valid below minimum in Jul-Sep	Cloud Score+ masking, valid counts as features, SAR cross-checks, causal fill with age counters	Use MODIS/Landsat seasonal composites and S1 for wet season
R4	Incomplete historical coverage (S2 from 2017, GSW to 2021, S1B end, GRACE gap)	Cert.	Med	Gaps visible in completeness matrices	Backbone from MODIS/Landsat/CHIRPS (2001-); sensor-specific periods; source flags	Restrict specific targets (flood) to 2016-2025
R5	Earth Engine processing limits and quotas	Med	High	Tasks fail/time out; quota warnings	Chunking, yearly stacks, tileScale, retries, overnight jobs, quota log	Reduce to resource zone/pilot districts; download rasters and compute locally
R6	Large computational requirements (chips, GNN, ensembles)	Med	High	Epoch time or memory exceeds plan	Subset chips; mixed precision; two-stage GAT; ensembles of 3; Kaggle/Colab	Tier 2/1/0 (Section 10)
R7	Missing ground truth	Cert.	High	No village-level reference observations	Use aggregated PBS tables, published event maps, blind visual validation, gauges; claim only agreement	State limitation; no accuracy claims on proxies
R8	Model overfitting	Med	Med	Train-validation gap; unstable seeds	Blocked validation, early stopping, weight decay, small models, 1-SE rule, ensembles	Choose simpler tier
R9	Spatial and temporal leakage	Med	V.high	Random-split score far above blocked score	Blocks + buffers from variogram range; train-only fits; availability dates; automated tests; audit	Withdraw affected results; rerun with corrected split
R10	Insufficient time (100 days, single researcher)	High	High	More than 1 day behind at a weekly review	Cut list; automate; protect core deliverables; early long-running jobs	Tier 0/1 paper plus explicit future work
R11	Non-causal or revised data products	Med	High	Inputs differ across product versions; GF products	Availability = window end; exclude GF inputs; record versions; archive extracts	Descriptive use only
R12	Flood false positives (paddies, wind) and short flood record	High	Med	Implausible flood frequency in rice areas	HAND/cropping masks; relative thresholds; validate vs 2022 products; low-confidence pre-2015 layers	Regress water-fraction anomaly instead of event classification
R13	Climate input too coarse for village differences	High	Med	Village differences collapse to the same pixel	Cell-id covariate; local static covariates; interpret region-scale skill	Report skill at taluka scale
R14	Misuse of outputs (aid targeting, stigma)	Med	High	Requests to use outputs for decisions	Ethics statement; AoA masking; uncertainty always shown; no household data; no welfare claims	Restrict release to aggregated maps if needed

16. Novelty assessment and literature positioning

No 'first of its kind' claim is made. The table lists what was found during preparation (items marked in the reference list were opened and checked). A systematic, logged search (Days 2-4) must establish what else exists before any novelty statement is written. The manuscript will say 'to our knowledge, within the sources searched up to [date]'.

Topic	What exists (examples found)	What remains unresolved	VISTA contribution to be tested
Flood mapping / hazards in Sindh and Pakistan	S1-based 2022 flood extent for Pakistan (Chen et al. 2024, RSE); S1 change detection in Sindh (Ghouri et al. 2024); S1/S2 flood and Rabi-2023 impacts (arXiv 2304.10603); causes of the 2022 event (Nanditha et al. 2023). A broader flood-susceptibility literature exists and is to be tabulated on Day 3.	Event mapping dominates; village-scale multi-season forecasting under spatial and temporal hold-out is not demonstrated in these sources.	Flood indicators and a next-monsoon inundation target evaluated under blocked validation.
Rural poverty mapping in Sindh	Ensemble CNN on satellite imagery predicts chronic poverty at 1 km cells in rural Sindh with district hold-out and an independent survey (Agyemang et al. 2023, PLOS ONE).	Static welfare mapping; environmental dynamics and persistence/recovery are not its target.	Complementary: VISTA forecasts environmental stress proxies, not welfare; no linkage claimed.
Soil salinity	Spatial-temporal salinity in Sujawal combining field and satellite data, 1990-2017 (Agronomy 2019, 9:869). Other Sindh salinity studies to be tabulated.	District-scale static or trend maps; no province-wide forecasting with transfer tests.	Salinity-sensitive indices as optional covariates; PBS brackish-groundwater share as aggregated reference only; no salinity claims.
Wetland monitoring	Global water products (JRC GSW; Pekel et al. 2016). Sindh wetland-specific studies not compiled in this plan (search on Day 3).	To be determined by systematic search.	Water persistence indicators checked on known wetlands.
Drought and environmental degradation	Rainfall-NDVI drought monitoring for Tharparkar using CHIRPS (Remote Sensing 2020, 12:580); NDVI+SPI in the Thar (Bilal et al. 2017).	Region-specific and descriptive; no multi-region village-level forecasting with spatial transfer.	Province-wide persistence/recovery indicators and forecasts across ecoregions.
Spatiotemporal forecasting	Temporal Fusion Transformer (Lim et al. 2021); LSTM in hydrology (Kratzert et al. 2018); graph forecasting STGCN and Graph WaveNet (traffic).	Generic methods; application to village-level environmental stress with leakage-audited validation is the open question.	Benchmark with strong baselines and ablations.
Graph-based remote sensing	GNN-RNN for crop yield using geospatial and temporal information (Fan et al. 2022); GAT/GATv2/PyG toolkits.	Whether graph attention adds skill beyond neighbour-averaged features at village scale in Sindh is untested here.	Edge-type ablations and edge-shuffle controls (H6).
Multimodal satellite AI	Pixel-set encoder with temporal attention (Garnot et al. 2020); MOSAIKS (Rolf et al. 2021); Prithvi (Jakubik et al. 2023); AlphaEarth embeddings (Google DeepMind 2025).	Embeddings and foundation models are strong competitors; leakage and transfer behaviour for village forecasting need testing.	Cross-modal fusion tested against concatenation and lagged-embedding baselines.
Spatial validation and uncertainty	Spatial CV (Roberts et al. 2017; Ploton et al. 2020); area of applicability (Meyer and Pebesma 2021, 2022); critique of spatial CV (Wadoux et al. 2021); conformal prediction (Angelopoulos and Bates 2021).	Seldom combined with village-scale forecasting, event-withheld tests and AoA masking in South Asian settings (to be confirmed by search).	Integrated validation framework, released splits and tests.

16.1 Positioning as a methodological contribution

- **Benchmark contribution.** A leakage-audited, village-scale forecasting benchmark with released splits, baselines and tests; generalisation to unseen blocks, districts, regions and extreme years is the headline.
- **Scientific contribution.** Evidence on persistence and recovery of environmental stress (event-based, indicator by indicator), and on whether graph and multimodal components add skill (including honest null results).
- **Methodological detail.** Practical protocol for hindcast integrity with products that incorporate future information; uncertainty and area-of-applicability reporting.

- **Resource contribution.** Open village inventory with quality classes and code; not a regional application only.

16.2 Search protocol (summary)

Databases: Scopus/Web of Science (or OpenAlex). Period 2010-2026. Strings combine geography (Sindh, Pakistan, Indus basin) with topics (flood, salinity, drought, wetland, poverty, forecasting) and methods (graph neural network, transformer, multimodal, remote sensing). Record queries, dates, hit counts, screening decisions. Add grey literature (PBS, PDNA 2022, UN agencies). Maintain evidence_table.csv and methods_matrix.csv; verify every DOI.

17. Research deliverables, feasibility and manuscript plan

17.1 What is completed within 100 days versus future extensions

Completed (target of the 100 days)	Future extensions
Inventory v1.0 with measured coverage and quality classes; zones; atlas.	Deh-level linkage if boundaries are obtained later; household/community surveys.
2001-2025 environmental store; indicators with documented reliability.	Higher-resolution (sub-metre) imagery, commercial data, and additional sensors.
Eight-model comparison, ablations, spatial/temporal/event validation, calibration, OOD/AoA.	Prospective operational forecasting with versioned real-time inputs.
Provincial and village maps with uncertainty; reproducible release; manuscript draft.	Causal analysis of interventions; field validation of salinity, water quality, yield (not inferable from this data).

17.2 Manuscript blueprint

Section	Words (approx.)	Content
Abstract	250	Problem, data, methods, key quantitative findings with uncertainty, limits
Introduction	900	Environmental stress in Sindh; gap; objective; contributions (non-first-claim)
Related work	700	Novelty matrix (Section 16) in prose
Study area and data	800	Inventory, datasets, epistemic status
Methods	2,500	Zones, indicators, trajectories, models, validation, uncertainty
Results	2,500	Inventory validation; indicators; persistence/recovery; model comparison; ablations; generalisation; uncertainty; independent validation
Discussion and limitations	1,200	Interpretation, negative results, proxies, retrospective design, ethics
Conclusion; availability	300	Claims limited by claims matrix; code/data/licences

Target journals to check against current scope, length and fees: Remote Sensing of Environment, ISPRS Journal of Photogrammetry and Remote Sensing, International Journal of Applied Earth Observation and Geoinformation; fallback cascade: Remote Sensing, Science of Remote Sensing, Environmental Research Letters (check suitability on Day 100). Acceptance cannot be guaranteed within or after 100 days.

17.3 Planned figures and tables

Figures	Tables
1 Study area and ecoregions; 2 Workflow; 3 Inventory validation; 4 Indicator examples (2022 flood, Thar drought); 5 Persistence/recovery typology; 6 Architecture; 7 Main model comparison with CIs; 8 Ablation forest plot; 9 Spatial/temporal generalisation; 10 Calibration and AoA; 11 Province map; 12 District maps.	1 Datasets and epistemic status; 2 Indicator definitions; 3 Inventory coverage and accuracy; 4 Baselines and models; 5 Generalisation experiments; 6 Ablations; 7 Uncertainty; 8 Independent validation; supplement: hyper-parameters, tuning logs, per-district results.

18. References and useful links

Items marked [verified] were opened during preparation of this plan. Items marked 'found in a cited-reference list' were seen only as citations in another paper; open them and confirm bibliographic details and DOIs on Day 3/98. DOIs for the other entries (mostly classical methods papers) were not individually re-opened during preparation; resolve every DOI with the Day-3 DOI script before relying on it.

[1] Gorelick N. et al. (2017). Google Earth Engine: planetary-scale geospatial analysis for everyone. Remote Sens. Environ. 202:18-27. doi:10.1016/j.rse.2017.06.031

[2] Drusch M. et al. (2012). Sentinel-2: ESA's optical high-resolution mission for GMES operational services. Remote Sens. Environ. 120:25-36. doi:10.1016/j.rse.2011.11.026

[3] Torres R. et al. (2012). GMES Sentinel-1 mission. Remote Sens. Environ. 120:9-24. doi:10.1016/j.rse.2011.05.028

[4] Roy D.P. et al. (2016). Characterization of Landsat-7 to Landsat-8 reflective wavelength and NDVI continuity. Remote Sens. Environ. 185:57-70. doi:10.1016/j.rse.2015.12.024

[5] Wulder M.A. et al. (2019). Current status of Landsat program, science, and applications. Remote Sens. Environ. 225:127-147. doi:10.1016/j.rse.2019.02.015

[6] Brown C.F. et al. (2022). Dynamic World, near real-time global 10 m land use land cover mapping. Sci. Data 9:251. doi:10.1038/s41597-022-01307-4

[7] Pekel J.-F. et al. (2016). High-resolution mapping of global surface water and its long-term changes. Nature 540:418-422. doi:10.1038/nature20584

[8] Funk C. et al. (2015). The climate hazards infrared precipitation with stations (CHIRPS). Sci. Data 2:150066. doi:10.1038/sdata.2015.66

[9] Munoz-Sabater J. et al. (2021). ERA5-Land: a state-of-the-art global reanalysis dataset for land applications. Earth Syst. Sci. Data 13:4349-4383. doi:10.5194/essd-13-4349-2021

[10] Landerer F.W., Swenson S.C. (2012). Accuracy of scaled GRACE terrestrial water storage estimates. Water Resour. Res. 48:W04531. doi:10.1029/2011WR011453

[11] Rodell M. et al. (2018). Emerging trends in global freshwater availability. Nature 557:651-659. doi:10.1038/s41586-018-0123-1

[12] Huete A. et al. (2002). Overview of the radiometric and biophysical performance of the MODIS vegetation indices. Remote Sens. Environ. 83:195-213. doi:10.1016/S0034-4257(02)00096-2

[13] McFeeters S.K. (1996). The use of the Normalized Difference Water Index (NDWI) in the delineation of open water features. Int. J. Remote Sens. 17:1425-1432. doi:10.1080/01431169608948714

[14] Xu H. (2006). Modification of NDWI to enhance open water features in remotely sensed imagery. Int. J. Remote Sens. 27:3025-3033. doi:10.1080/01431160600589179

[15] Vicente-Serrano S.M. et al. (2010). A multiscalar drought index sensitive to global warming: the SPEI. J. Climate 23:1696-1718. doi:10.1175/2009JCLI2909.1

[16] AlphaEarth Foundations team (2025). AlphaEarth Foundations helps map our planet in unprecedented detail (Satellite Embedding dataset in Earth Engine). Google DeepMind blog: deepmind.google/discover/blog/alphaearth-foundations-helps-map-our-planet-in-unprecedented-detail/ [verified]

[17] Verbesselt J. et al. (2010). Detecting trend and seasonal changes in satellite image time series. Remote Sens. Environ. 114:106-115. doi:10.1016/j.rse.2009.08.014

[18] Hamed K.H., Rao A.R. (1998). A modified Mann-Kendall trend test for autocorrelated data. J. Hydrol. 204:182-196. doi:10.1016/S0022-1694(97)00125-X

[19] Anselin L. (1995). Local indicators of spatial association - LISA. Geogr. Anal. 27:93-115. doi:10.1111/j.1538-4632.1995.tb00338.x

[20] Scheffer M. et al. (2009). Early-warning signals for critical transitions. Nature 461:53-59. doi:10.1038/nature08227

[21] Dakos V. et al. (2012). Methods for detecting early warnings of critical transitions in time series illustrated using simulated ecological data. PLoS ONE 7:e41010. doi:10.1371/journal.pone.0041010

[22] Vaswani A. et al. (2017). Attention is all you need. arXiv:1706.03762

[23] Velickovic P. et al. (2018). Graph attention networks. ICLR. arXiv:1710.10903

[24] Brody S., Alon U., Yahav E. (2022). How attentive are graph attention networks? ICLR. arXiv:2105.14491

[25] Fey M., Lenssen J.E. (2019). Fast graph representation learning with PyTorch Geometric. arXiv:1903.02428

[26] Hochreiter S., Schmidhuber J. (1997). Long short-term memory. Neural Comput. 9:1735-1780. doi:10.1162/neco.1997.9.8.1735

[27] Breiman L. (2001). Random forests. Mach. Learn. 45:5-32. doi:10.1023/A:1010933404324

[28] Chen T., Guestrin C. (2016). XGBoost: a scalable tree boosting system. KDD. doi:10.1145/2939672.2939785

[29] Lim B. et al. (2021). Temporal Fusion Transformers for interpretable multi-horizon time series forecasting. Int. J. Forecast. 37:1748-1764. doi:10.1016/j.ijforecast.2021.03.012

[30] Yu B., Yin H., Zhu Z. (2018). Spatio-temporal graph convolutional networks. IJCAI. arXiv:1709.04875

[31] Wu Z. et al. (2019). Graph WaveNet for deep spatial-temporal graph modeling. IJCAI. arXiv:1906.00121

[32] Fan J. et al. (2022). A GNN-RNN approach for harnessing geospatial and temporal information: application to crop yield prediction. AAAI. arXiv:2111.08900

[33] Sainte Fare Garnot V. et al. (2020). Satellite image time series classification with pixel-set encoders and temporal self-attention. CVPR. arXiv:1911.07757

[34] Kratzert F. et al. (2018). Rainfall-runoff modelling using LSTM networks. Hydrol. Earth Syst. Sci. 22:6005-6022. doi:10.5194/hess-22-6005-2018

[35] Reichstein M. et al. (2019). Deep learning and process understanding for data-driven Earth system science. Nature 566:195-204. doi:10.1038/s41586-019-0912-1

[36] Ma L. et al. (2019). Deep learning in remote sensing applications: a meta-analysis and review. ISPRS J. Photogramm. Remote Sens. 152:166-177. doi:10.1016/j.isprsjprs.2019.04.015

[37] Zhu X.X. et al. (2017). Deep learning in remote sensing: a comprehensive review and list of resources. IEEE Geosci. Remote Sens. Mag. 5:8-36. doi:10.1109/MGRS.2017.2762307

[38] Rolf E. et al. (2021). A generalizable and accessible approach to machine learning with global satellite imagery. Nat. Commun. 12:4392. doi:10.1038/s41467-021-24638-z

[39] Jean N. et al. (2016). Combining satellite imagery and machine learning to predict poverty. Science 353:790-794. doi:10.1126/science.aaf7894

[40] Jakubik J. et al. (2023). Foundation models for generalist geospatial artificial intelligence (Prithvi). arXiv:2310.18660

[41] Akiba T. et al. (2019). Optuna: a next-generation hyperparameter optimization framework. KDD. arXiv:1907.10902

[42] Roberts D.R. et al. (2017). Cross-validation strategies for data with temporal, spatial, hierarchical, or phylogenetic structure. Ecography 40:913-929. doi:10.1111/ecog.02881

[43] Ploton P. et al. (2020). Spatial validation reveals poor predictive performance of large-scale ecological mapping models. Nat. Commun. 11:4540. doi:10.1038/s41467-020-18321-y

[44] Meyer H., Pebesma E. (2021). Predicting into unknown space? Estimating the area of applicability of spatial prediction models. Methods Ecol. Evol. 12:1620-1633. doi:10.1111/2041-210X.13650

[45] Meyer H., Pebesma E. (2022). Machine learning-based global maps of ecological variables and the challenge of assessing them. Nat. Commun. 13:2208. doi:10.1038/s41467-022-29838-9

[46] Wadoux A.M.J.-C. et al. (2021). Spatial cross-validation is not the right way to evaluate map accuracy. Ecol. Model. 457:109692. doi:10.1016/j.ecolmodel.2021.109692

[47] Gal Y., Ghahramani Z. (2016). Dropout as a Bayesian approximation. ICML. arXiv:1506.02142

[48] Lakshminarayanan B. et al. (2017). Simple and scalable predictive uncertainty estimation using deep ensembles. NeurIPS. arXiv:1612.01474

[49] Kendall A., Gal Y. (2017). What uncertainties do we need in Bayesian deep learning for computer vision? NeurIPS. arXiv:1703.04977

[50] Guo C. et al. (2017). On calibration of modern neural networks. ICML. arXiv:1706.04599

[51] Romano Y., Patterson E., Candes E. (2019). Conformalized quantile regression. NeurIPS. arXiv:1905.03222

[52] Angelopoulos A.N., Bates S. (2021). A gentle introduction to conformal prediction and distribution-free uncertainty quantification. arXiv:2107.07511

[53] Ovadia Y. et al. (2019). Can you trust your model's uncertainty? Evaluating predictive uncertainty under dataset shift. NeurIPS. arXiv:1906.02530

[54] Gneiting T., Raftery A.E. (2007). Strictly proper scoring rules, prediction, and estimation. J. Am. Stat. Assoc. 102:359-378. doi:10.1198/016214506000001437

[55] Saito T., Rehmsmeier M. (2015). The precision-recall plot is more informative than the ROC plot when evaluating binary classifiers on imbalanced datasets. PLoS ONE 10:e0118432. doi:10.1371/journal.pone.0118432

[56] Chen F. et al. (2024). Pakistan's 2022 floods: spatial distribution, causes and future trends from Sentinel-1 SAR observations. Remote Sens. Environ. 304:114055. doi:10.1016/j.rse.2024.114055 [found in a cited-reference list; open and verify on Day 3]

[57] Ghouri A.Y. et al. (2024). Flood mapping using the Sentinel-1 SAR dataset and application of the change detection approach technique (CDAT) to the Google Earth Engine in Sindh Province, Pakistan. Ecol. Questions 35(2). doi:10.12775/EQ.2024.024 [verified]

[58] Analyzing the knock-on impacts of 2022 floods on Rabi 2023 using remote sensing and field surveys. arXiv:2304.10603 [verified]

[59] DeVries B. et al. (2020). Rapid and robust monitoring of flood events using Sentinel-1 and Landsat data on the Google Earth Engine. Remote Sens. Environ. 240:111664. [found in a cited-reference list; DOI to resolve on Day 98]

[60] Nanditha J.S. et al. (2023). The Pakistan flood of August 2022: causes and implications. Earth's Future 11:e2022EF003230. doi:10.1029/2022EF003230 [verify on Day 3]

[61] Agyemang F.S.K., Memon R., Wolf L.J., Fox S. (2023). High-resolution rural poverty mapping in Pakistan with ensemble deep learning. PLoS ONE 18:e0283938. doi:10.1371/journal.pone.0283938 [verified]

[62] An assessment of the spatial and temporal distribution of soil salinity in combination with field and satellite data: a case study in Sujawal District. Agronomy 2019, 9(12):869. doi:10.3390/agronomy9120869 [verified; add authors from the DOI record]

[63] A spatio-temporal analysis of rainfall and drought monitoring in the Tharparkar region of Pakistan. Remote Sens. 2020, 12(3):580. doi:10.3390/rs12030580 [verified; add authors from the DOI record]

[64] Bilal M., Liaqat M.U., Cheema M.J.M., Mahmood T., Khan Q. (2017). Spatial drought monitoring in Thar Desert using satellite-based drought indices and geo-informatics techniques. Proceedings 2(5):179. doi:10.3390/ecws-2-04948 [verified]

Useful links

Resource	URL
PBS Mouza Census 2020 (hub)	https://www.pbs.gov.pk/mouza-census/
PBS Sindh Mouza Census 2020 provincial report (PDF)	https://www.pbs.gov.pk/sites/default/files/agriculture/publications/mouza_census_2020/mouza_census_2020_sindh_report.pdf
PBS Mouza Census 2020 country report (PDF)	https://www.pbs.gov.pk/wp-content/uploads/2020/07/mouza_census_2020_country_report.pdf
Earth Engine data catalogue	https://developers.google.com/earth-engine/datasets
Geofabrik Pakistan extract	https://download.geofabrik.de/asia/pakistan.html
GeoNames dump	https://download.geonames.org/export/dump/
HDX (OCHA boundaries)	https://data.humdata.org/
Microsoft building footprints	https://github.com/microsoft/GlobalMLBuildingFootprints
Overture Maps	https://overturemaps.org
JRC GHSL	https://ghsl.jrc.ec.europa.eu/
WorldPop	https://www.worldpop.org/
CHIRPS	https://www.chc.ucsb.edu/data/chirps
JRC Global Surface Water	https://global-surface-water.appspot.com/
LP DAAC MODIS products	https://lpdaac.usgs.gov/
PyTorch Geometric	https://pytorch-geometric.readthedocs.io/

Appendix A. Project charter template

Field	Content (fill on Day 1)
Project / version / date	VISTA-Sindh; charter v1.0; ____
Primary question	(copy from Section 3.1)
Sub-questions Q1-Q3	(falsifiable statements)
Unit of analysis	Settlement-based village (VID); deh as admin link; taluka/district as strata
Study period / region	2001-2025; rural Sindh; urban cores excluded
In scope	Inventory, indicators, forecasting, validation, maps, manuscript
Out of scope (non-goals)	Drinking-water quality; migration/displacement; yield; household welfare; causal attribution; operational real-time service; field campaigns
Success criteria (process)	Reproducible pipeline; leakage audit signed; all claims backed by evidence; no accuracy targets
Constraints	Free data; single researcher; 5-7 h/day; Earth Engine and GPU quotas
Risks (top 5)	R1, R5, R7, R9, R10 (Section 15)
Decision rights	Researcher decides; deviations logged in DECISIONS.csv

Appendix B. Asset verification list (Day 8)

All IDs below are hypotheses to be confirmed by scripts/verify_assets.py. Record outcomes in assets_report.md and replace any missing ID with its successor or an uploaded asset.

Asset ID	Check
COPERNICUS/S2_SR_HARMONIZED	Start date over Sindh; band names; scale
GOOGLE/CLOUD_SCORE_PLUS/V1/S2_HARMONIZED	Band cs_cdf; linkCollection works
COPERNICUS/S1_GRD	IW VV/VH counts per orbit; S1B end
LANDSAT/LT05|LE07|LC08|LC09/C02/T1_L2	Scale factors; QA bits
GOOGLE/DYNAMICWORLD/V1	First image over Sindh
MODIS/061/MOD13Q1, MOD11A2, MYD11A2, MOD16A2GF, MOD17A2HGF	Versions; QA bands
JRC/GSW1_4/MonthlyHistory, GlobalSurfaceWater	Last month (expected 2021-12)
UCSB-CHG/CHIRPS/DAILY	Last date; units mm/day
ECMWF/ERA5_LAND/MONTHLY_AGGR	Band names; PET sign
NASA/GRACE/MASS_GRACE_LAND_V04	Existence; gap months (regional context only)
USGS/SRTMGL1_003; COPERNICUS/DEM/GLO30; MERIT/Hydro/v1_0_1	Vertical datum; bands elv, upa, hnd
JRC/GHSL/P2023A/GHS_POP, GHS_BUILT_S, GHS_SMOD	Epochs available
WorldPop/GP/100m/pop	Years available
GOOGLE/SATELLITE_EMBEDDING/V1/ANNUAL (optional)	Years; band names
NOAA/VIIRS/DNB/ANNUAL_V21; Oxford/MAP/accessibility_to_cities_2015_v1_0 (optional)	Existence; years
GOOGLE/Research/open-buildings/v3/polygons (optional)	Pakistan coverage